PDQUEST™

User Guide for Version 6

Windows

P/N 4000129 RevA

PDQUEST User Guide

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Bio-Rad is not responsible for the misinterpretation of results obtained by following the instructions in this guide. Whenever possible, you should contact the Technical Services Department at Bio-Rad to discuss your results. As with all scientific studies, we recommend that you repeat your experiment at least once before making any significant conclusions for presentation or publication.

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Preface

1. About This Document

This user guide is designed to be used as a reference in your everyday use of PDQUEST™. It provides detailed information about the tools and commands of PDQUEST for the Windows 95 / 98 / NT 4.0 platforms.

This guide assumes you have a working knowledge of your Windows operating system and its conventions, including how to use a mouse and standard menus and commands, and how to open, save, and close files. For help with any of these techniques, see the documentation that came with your computer.

This guide uses certain text conventions to describe specific commands and functions.

Example	**Indicates**
File > Open	Choosing the Open command under the File menu.
Dragging	Positioning the cursor on an object and holding down the left mouse button while you move the mouse.
Control+s	Holding down the Control key while typing the letter *s*.
Right-click / Left-click / Double-click	Clicking the right mouse button / Clicking the left mouse button / Clicking the left mouse button twice.

2. Bio-Rad Listens

The staff at Bio-Rad are receptive to your suggestions. Many of the new features and enhancements in this version of PDQUEST are a direct result of conversations with our customers. Please let us know what you would like to see in the next version of PDQUEST by faxing, calling, or e-mailing our Technical Services staff. You can also use Solobug (installed with PDQUEST) to make software feature requests.

1. Introduction

1.1. An Overview of PDQUEST

PDQUEST is a software package for imaging, analyzing, and databasing 2-D electrophoresis gels.

The software runs in a Windows environment and has an easy-to-use graphical interface with standard pull-down menus, toolbars, and keyboard commands.

PDQUEST collects data from gels scanned with one of several Bio-Rad imaging systems. An image of a sample is captured using the controls in the imaging device window and displayed on your computer monitor. Image processing and analysis operations are performed using commands from the menus and toolbars.

Once a gel has been scanned, advanced algorithms are available to remove background noise, gel artifacts, and horizontal or vertical streaking from the image. PDQUEST then uses a spot segmentation facility to detect and quantify protein spots. Protein spots that change over time can be traced, quantified, displayed on-screen, and exported to other applications for statistical analysis.

The software can be used to simultaneously analyze up to 10,000 spots on a given 2-D gel. PDQUEST can track and report all of the protein pattern and concentration changes in the sample, making feasible both qualitative and quantitative analysis of the data generated from 2-D gels.

The spot matching and databasing facilities make it possible to objectively compare hundreds of different gels. Histograms allow you to quickly compare the amount of the same spot in each gel within an experimental series. Spots can be grouped into user-defined sets. Higher level comparisons and analysis can be performed on master images that you generate from your experimental gels.

Scan files can be transferred into or exported out of Discovery Series applications. Scans can be converted into TIFF format for easy compatibility with other applications. Match data can be exported into spreadsheets for easy analysis.

1.2. A Brief Overview of 2-D Gel Electrophoresis

2-D gel electrophoresis combines the techniques of isoelectric focusing with SDS polyacrylamide gel electrophoresis. Since these two separation methods rely on independent properties of proteins, chemical (IEF) and physical (SDS), this procedure can resolve complex biological samples and polypeptide mixes. The extraordinary resolution capabilities of SDS-PAGE make it useful for studying complex systems in great detail.

2-D gels generate high resolution separation of proteins. The isoelectric focusing (IEF) dimension is most often run in cylindrical tube gels. After focusing, the cylindrical gel is placed on top of an SDS-PAGE slab gel which is then run as usual. Proteins can be visualized by metabolic radiolabeling or a variety of staining methods.

Gene Expression Studies

2-D gel electrophoresis is an excellent technique for the study of differential gene expression under various growth conditions. Since the expression and regulation of individual proteins can be detected, 2-D gels are an indirect way to monitor gene activity. They allow for the investigation of quantitative as well as qualitative changes in cellular protein expression.

The environmental conditions of a cell can be changed in order to determine optimal growth conditions as well as monitor the cell's response to different stresses. Environmental conditions that can stress the cell include changes in temperature, pH and nutrient availability. Some examples of the chemical stresses that can be placed on a cell are drug and hormone treatments. Since protein structure and function are the direct result of gene expression, the loss or change of a protein as detected by a 2-D gel can be extrapolated back to events occurring at the DNA level.

Many questions encountered when genes are inserted and expressed in bacteria, yeast, and other cell types can be answered with 2-D analysis: Is the cell making the protein? Is the cell's progeny making the protein? Is the protein being made but not secreted? Have mutations occurred?

Medical Applications

2-D gels can also have important applications in medical research. For example, this technique can be used to verify the presence of specific protein markers that are linked to genetic diseases and disease states. Used in conjunction with other tests, 2-D gels can be part of medical screening procedures associated with mutations and teratology linked to genetic damage.

Growth factors are being studied for their role in the regulation of cell growth. 2-D gels can be used to evaluate quantitative and qualitative changes in cellular proteins in response to growth factor stimulation.

2-D gels allow the visualization of proteins whose expression is altered as the result of cell transformation, introducing oncogenes into the host genome. Assessment of phosphorylation, sulfation, or other secondary modifications could reveal functional protein pathways affected by oncogene expression. This information could contribute to a better understanding of cell growth and regulation.

Sample Experiments

Valuable information can be learned by exposing cells to a set of specific experimental conditions and subsequently examining their biological response.

A preliminary in vitro experiment that is useful when beginning 2-D gel work is to radiolabel a cell's proteins to steady-state with ^{35}S-methionine. Running gels of these proteins will help to determine the commonly expressed proteins in the cell under normal conditions.

Subsequently, other amino acids such as ^{3}H-leucine, ^{3}H-proline, ^{3}H-lysine, etc., can be used to radiolabel the proteins to steady-state. This will label any proteins that do not contain methionine and were, therefore, not detected in the first experiment, and will provide preliminary information on the amino acid composition.

Other experiments complementary to 2-D gels include: cell fractionation procedures, and post-translational modifications (phosphorylation, methylation, etc.). Data from these experiments can be added to a database, accumulating information about these proteins.

1.3. Digital Data and Signal Intensity

The Bio-Rad imaging devices supported by PDQUEST are light and/or radiation detectors that convert signals from biological samples into digital data. PDQUEST then displays the digital data on your computer screen, in the form of gray scale or color images.

A data object as displayed on the computer is composed of tiny individual screen pixels. Each pixel has an X and Y coordinate, and a value Z. The X and Y coordinates are the pixel's horizontal and vertical positions on the image, and the Z value is the signal intensity of the pixel.

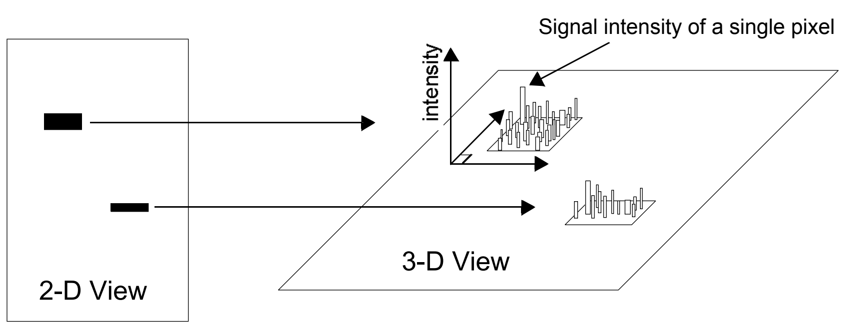

Fig.1-1. Representation of the pixels in two digitally imaged bands in a gel.

For a data object to be visible and quantifiable, the intensity of its clustered pixels must be higher than the intensity of the pixels that make up the background of the image. The total intensity of a data object is the sum of the intensities of all the pixels that make up the object. The mean intensity of a data object is the total intensity divided by the number of pixels in the object.

The units of signal intensity are Optical Density (O.D.) in the case of the GS-700 and 710 densitometers, the Gel Doc with a white light source, or the Fluor-S and Fluor-S MAX MultiImagers with white light illumination. Signal intensity is expressed in counts when using the Personal FX or FX, or in the case of the Gel Doc, Fluor-S, or Fluor-S MAX when using the UV light source.

1.4. Steps Involved in PDQUEST

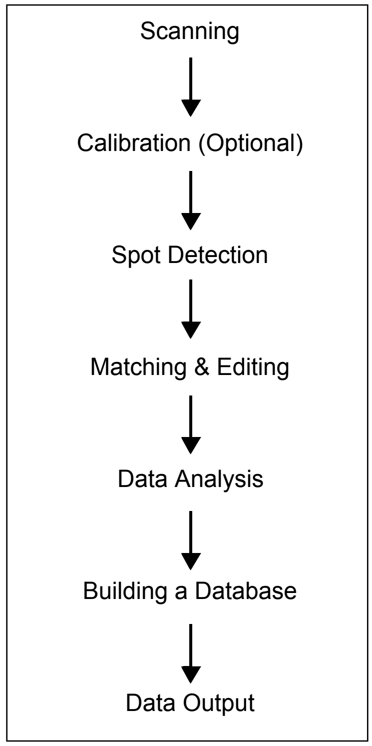

Fig.1-2. Steps in the Analysis of 2-D Gels.

Scanning

Before you can use PDQUEST to analyze a gel, you need to capture its image into your computer using a Bio-Rad imaging system. The resulting images can be stored on your hard disk, network file server, or removable storage media.

PDQUEST supports the imaging of the following types of gels and films:

Radiolabeled (^{35}S, ^{3}H, ^{14}C, ^{32}P, ^{33}P, ^{125}I) gel films, wet, silver, and Coomassie-stained gels, and duplicate contact films from stained gels.

Calibration (Optional)

Calibration strips can be used by PDQUEST to create a calibration curve that establishes the relationship between film density and radioactivity per unit area in the gel.[1] Calibration strips consisting of segments of known radioactivity, are processed simultaneously with ^{35}S, ^{3}H, or ^{14}C radiolabeled gels. The data from each calibration strip is used by PDQUEST to compare the average optical density of the individual segments of the calibration strip with the average disintegrations per minute (DPM) of each sampled segment, and assemble a calibration curve based on that data. If the curve is acceptable, you save the experiment and thereby save the associated calibration information.

Spot Detection

Once your gel has been scanned, the next step is the detection and quantitation of protein spots. If you have multiple exposures of the same gel image, and those exposures are calibrated, the spot detection routine in PDQUEST “merges” the available information in each exposure to produce a single **Gel Image** . If you have uncalibrated gels or single exposures, a Gel Image is still made, but the information comes from a single 2-D scan.

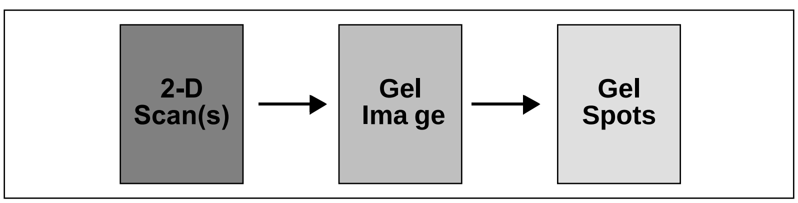

Fig.1-3. Images created and processed during spot detection.

The advantage of having multiple gel exposures is that you can obtain more quantitative information from the proteins on the gel. On a dark exposure, faint spots are quantifiable but dark spots are saturated and cannot be quantitated. On a light exposure, faint spots may not be detectable. Merging

1. See Garrels, J.I., Farrar, J.T., and Burwell, C.B., *J. Biol. Chem.*, **254**, 7961-7977 [1979]

lets you quantitate faint spots from the dark exposures and uses light exposures to quantitate dark spots.

The merged Gel Image is processed to remove streaks and background noise. Spots are then detected and quantified from the processed Gel Image, and a new, synthetic, image called a **Gel Spots** file is created. Spot quantitation is stored as part of the Gel Spots file.

Spot detection and quantitation are performed using 2-dimensional Gaussian modeling. This method of detection will accurately detect and quantify overlapped spots as well as isolated spots. The end result is a Gel Spots file that represents each detected spot as an ideal Gaussian with associated DPM or PDU (protein data unit) values.

Spot detection reports image data in units of optical density (O.D.). O.D. is calculated by dividing the incident light intensity by the transmitted light intensity, and taking the log (base 10) of this quotient:

Optical Density (O.D.) = $\log_{10}$(Intensity $\boldsymbol{i}$ / Intensity $\boldsymbol{t}$), where:

> Intensity $\boldsymbol{i}$ = incident light intensity, and
> Intensity $\boldsymbol{t}$ = transmitted light intensity.

Thus, an O.D. of 0.0 means that all the light is transmitted through the film, whereas an O.D. of 2.0 means that only 1 percent ($\log_{10}$ [100 / 1]) of the light is transmitted through the film.

Spot detection requires several steps—detecting spot centers and quantifying spots are the last two steps in the overall process. The Auto-Detect Spots function (Quantify > Auto-Detect Spots) performs all the necessary steps automatically. These steps are outlined below:

Create Gel Image — Using calibration strip data, you can merge multiple scans derived from different film exposures of the same gel into a single gel image. This increases the effective dynamic range of the film such that both very faint and very dark spots can be accurately quantified.

Each pixel of a 2-D scan is originally assigned an O.D. value based upon a step tablet calibration. For radiolabeled gels that include calibration strips,

	O.D. values are converted to DPMs; for uncalibrated or non-radiolabeled gels, linear interpolations of O.D.s are used to express spot quantitation. These images are called O.D. Gels.
Initial Smoothing	Gel Images are passed through an initial smoothing algorithm to remove densitometer noise.
Background Subtraction	After the initial smoothing pass, background density is subtracted from the image. This includes vertical and horizontal streaks as well as film fog and overall background density caused by the carrier medium (film, gel matrix, or blot matrix).
Final Smoothing	One or more additional smoothing passes are necessary to remove extraneous spots at or near the background level. The exact number of passes is empirically determined (e.g., 2 passes for radiolabeled gels).
Locate Spots in Gel Image	This step locates the center and position of each recognizable spot in the Gel Image. This is the spot detection step of the process.
Fit and Quantify Spots	This step fits ideal Gaussian distributions to spot centers. Quantitation for each spot is determined by calculating the volume under the Gaussian.

The final fitting step of spot detection produces a gel spots file for each scan. A gel spots file is a list of all the detected spots, their X and Y coordinates, and their associated Gaussian parameters (X-sigma, Y-sigma, and height). PDQUEST uses this information to store gel data and to display a synthetic representation of the gel image on the computer screen.

Matching and Editing

After you have spot-detected a set of gels, you will probably want to compare the protein spots from one gel with those from another to examine changes in protein levels under experimental conditions. Groups of gels can be edited and matched to one another in a "matchset." Matchsets consist of gel spots files and can include gel images. In a matchset, protein spots are matched to each other, enabling you to compare their quantities.

One member of your matchset can be used as a template for creating a matchset Standard. The Standard combines all the unique, matchable spots in a matchset into a single file.

Data Analysis

PDQUEST provides a variety of analytical tools to help you determine which sets of protein spots are statistically and scientifically meaningful.

Since it is virtually impossible to analyze all your experimental data at once, PDQUEST offers methods for breaking down the information into smaller, more manageable pieces. This is done by creating sets of spots called “user sets.” All the spots in a user set satisfy qualitative, quantitative, or statistical criteria that you specify.

Building a Database

As the amount of protein data that you collect increases, it will become important to store that information in an organized and accessible format. PDQUEST provides several functions for building and analyzing a protein spot database, enabling you to store and compare data from various experiments using higher level matchsets.

Data Output

When your analysis is complete, you can print your experimental data or export it to another system for further analysis.

1.5. System Requirements

The following are the **minimum** system requirements for installing and running PDQUEST on a PC:

Operating system:	Windows NT 4.0 recommended. Windows 95/98 acceptable.

Processor:	Pentium 166.
RAM:	64 MB for Gel Doc and Fluor-S imaging systems. 128 MB for FX, Personal FX, and densitometers.
Hard disk space:	3 GB. Recommended: Removable storage media (such as an Iomega Jaz drive) or a network file server.
Monitor:	17" monitor, 1024 x 768 resolution, 256 colors.
SCSI:	Required to run all Bio-Rad imaging devices except the Gel Doc. Adaptec SCSI card with EZ-SCSI software.
Printer:	Optional.

1.6. Installing PDQUEST

Before attempting to install and use PDQUEST, you should be familiar with your Windows operating system.

PDQUEST can be installed from a CD-ROM or you can download the installation program from the Internet. Make sure that Windows is up and running on your computer before attempting to install PDQUEST. We recommend that you check your hard disk with a disk utility program (such as Norton Disk Doctor) before loading PDQUEST.

Installing PDQUEST from a CD-ROM

Insert the Discovery Series™ CD-ROM into the CD-ROM drive on your computer.

Downloading from the Internet

You can download PDQUEST from the Internet using Bio-Rad's FTP site. Using your browser, go to ftp://ftp.genetics.bio-rad.com/Public/Discovery Series/Windows/Pdquest 6.0 and click on Pdquest.exe. This will download the installer onto your computer. Then you can double-click on the Setup.exe icon on your desktop to begin running the installer.

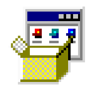

Fig.1-4. Setup.exe icon.

The Installation Program

The PDQUEST installation program will guide you through a series of screens. The installer will create a default directory tree under Program Files on your hard disk called Bio-Rad / The Discovery Series / Bin (you can select your own directory if you wish). The main PDQUEST program will be placed in the Bin directory. Additional directories for storing user profiles and sample images will also be created

The installer will place a PDQUEST icon on your desktop and create a Discovery Series folder in the Programs folder on your Windows Start menu.

Finally, after installation is complete, the installer will ask you if you want to open PDQUEST.

1.7. Starting PDQUEST

A hardware protection key (HPK) has been provided with your PDQUEST software.

Most computers will not need an HPK to run PDQUEST. However, if you attempt to start PDQUEST and receive an “Unable to obtain authorization” message, turn off your computer, attach the HPK, and restart the software.

Note: Some parallel port devices such as zip drives may be incompatible with HPKs. Please check with your peripherals vendor.

1.7.a. Attaching the Hardware Protection Key

Before attaching the HPK, you should first attempt to start PDQUEST without the HPK. If PDQUEST opens successfully, you can skip this section.

The code for the HPK is EYYCY. The code is printed in the lower left corner of the HPK label.

Before proceeding, please turn off your computer. If you have a printer attached to your computer's parallel port, please turn that off as well.

The HPK attaches to the parallel port on the back of your PC. If a printer cable is attached to this port, disconnect it. After you have attached the HPK, you can attach the printer cable to the key itself and restart your computer and printer.

You will also need to install the system driver that allows your PDQUEST software to "read" the HPK.

Note: Windows NT users must be in the local administrator group to install the HPK driver.

To install the driver, click on your Windows Start menu and select Programs > The Discovery Series. (This directory will not exist until you run the PDQUEST installer.) Select Install HASP HPK driver to begin installation. After the driver is installed, a message box will appear confirming the installation.

1.7.b. Opening PDQUEST

The installation program creates a PDQUEST application icon on your desktop. To start the program, double-click on this icon.

Fig.1-5. PDQUEST application icon.

The installer also creates a Discovery Series folder in the Programs folder on your Windows Start menu. You can start PDQUEST by double-clicking on the icon in this folder.

1.8. Registering PDQUEST

When you open PDQUEST, the Main Registration Screen will pop up welcoming you to PDQUEST.

Note: If you receive a message saying "Unable to obtain authorization," attach the HPK if you haven't already (refer to the previous section). If you have already attached your HPK, make sure that it is attached correctly and your HPK driver is installed. If you continue to have problems, contact Bio-Rad technical service at the numbers listed in section 1.9., Contacting Bio-Rad, below.

The type of screen you see will depend on whether you have an HPK attached or not.

Without HPK

If you do not have an HPK attached, you can register to receive a 30-day free trial of the software.

Fig.1-6. Main Registration Screen without HPK

Click on the Free Trial button. This will open the Software License Registration Form.

With HPK

If you have an HPK attached, the Main Registration Screen will reflect the fact that you receive a 30-day temporary license with your HPK ("Your license will expire on _______").

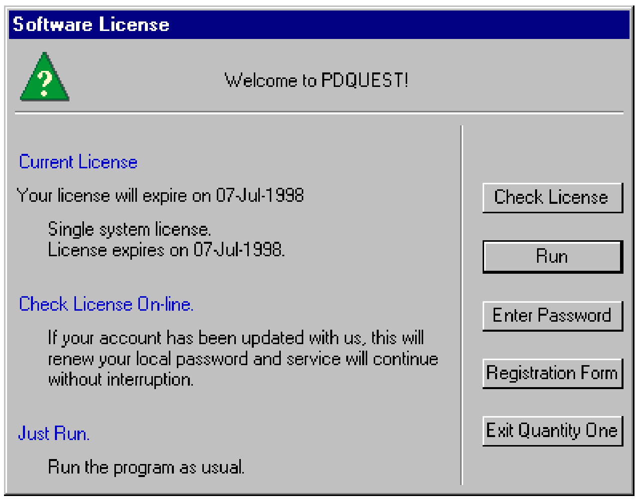

Fig.1-7. Main Registration Screen with HPK.

Click on the Run button to begin using the software.

Some time during the 30-day trial period, click on the Registration Form button to register your software. If your 30-day period has expired, a Free Trial button will appear when you open the software. Click on this button to register your software.

This will open the Software License Registration Form.

1.8.a. Software License Registration Form

To obtain your software license, you will need to fill out in the information in the Software License Registration Form.

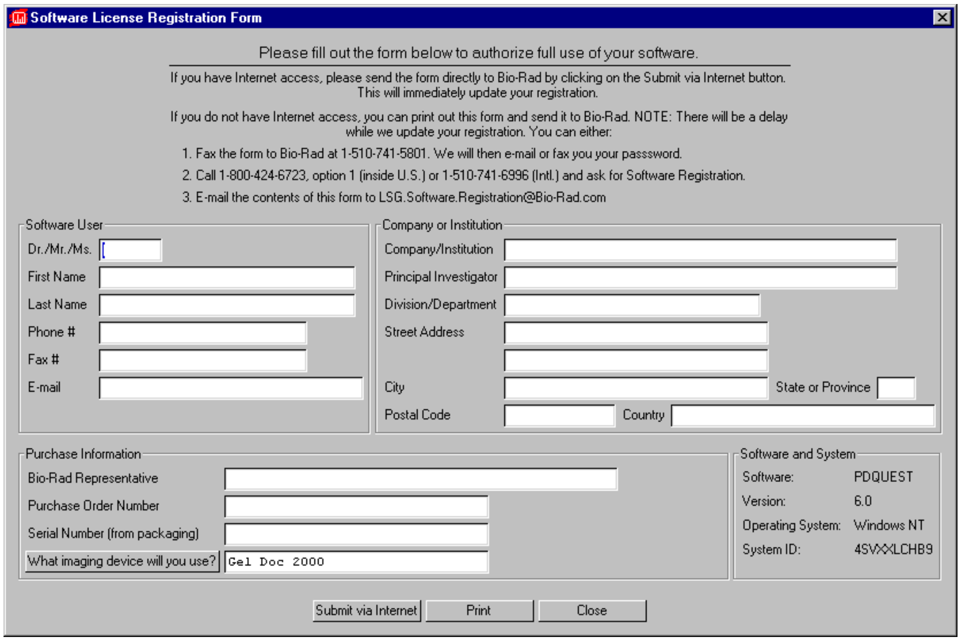

Fig.1-8. Software License Registration Form.

Note: If you do not yet have a Purchase Order Number or Software Serial Number, you may leave these fields blank to receive a trial license.

Submitting the Registration Form by Internet

If you have Internet access on your computer (the same computer on which you loaded PDQUEST), you can register quickly and easily.

In the Software License Registration Form, click on the Submit via Internet button.

Your information will be submitted automatically over the Internet, and a password will be generated automatically and sent back to your computer. Simply continue to run PDQUEST as before.

This password will be good for 30 days. During this period, if you have already submitted your Software Serial Number, click on Check License in the Main Registration Screen to update your license. (To access the Main Registration Screen from within PDQUEST, select Help > Register.)

If you have not yet submitted your Software Serial Number, open the form again, enter the serial number, and resubmit it over the Internet. In 1–2 days, click on the Check License button to update your license.

Submitting the Registration Form by Fax or E-mail

If you do not have Internet access, click on the Print button in the Software License Registration Form and fax the form to Bio-Rad at the number listed on the form. Alternatively, you can enter the contents of the form into an e-mail and send it to Bio-Rad at the address listed in the Registration Form.

If you include your Software Serial Number in the form, Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

If you do not include your Software Serial Number in the form, Bio-Rad will contact you in 2–3 days with a temporary 30-day license. Then, some time during the 30-day period, enter the serial number in the form and fax or e-mail the revised form to Bio-Rad. Bio-Rad will contact you by fax or e-mail in 2–3 days with a full license.

1.8.b. Entering a Password

If you submit your registration information over the Internet, you do not need to enter a password; it is done automatically.

If you fax your registration information, you will receive a password from Bio-Rad. You must enter this password manually.

To enter your password, click on Enter Password in the Main Registration Screen. If you are not currently in the Main Registration Screen, select Register from the Help menu in PDQUEST.

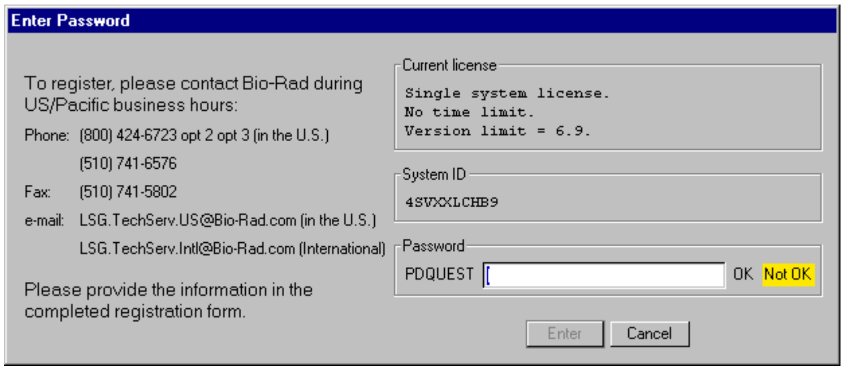

Fig.1-9. Enter Password screen.

In the Enter Password screen, type in your password next to the PDQUEST prompt.

Once you have typed in the correct password, the OK light next to the password field will change to green and the Enter button will activate. Click on Enter to run PDQUEST.

1.9. Contacting Bio-Rad

If you have problems with the HPK/password system, please contact Bio-Rad technical service. Technical service hours are from 8:00 a.m. to 4:00 p.m., Pacific Standard Time in the U.S.

Phone:	(800) 424-6723, option 2, option 3
	(510) 741-6576
Fax:	(510) 741-5802
E-mail:	LSG.TechServ.US@Bio-Rad.com (in the U.S.)
	LSG.TechServ.Intl@Bio-Rad.com (International)

For software registration, phone:

(800) 424-6723, option 1, or (510) 741-6996

2. Graphical Interface

2.1. The Graphical Interface

PDQUEST has an easy-to-use graphical interface featuring a menu bar, toolbars, and a Quick Guide.

2.1.a. Menu Bar

PDQUEST has a standard menu bar with 10 pulldown menus that contain all the major features and functions available in the software.

Fig.2-1. Menu bar.

The pulldown menus contain groups of related functions:

- **File** —General file commands (Open, Save, Print), imaging device acquisition windows (Gel Doc, GS-700, GS-710, Fluor-S, Fluor-S MAX, Personal FX, FX).
- **Edit** —Preferences.
- **View**—General viewing tools (Zoom Box, Grab), quick analysis tools (Density in Box, Plot Density), image assign functions, image contrast and color controls.
- **Image**—Image processing tools (Crop, Flip, Subtract Background).
- **Quantify** —Spot detection and calibration functions.
- **Match**—Matchset functions.
- **Database** —User set and annotation functions, standards and replicate groups functions, spot finding tools.
- **Reports**—Graphs and reports.
- **Window**—Tile windows, imitate zoom, configure subwindows.
- **Help** —On-line help, keyboard layout, software registration.

Below the menu bar is the main toolbar, containing some of the most commonly-used commands. Next to the main toolbar are the status boxes, which provide information about cursor selection and toolbar buttons.

2.1.b. Main Toolbar

The main toolbar in PDQUEST appears below the menu bar. It includes buttons for the main file commands (Open, Save, Print) and essential viewing tools (Zoom Box, Grab, etc.), and buttons to open the secondary toolbars.

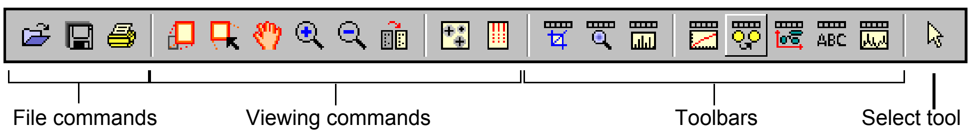

Fig.2-2. Main toolbar.

Tool Help

If you hold the cursor over a toolbar icon, the name of the command will pop up below the icon. This utility is called Tool Help. Tool Help appears on a time delay basis that can be specified in the **Pref erences** dialog box under **Edit** > Preferences. You can also specify how long the Tool Help will remain displayed.

2.1.c. Status Boxes

There are two status boxes in PDQUEST. These appear to the right of the main toolbar.

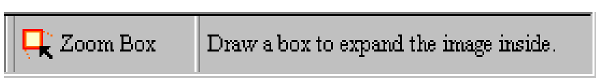

Fig.2-3. Status boxes.

The first box displays any function that is assigned to the mouse (see section 2.2., Mouse-assignable Tools). If you select a command such as Zoom Box, the name and icon of that command will appear in this status box and remain there until another mouse function is selected or the mouse is deassigned.

The second status box is designed to supplement Tool Help (see above). It provides additional information about the toolbar buttons. If you hold your cursor over a button, a short explanation about that command will be displayed in this second status box.

2.1.d. Secondary Toolbars

PDQUEST has eight secondary toolbars, which contain icons for groups of associated functions. You can open these toolbars from the main toolbar or from the **Vie w** > Toolbars pulldown menu.

The secondary toolbars can be toggled between vertical, horizontal, and expanded formats by clicking on the resize button on the toolbar itself.

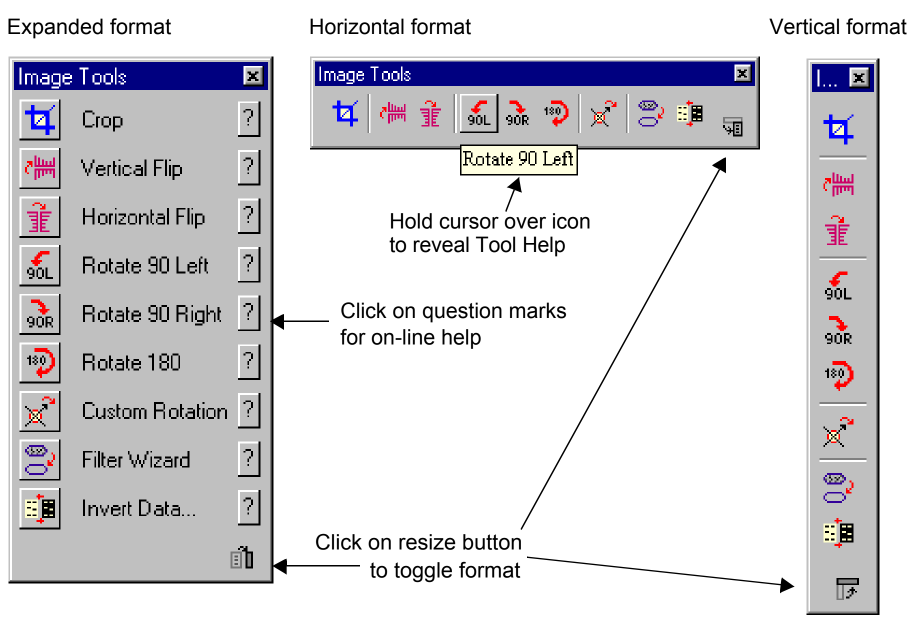

Fig.2-4. Secondary toolbar formats and features.

The expanded toolbar format shows the name of each of the commands. Click on the "?" icon next to the name to display on-line Help for that command.

2.2. Mouse-assignable Tools

PDQUEST has a number of commands that don't perform actions right away, but instead "assign" a function to your mouse (e.g., Zoom Box, Density at Cursor, Make Spot Crosshair). You first select one of these tools from a menu or toolbar, then execute the command by clicking or dragging on the image.

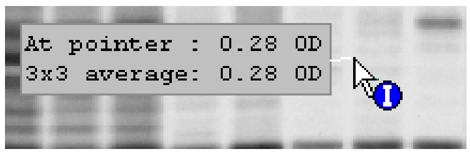

Fig.2-5. Example of a mouse-assignable tool: Density at Cursor

When a "mouse-assignable" function is selected, the cursor appearance changes. The name and icon of the function appear in the status box next to the main toolbar (see section 2.1.c).

Note: To *deassign* a function from the mouse, click on the toolbar button of the assigned function or click in the status box displaying the assigned function. You can also double-click on the Hide Overlays button.

2.3. Keyboard Shortcuts

Many of the functions in PDQUEST may be executed using keyboard shortcuts. These are listed in the **Help** menu under Keyboard Layout, as well as in Appendix H of this manual.

The pulldown menus also list the shortcut keys for the menu commands.

2.4. Windows and Subwindows in PDQUEST

PDQUEST handles image windows differently than other, standard Windows applications.

In a standards Windows application, you open a single file and it appears in a single window. But in PDQUEST, you can "load" more than one image file into the same window. You can then display all your images in subwindows of the main window.

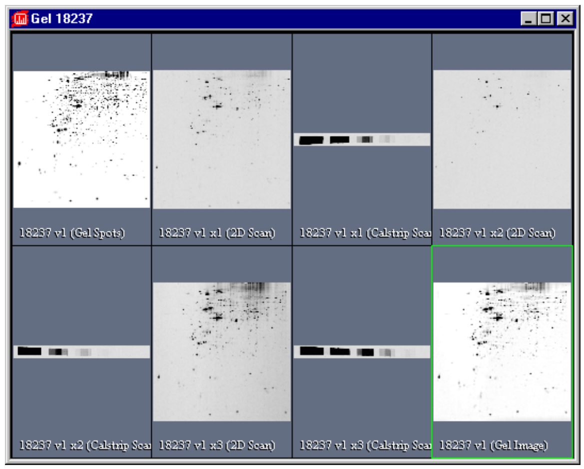

Fig.2-6. This image window has seven subwindows.

This grows out of the fact that, in PDQUEST, a single gel can have multiple images, all of which have the same root name. Subwindows are also necessary when working with matchsets, which are composed of multiple images.

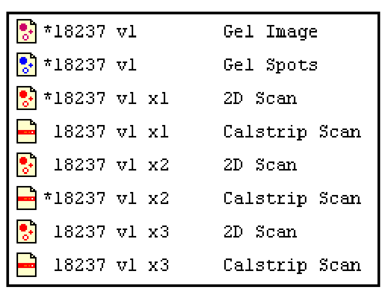

Fig.2-7. Gel 18237 has various image files, all of which can be loaded into the same window.

You start with a 2D Scan, which you convert into a Gel Image, which you convert into a Gel Spots image. You can also have multiple exposures of a 2D Scan, as well as Calstrip Scans associated with those exposures. All of these files can be loaded into a single window.

In addition, related images can be grouped into matchsets, which are displayed in one window. And the individual images in those matchsets can consist of both Gel Spots images and Gel Images.

You start by loading each image individually into a single window. Then PDQUEST has various functions that allow you to view the different images that are loaded in that window.

The Assign command under the **View** menu allows you view and select from the list of images that can be displayed in a given window.

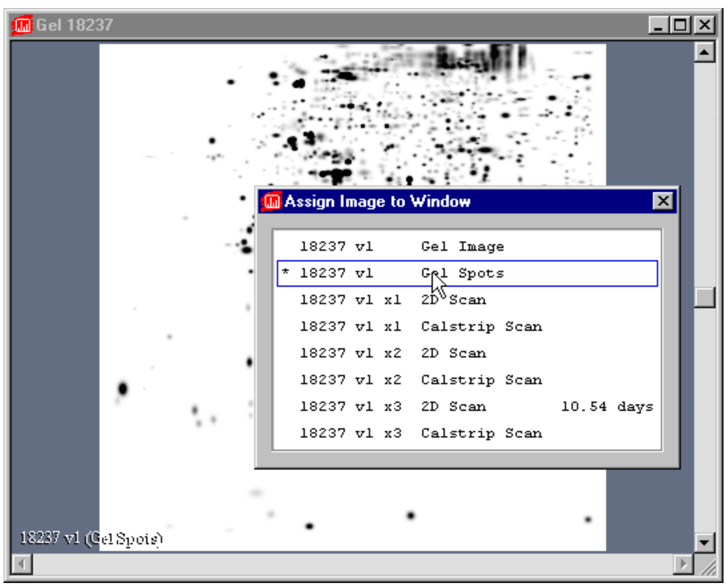

Fig.2-8. Assign Image lists the images loaded in an image window and allows you to select the one to display.

The Configure Subwindows command under the **Window** menu allows you to display the different images assigned to a given window in different combinations of subwindows.

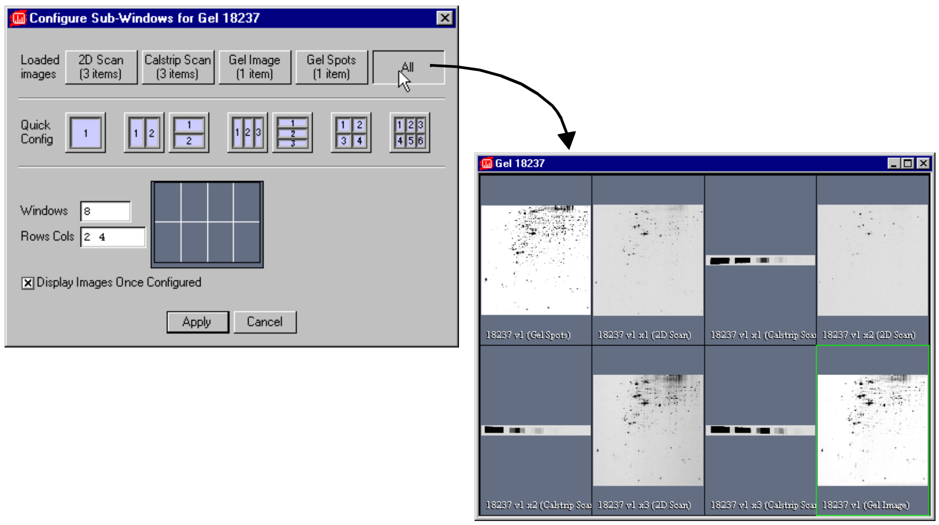

Fig.2-9. Configuring subwindows.

These commands are described in greater detail in Chapter 3.

2.5. File Commands and Functions

This section describes the basic file commands and functions of PDQUEST. These are selected from the **File** menu.

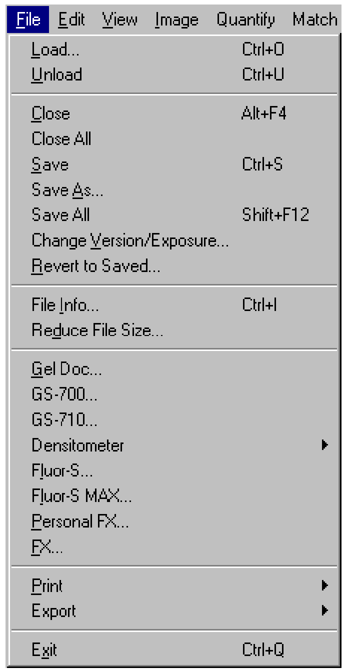

Fig.2-10. File menu.

Load

You use the **File** > Load command to load a previously saved image or a matchset into a window. It is similar to "Open" in a standard Windows application, except that it doesn't always open a new window for each image. Images with the same root name are loaded into the same window (see previous section).

You can also click on the Load button on the main toolbar.

Note: This version of PDQUEST will load any images and matchsets created with earlier versions of PDQUEST.

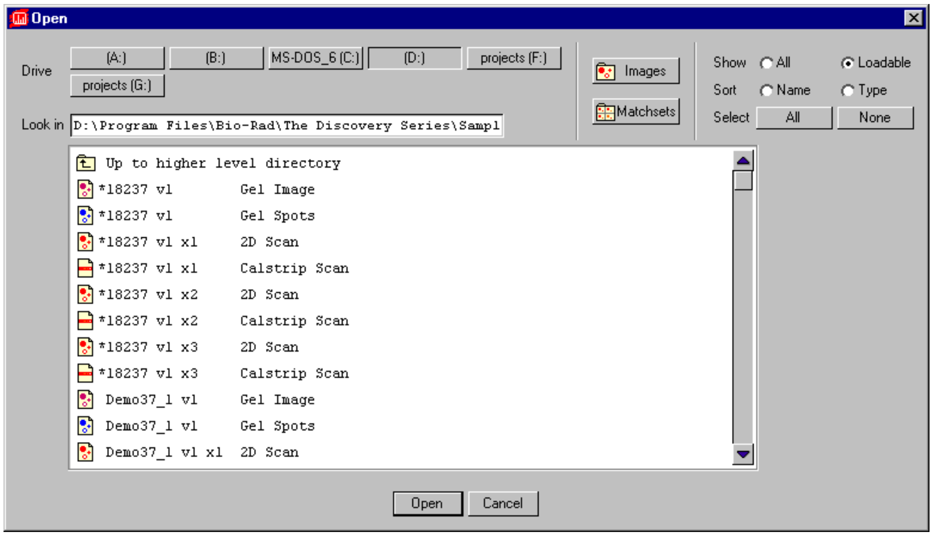

Fig.2-11. Load dialog box.

The **Load** dialog box differs from the standard Windows **Open** dialog in several key ways:

- Loaded images are marked with an asterisk.
- File types are listed next to each image name.
- Matchsets are listed as single files, even though they are really collections of images.

In the dialog box, load an image by double-clicking on its name or selecting the file name and clicking on the Open button.

To look for files in other directories, click on the appropriate folder name or click on the Up to higher level directory icon. You can also type the full file path into the Look in: field. To look for files on other drives, click on the appropriate drive letter button at the top of the dialog.

Note: The drive buttons at the top of the dialog box are only for mapped drives. If you need to access an unmapped (network) drive, you can type the drive name directly into the Look in: field using UNC notation (e.g., \\remote server name\shared directory name\...\file name).

To go directly to the folder where your images are stored, click on the Images button. To go to the folder containing your matchsets, click on Matchsets. The locations of these folders are specified in the **File Paths** dialog box, accessed under the **Edit** menu.

You can show all the files or only loadable files by selecting the appropriate button next to the Show prompt. You can also Sort the files by name or type.

To load several images at once, click on each file name once to select it, then click on Open when all the desired images are highlighted. Remember that files with the same root name will be loaded into the same window.

To select all the images in a folder, click on the All button next to the Select prompt.

Loading TIFF Images

The Load command can also be used to import TIFF images from other software applications.

TIFF images do not contain all of the tagged information that would normally be included in a Discovery Series image file (e.g., scanner type, date, color, etc.). For this reason, the File Info and Scan Report commands will not display all such tagged information from TIFF files.

There are many types of TIFF formats that exist on the market. Not all are supported by the Discovery Series. There are two broad categories of TIFF files that are supported:

1. 8-bit Grayscale. Most scanners have an option between line art, full color, and grayscale formats. Select grayscale for use with the Discovery Series software. In a grayscale format, each pixel is assigned a value from 0 to 255, with each value corresponding to a particular shade of gray. Hewlett Packard™, Microtek™, and Sharp™ each make scanners that produce compatible 8-bit grayscale images.
2. 16-bit Grayscale. Bio-Rad's Molecular Imager (storage phosphor) systems use 16-bit pixel values to describe intensity of scale. Molecular

Dynamics™ imagers also use 16-bit pixel values. The Discovery Series understands these formats and can interpret images from both Bio-Rad and Molecular Dynamics storage phosphor systems.

Note: PDQUEST can import 8- and 16-bit TIFF images from both Macintosh and PC platforms.

TIFF files that are *not* supported include:

1. 1-bit Line Art. This format is generally used for scanning text for optical character recognition or line drawings. Each pixel in an image is read as either black or white. Because the software needs to read continuous gradations to perform gel analysis, this on-off pixel format is not used.
2. 24-bit Full Color or 256 Indexed Color. These formats are frequently used for retouching photographs and are currently unsupported in the Discovery Series, although most scanners that are capable of producing 24-bit and indexed color images will be able to produce grayscale scans as well.
3. Compressed Files. The software does not read compressed TIFF images. Since most programs offer compression as a selectable option, files intended for compatibility with the Discovery Series should be formatted with the compression option turned off.

Unload

To close an image or scan, select **File** > Unload. If the file has changed since it was last saved, a message box will ask you if you want to save the changes before closing.

Close All

File > Close All closes all open images. If you have made changes to any of the images, a message box will open for each unsaved image giving you the option of saving those changes.

Save

Save will save a new or old scan, scan set, or matchset to your hard disk, network drive, or other storage media.

In Windows, new PDQUEST images will be given a .gsc extension when they are first saved.

Save As

Save As can be used to save scans, scan sets, and matchsets under different names and different directories.

Note that in the case of scan sets (i.e., multiple exposures, versions, and / or calstrips of the same gel, all with the same root name), all the different files in the set must be loaded when you select Save As if you want the root name changed for all of them.

Save All

Save All on the main toolbar or **File** menu saves all images, scan sets, and / or matchsets that are currently open.

Change Version/Exposure

Change Version/Exposure allows you to change the version or exposure number of a scan file. You can also change the type of the file (to 1-D, 2-D, Calstrip, or DNA scan).

Revert to Saved

Revert to Saved will reload the last saved version of the image you are working on. This is a quick way to undo any alterations you may have made to an image since you last saved it.

Any changes you have made since last saving the file will be lost. (Also, all open dialog boxes will be closed.) A message box will warn you before completing the command.

File Info

File Info displays general information about your image, including the scan date, scan area, number of pixels in the image, data range, and the size of the file. Gel Spots files include the number of spots, total data in the spots, and other information. There is also a field where you can type in a file description or comments.

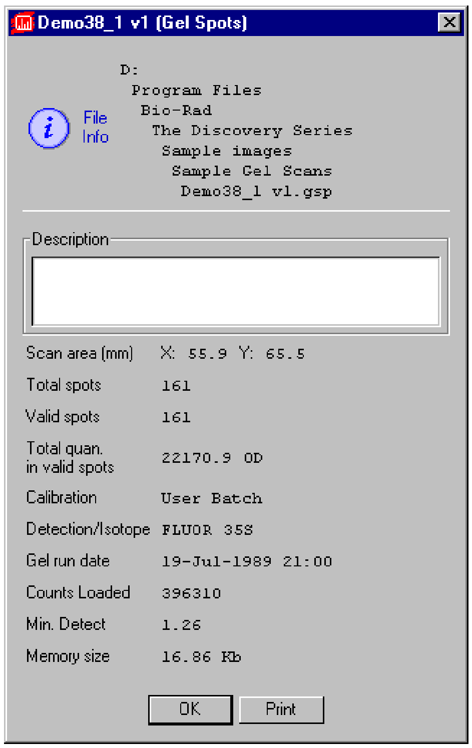

Fig.2-12. File Info box.

Reduce File Size

Scan files can be quite large, and computer systems do not have unlimited memory or storage space. If you are having difficulty loading or storing a particular scan, you may want to reduce the size of the file by reducing the number of pixels in the image. (You can also trim unneeded parts of an image to reduce its memory size. See section 3.9., Cropping Images.)

Note: You cannot reduce the file size of Gel Images or Gel Spots files

This function is comparable to scanning at a lower resolution, in that you are increasing the size of the pixels in the image, thereby reducing the total number of pixels and thus memory size.

Note: Reducing the file size of an image will result in some loss of resolution. In most cases this will not affect quantitation. In general, as long as the pixel size remains less than 10 percent of the size of the objects in your image, changing the pixel size will not affect quantitation.

Select **File** > Reduce File Size to open the **Reduce File Size** dialog box.

The dialog box shows you the size of the pixels in the image (Pixel Size: X by Y microns), the number of pixels in the image (Pixel Count: X by Y pixels), and the Memory Size of the image.

As you increase the size of the pixels, the pixel count will decrease, as will the memory size. You can increase the pixel size in either dimension (see the following figure for an example).

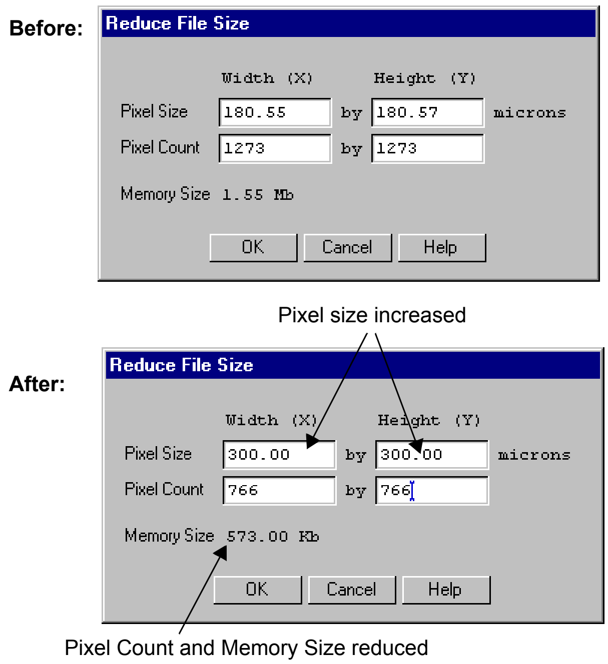

Fig.2-13. Reduce File Size dialog box, before and after pixel size increase.

Note: Asymmetric pixel reduction (i.e., making pixels smaller in one dimension than the other) is not recommended for 2-D gels, since you are concerned with resolving spots in both dimensions.

When you are finished, click on the OK button.

A pop-up box will give you the option of reducing the file size of the displayed image or making a copy of the image and then reducing the copy's size.

Reducing the file size is an irreversible process. For that reason, we suggest you make a copy of the image and reduce its file size. That way, if you lose too much resolution, you can simple delete the copy and try again. Once you are happy with the reduced image, don't forget to delete the original. The goal is to save space!

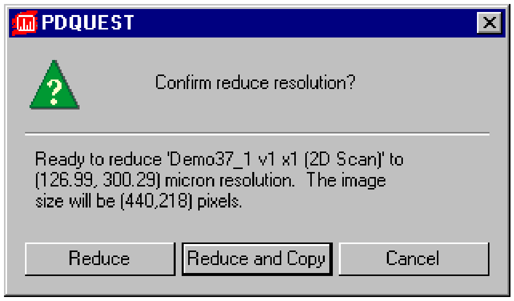

Fig.2-14. Confirm Reduce File Size pop-up box.

If you choose to make a copy of the image, you will be asked to enter a name for the new copy before the operation is performed.

Imaging Device Acquisition Windows

The **File** menu contains a list of Bio-Rad imaging devices supported by the Discovery Series software. These are:

1. Gel Doc 1000 / 2000
2. GS-700 Imaging Densitometer
3. GS-710 Imaging Densitometer
4. Fluor-S MultiImager
5. Fluor-S MAX MultiImager
6. Personal Molecular Imager FX
7. Molecular Imager FX

Selecting a name in the **File** menu will open the acquisition window that allows you to scan using that instrument.

See the individual appendices on the imaging devices for more details.

Exit

File > Exit quits the application.

2.6. Preferences

The **Pref erences** dialog box (accessed under the **Edit** menu) allows you to customize basic features of your system, such as displays and toolbars.

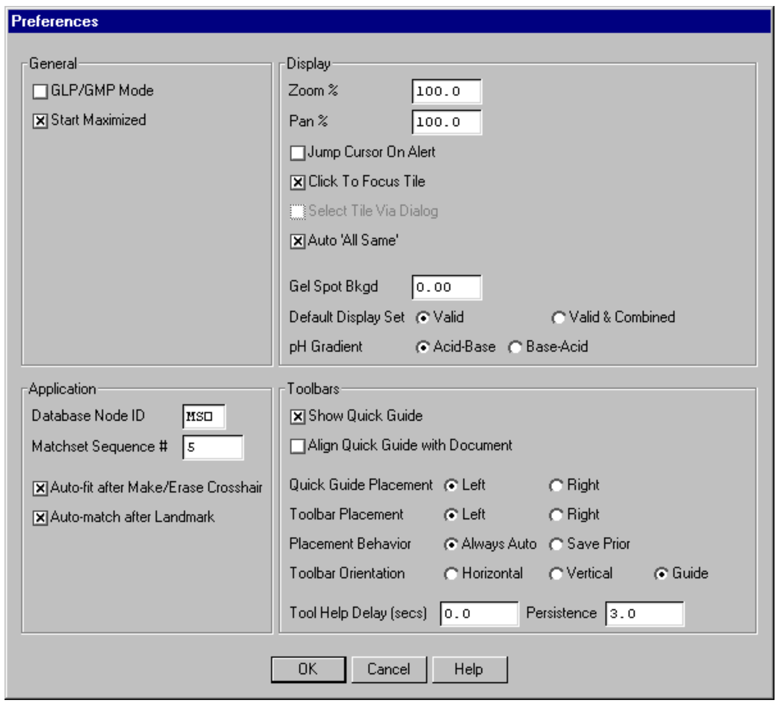

Fig.2-15.

After you have selected your preferences, click on OK to implement them. Some of your preferences will only be implemented by exiting and restarting the program. A message in a pop-up box will notify you if this is necessary.

2.6.a. General

GLP/GMP Mode

The GLP / GMP Mode checkbox allows you to prevent changes to an image that would change the raw image data. In GLP / GMP mode, the software will not allow the following operations to be performed (these are all located on the Image menu):

- Custom Rotation
- Filters
- Invert Data

If a user attempts to use any of these functions in GLP / GMP mode, he will receive a message that that function is not available.

To set GLP / GMP mode, click on the checkbox. A dialog box will pop up asking you to enter a password.

After you enter a password and click on OK, another dialog will ask you to reenter the password for confirmation. Retype the password and click on OK again.

To disable GLP / GMP mode, click on the checkbox to deselect it. The dialog box will pop up asking you to enter the password. When you click on OK, GLP / GMP Mode will be disabled.

Start Maximized

In the Windows version, the Start Maximized checkbox determines whether PDQUEST occupies the entire computer screen when first opened. If this is unchecked, PDQUEST’s menu and status bars will appear across the top of the screen and any toolbars will appear “floating” on the screen.

2.6.b. Display

The Zoom % field allows you to specify the percentage by which an image moves closer or farther away when you use the Zoom functions. This percentage is based on the size of the image.

Pan % determines the percentage by which the image moves side to side or up and down when you use the arrow keys. This percentage is based on the size of the image.

When an alert box pops up in PDQUEST, you can set your cursor to automatically go to the OK button in the box by selecting Jump Cursor on Alert.

When you have multiple subwindows open in a window, you can activate a subwindow by clicking on it (Click to Focus Tile selected) or you can activate a subwindow by simply moving your cursor over it (Click to Focus Tile deselected).

Auto 'All Same' duplicates the zoom and positioning commands made in one subwindow in all displayed subwindows automatically.

If you want to apply a uniform background to Gel Spots images so they more closely resemble the original gels, enter a background optical density (O.D.) value in the text box next to the Gel Spot Bkgd prompt.

Specify whether you want to see only valid spots or valid spots along with combined spots by clicking on the appropriate button next to the Default Display Set prompt.

Finally, indicate the orientation in which your gels are run by clicking on either the Acid-Base or Base-Acid button next to the pH Gradient prompt.

2.6.c. Application

Enter up to three letters for the Database Node ID and a number for the Matchset Sequence #.

Each matchset that you create will be identified by a Node ID followed by a unique Sequence Number. The Node ID remains the same for all the matchsets while the Sequence Number increases by 1 each time a new matchset is created.

Once the Node ID and the Sequence Number have been initially set, you probably will not need to change them.

If Auto-fit after Make/Erase Crosshair is selected, PDQUEST will automatically apply Gaussian fitting to each spot you mark on an image.

If Auto-match after Landmark is selected, PDQUEST will automatically attempt to match the spots in your images when you create a landmark.

2.6.d. Toolbars

These settings determine the behavior and positioning of the secondary toolbars and Quick Guide.

If the Show Quick Guide checkbox is selected, the Quick Guide will open automatically when you open PDQUEST.

If Align Quick Guide with Document is selected, the Quick Guide will pop up flush with the edge of your documents. Otherwise, it will appear flush with the edge of the screen.

Quick Guide Placement and Toolbar Placement determine which side of the screen the Quick Guide and toolbars will first open.

The Placement Behavior setting determines whether a Quick Guide or toolbar will always pop up in the same place and format (Always Auto), or whether they will pop up in the last location they were moved to and the last format selected (Save Prior).

The Toolbar Orientation option buttons specify whether toolbars will first appear in a vertical, horizontal, or expanded format when you open PDQUEST.

Tool Help Delay allows you to specify the amount of time (in seconds) the cursor must remain over a toolbar icon before the Tool Help appears. First-time users may want to specify a short delay to learn the names of the toolbar functions, while experienced users can specify a longer delay once they are familiar with the icons.

Persistence determines how long the Tool Help will linger on the screen after you move the cursor off a toolbar icon.

2.7. File Paths

This function lets you set the shortcut paths (to the location on your hard disk or file server) for the Matchsets, Images, Local, Temporary, and Log directories.

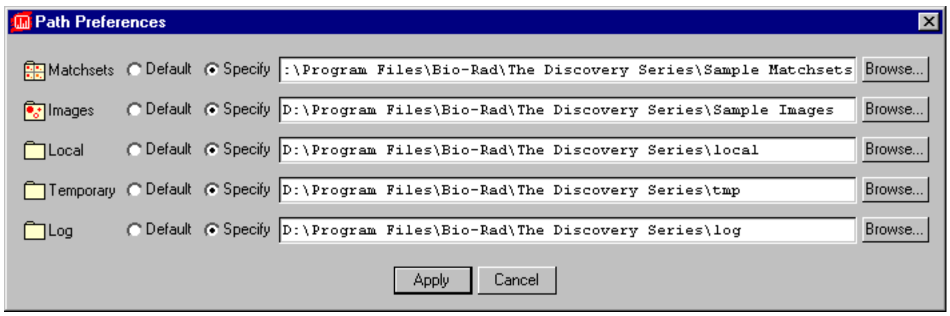

Fig.2-16. Path Preferences dialog box.

Shortcut buttons to these directories can be found in the Load dialog box and in other locations throughout PDQUEST.

To change a default path, click on the Specify button and type the desired path in the text box, or click on the Browse button to search the directories.

2.8. User Settings

If PDQUEST is on a workstation that has multiple users, each user can have his or her own set of PDQUEST preferences and settings.

In multiple-user situations, PDQUEST's preferences and settings are associated with individual user names. On a PC with Windows 95, 98, or NT 4.0, your user name is the name you use to log onto the computer. On a Macintosh, your user name is the Owner Name under the File Sharing (OS 8.x) or Sharing Setup (OS 7.x) control panel.

Note: If you do not log onto your PC under Windows 95/98 or do not have a Owner Name on your Macintosh, then you do not have a user name and your preferences and settings will be saved in a generic file.

3. Viewing and Editing Images

This chapter describes PDQUEST's viewing tools and commands for configuring windows and subwindows. This chapter also describes the tools for cropping, flipping, and rotating images and filtering image noise.

These tools may be found on the View, Image, and Window menus.

3.1. Magnifying and Positioning Tools

The magnifying and positioning tools in PDQUEST are located on the View menu and toolbar; some of these functions are also found on the main toolbar.

These commands will only change how the image is displayed on your computer screen. *They will not change the underlying data.*

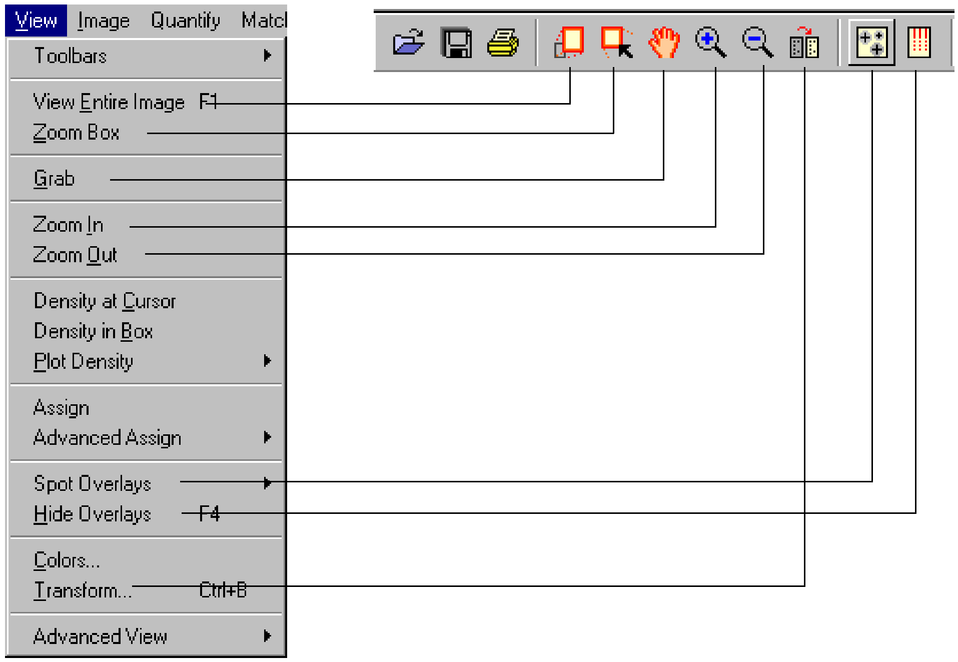

Fig.3-1. Viewing functions on View menu and main toolbar.

3.1.a. Zoom Box

Zoom Box allows you to select a small area of the image to magnify so that it fills the entire image window.

First, click on the Zoom Box button on your main toolbar or select View > Zoom Box. Your cursor arrow will change to a cross. Then drag the cursor on the image to enclose the area you want to magnify. The area of the image you selected will be magnified to fill the entire window.

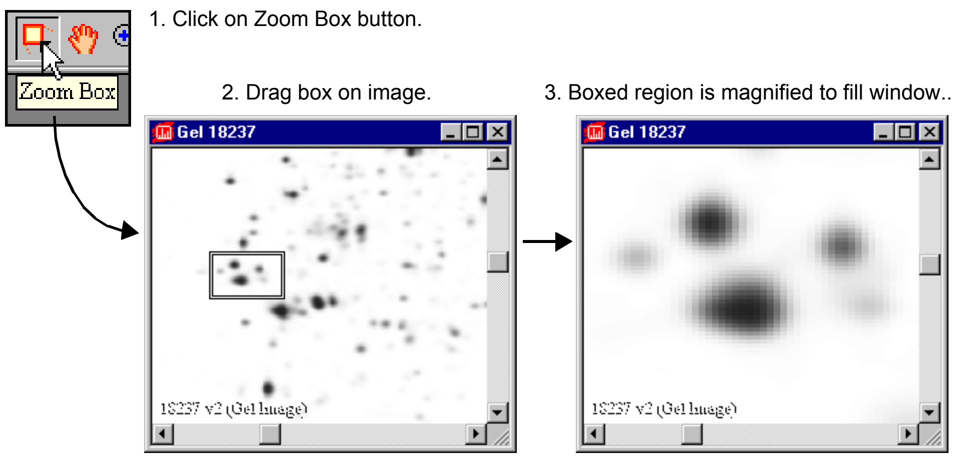

Fig.3-2. Zoom Box tool.

3.1.b. Zoom In/Zoom Out

These tools work like standard magnifying tools in other applications.

Click on the Zoom In or Zoom Out button on your Main toolbar (or select from the View menu). Your cursor will change to a magnifying glass. Click on an area of the image to zoom in or zoom out a defined amount, determined by the setting in the Edit > Preferences dialog box.

3.1.c. Grab

This tool allows you to change the position of your image in the image window. Select View > Grab or click on the Grab button. Your cursor will change to a “hand” symbol. Dragging the Grab symbol on the image will move the image in any direction.

You can also move your image inside the image window by using the ARROW keys.

3.1.d. View Entire Image

If you have magnified part of an image or moved part of an image out of view, View Entire Image returns to the original, full view of the image.

3.2. Advanced Viewing Tools

The Advanced View submenus contain a number of advanced positioning and viewing tools for use on individual subwindows or all subwindows.

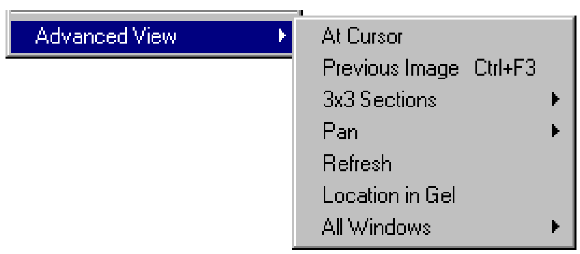

Fig.3-3. Advanced View submenu.

If you select At Cursor then click on the image, the image will be centered in the window at the point where you click.

Previous Image will "undo" the last zooming or positioning command.

3x3 Sections will magnify and position within the window one of nine different quandrants of the image.

Pan will move the image up, down, left, or right by increments within the window. You can also pan using the arrow keys on the keyboard.

Refresh cleans up the overlays on the screen and redisplays the image and overlays.

If you have magnified or repositioned a portion of the image, Location in Gel allows you to see which region of the image you are currently examining. Selecting Location in Gel displays two overlays in the window. The yellow overlay represents in the entire image, while the white overlay represents the region currently displayed within the window.

All Windows Viewing Commands

The All Windows submenu accesses advanced viewing features that can be applied to all subwindows within a window. Many of these commands are similar to those described above. Unique commands are described below.

Hide All Overlays conceals all the overlays in all the displayed subwindows.

All Same applies the magnification and position of a subwindow you click on to all the subwindows. First select All Same from the menu, then click on the subwindow whose view you want to duplicate. (See also Imitiate Zoom below.)

Note: All Same will only work on comparable images with similar dimensions.

All Entire Images returns all the subwindows to their unmagnified, unrepositioned display.

3.3. Assigning and Interchanging Images

Because in PDQUEST you can load more than one image into an image window, you can use the Assign and Interchange commands to select the which loaded image you want to display.

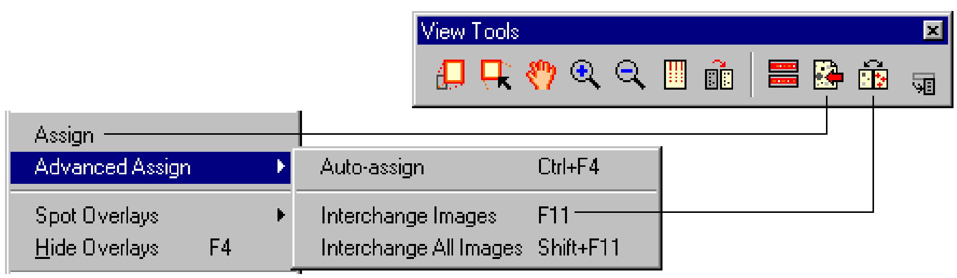

Fig.3-4. Assign commands on the View menu and toolbar.

Assigning Images

To assign an image to a window, select View > Assign (CTRL+A) and click on the desired window or subwindow. This will open a pop-up box listing all the images currently loaded.

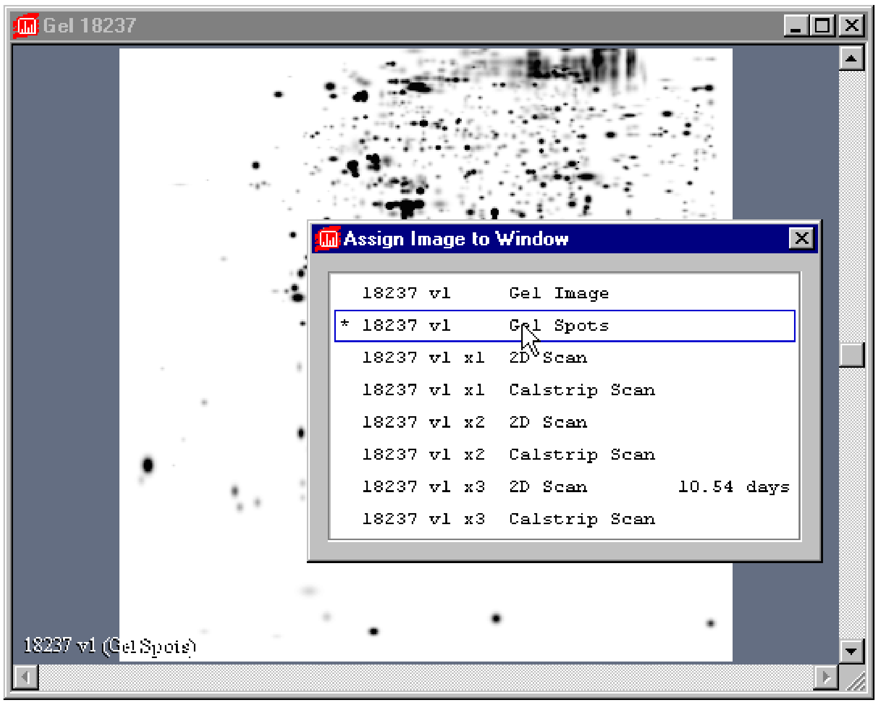

Fig.3-5. Example of an Assign pop-up list.

Select the image that you want to display. The image will appear in the window that you chose.

Auto-Assign opens a dialog box in which you can select the type of file to assign to a particular subwindow.

Fig.3-6. Auto-assign dialog box.

Interchanging Images

View > Advanced Assign > Interchange Images (F11) switches between Gel Images, Gel Spots images, and 2D scans within a particular subwindow.

View > Advanced Assign > Interchange All Images (SHIFT+F11) switches between Gel Images and Gel Spots images in all the displayed subwindows of a given matchset.

3.4. Spot Overlays

PDQUEST allows you to create a variety of image overlays. These commands are available on the View > Spot Overlays submenu.

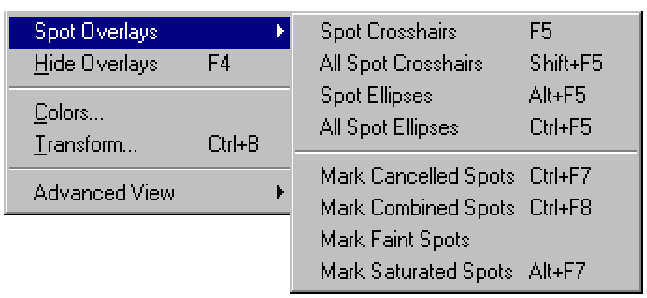

Fig.3-7. Spot Overlays submenu.

Spot crosshairs mark the exact centers of the spots that are generated during spot detection. The crosshairs in a Gel Image file are exact duplicates of those in the corresponding Gel Spots file.

Spot Crosshairs shows the spot crosshairs in the active subwindow.

All Spot Crosshairs shows the spot crosshairs in all the subwindows.

Spot Ellipses shows the shape of each spot generated by Guassian fitting in the active subwindow.

All Spot Ellipses shows the spot ellipses in all the subwindows.

Mark Cancelled Spots highlights those spots that have been pushed into the background using the Match > Edit Spots > Cancel command.

Mark Combined Spots highlights those spots that have been combined from spot clusters using the Match > Edit Spots > Combine command.

Mark Faint Spots highlights those spots whose signal falls within the bottom 5 percent of the intensity range of the image.

Mark Saturated Spots highlights those spots whose signal is saturated.

Hide Overlays

Hide Overlays on the main toolbar and View menu is useful for concealing any overlays you have created. It can also be used to erase plots, traces, and information boxes that are created by other functions.

Note: Clicking once on Hide Overlays will conceal the overlays. Clicking twice will deassign any function that has been assigned to the mouse.

3.5. Density Tools

The Density Tools (under the View menu and on the Density Tools toolbar) are designed to provide a quick measure of the signal intensity of the data in your scan or Gel Image files.

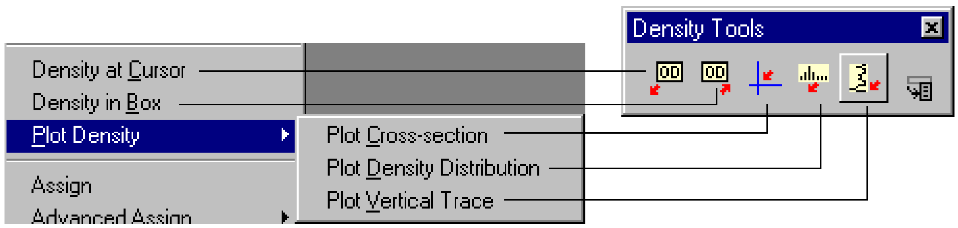

Fig.3-8. Density submenu and toolbar.

Density at Cursor

Density at Cursor displays the signal intensity of the pixel on the image where you click your mouse. It also shows the average intensity for a 3 x 3 pixel box centered on that point.

To use this tool, click on the Density at Cursor button on the Density toolbar or select View > Density at Cursor. Then click on the point of interest on the image.

Density in Box

Density in Box displays the average and total intensity within a boxed region on the image. Select Density in Box from the Density toolbar or View menu, then define the region you want to measure by dragging your cursor across the image.

Plot Density Distribution

Plot Density Distribution displays a histogram of the signal intensity distribution for the part of the image displayed in the image window. The average intensity is marked in yellow on the histogram.

The histogram will appear along the right side of the image. Use the Zoom functions to display the histogram for magnified regions of your image.

Plot Cross-section

Plot Cross-section displays an intensity trace of a cross-section of the image centered on the point where you click your mouse. The horizontal trace is displayed along the top of the image, and the vertical trace is displayed along the side of the image.

The intensity at the point you clicked on is displayed, as is the maximum intensity along the lines of the cross-section. Dragging the mouse with this function selected will continuously update the display.

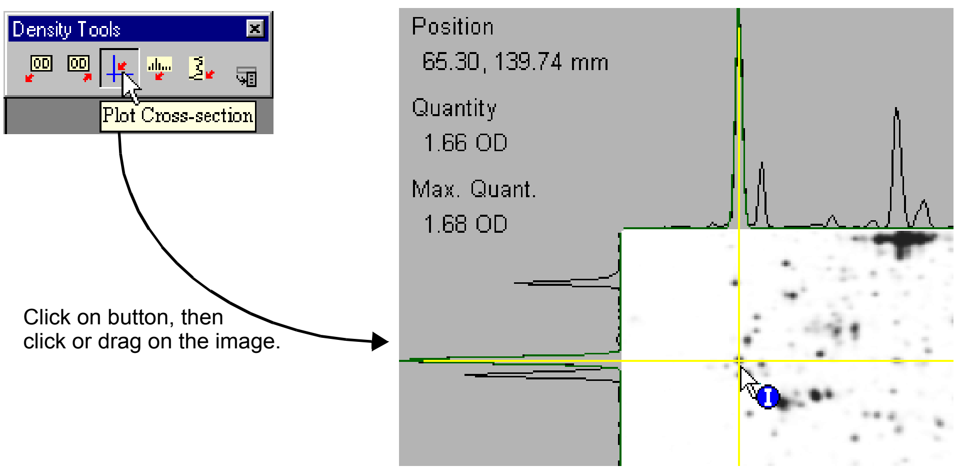

Fig.3-9. Plot Cross-section tool.

Plot Vertical Trace

Plot Vertical Trace plots an intensity trace of a vertical cross-section of the image centered on the point where you click your mouse. Dragging the mouse with this function selected will continuously update the display.

Note: The sampling width for the Density traces is one pixel.

3.6. Colors

View > Colors opens a dialog box in which you can adjust the colors of the image, windows, buttons, etc.

Fig.3-10. Colors dialog box.

Selecting a Color Group

Within the dialog box, the Color Group button allows you to select the colors of a particular group of objects (e.g., pop-up boxes, image colors, etc.). Click on the button to open the list of objects you can change.

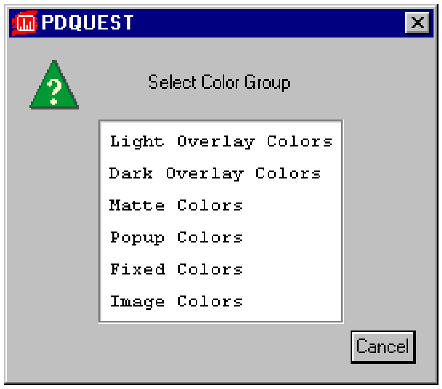

Fig.3-11. List of Color Groups.

Click on a color group in the list to select it.

Changing a Color

After you have selected the color group to change, click on the specific color button to change. The Color Edit dialog box will open, allowing you to adjust the red-green-blue (RGB) values of the color you selected.

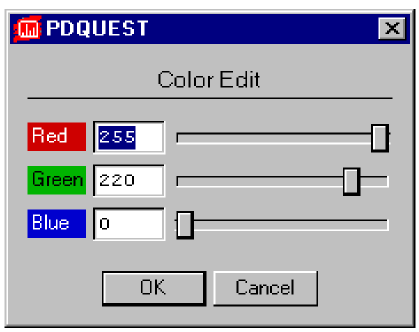

Fig.3-12. Color Edit dialog box.

Drag the sliders or enter a value in the fields. The color of the button will change with your adjustments.

Saving/Selecting a Defined Set of Colors

After you have changed the colors within color groups, you can save these settings for future use on other images. The Colormap Name field displays the name of a defined set of colors and color groups. There are several standard colormaps included with PDQUEST, or you can create your own.

To select a predefined colormap, click on the Load button.

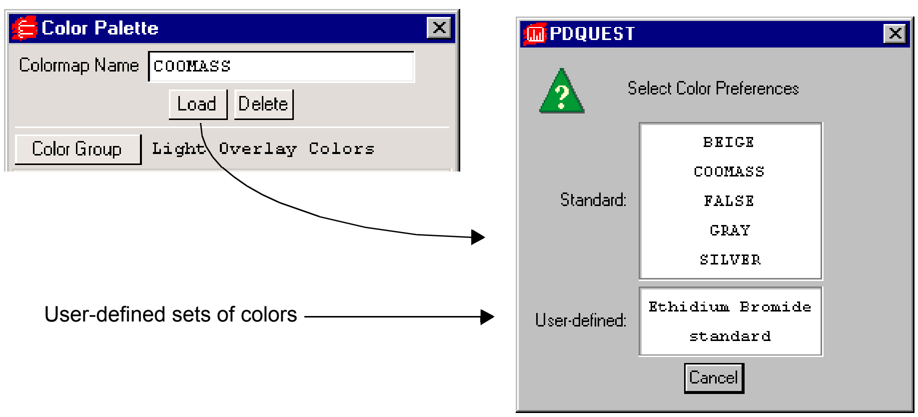

Fig.3-13. Selecting a Colormap.

From the list displayed, click on the set of colors you want to apply.

To create your own colormap, adjust the colors within the color groups as described above and type in a new colormap name. Click on OK to apply your changes.

To remove a colormap, click on the Delete button. Select the colormap to be deleted from the displayed list. A pop-up box will ask you to confirm the deletion.

If you change your mind about applying any changes you make, click on the Cancel button. If you want to return to the Standard colormap, click on the Reset button. All colors will be returned to their default values.

3.7. Transform

If features in your image are indistinct or fine details appear to be lost in background noise, you can use the Transform functions to optimize your displayed image. To open the Transform dialog box, select View > Transform.

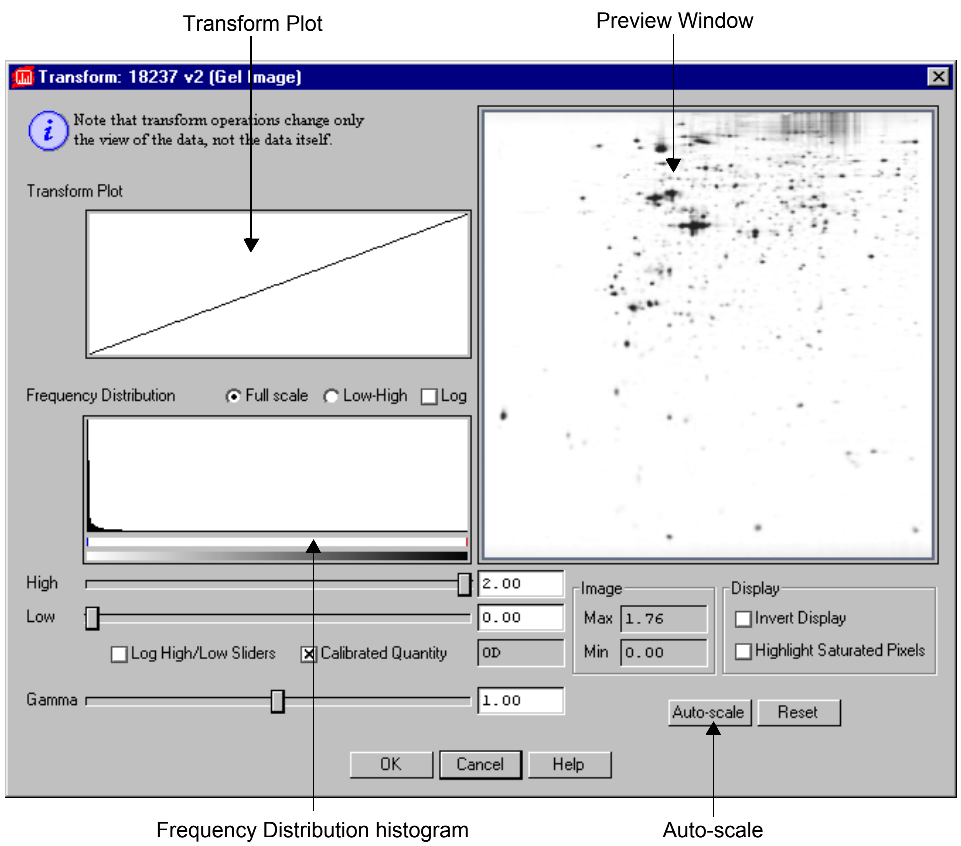

Fig.3-14. Transform dialog box.

The Transform dialog box contains a Preview Window, a Frequency Distribution histogram, a Transform Plot, and three main methods of optimizing your image: Auto-scale, High and Low sliders, and a Gamma slider. You can use these controls to adjust the way the software transforms raw data into visual data.

Note: Any changes made with the Transform controls will only affect how the image is displayed on your screen. *They will not affect the underlying data.*

3.7.a. Features of the Transform Dialog

Preview Window

The Preview Window in the Transform dialog shows a smaller view of the same image that is displayed in the main image window. Changes in the Transform controls are automatically reflected in the Preview Window. They are only applied to the main image when you click on OK.

You can use view tools like Zoom Box and View Entire Image in the Preview Window just as you can in the main image window, to focus on particular regions of interest. You can also use the ARROW keys to move the image within the Preview Window.

Frequency Distribution Histogram

The Frequency Distribution histogram shows the total data range in the image and the amount of data at each point in the range. In a typical scan, there is a signal spike at the left (“gray”) end of the histogram due to background noise.

Transform Plot

The Transform Plot is a logarithmic representation of how the raw pixel data are mapped to the pixels of your computer screen.

3.7.b. Transform Controls

Auto-scale

Clicking on the Auto-scale button will optimize your image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). This enhances minor variations in the image, making fine details easier to see. You can then “fine-tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn’t give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

As you drag the sliders, the slider markers on the Frequency Distribution histogram will move. Everything to the left of the Low marker will be remapped to minimum intensity, while everything to the right of the High marker will be remapped to maximum intensity. Using the histogram, you can position the markers at either end of the data range in your image, and use the low slider to cut off the “spike” of background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Log High / Low Sliders changes the feedback from the slider handles, so that when you drag them, the slider markers move a shorter distance in the histogram. This allows for finer adjustments when your data is in a narrow range.

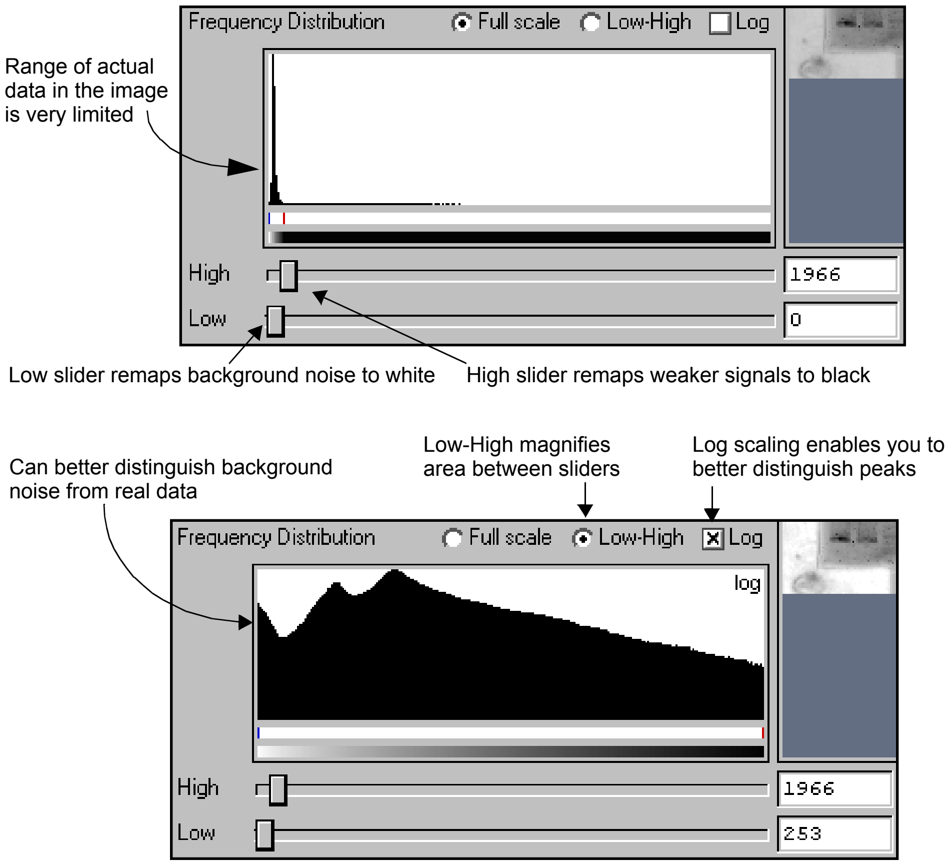

Fig.3-15. Two views of the Frequency Distribution histogram.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle expands or compresses the contrast range at the dark or light end of the range, and this is reflected in the Transform Plot and Preview Window.

3.7.c. Other Features

Full Scale/Low-High

The Full Scale / Low-High option buttons adjust how the range of data in the image is displayed in the Frequency Distribution histogram and Transform Plot. They do not change how the data is displayed in the image window.

Selecting Full Scale adjusts the Frequency Distribution and Transform Plot displays so they show the full intensity range of the image.

Selecting From Low to High magnifies the range between the Low and High sliders. This makes it easier to view your data if it does not occupy the full intensity range of the image.

Log

The Log checkbox changes the way the data is displayed in the histogram so you can better discern subtle changes in signal intensity.

Image Max/Min and Units

Image Max and Min display the range of signal intensity in the image.

The image units are determined by the type of scanner used to create the image. In the case of densitometers, you can select Calibrated Quantity to display your image units in O.D.s.

Invert Display

The Invert Display checkbox flips light bands on a dark background to dark bands on a light background, and visa versa. Once, again, the actual data will not change—only the image.

Highlight Saturated Pixels

When the Highlight Saturated Pixels checkbox is selected, areas of the image with saturated signal intensity are highlighted in red.

Reset

If at any time you want to return to an unmodified view of the scan data, click on Reset.

Transform All

If you have multiple subwindows open, you can transform all of them at once by clicking on the All button at the bottom of the dialog.

3.8. Windows and Subwindows Commands

The commands in the Window menu control how your windows and subwindows are configured.

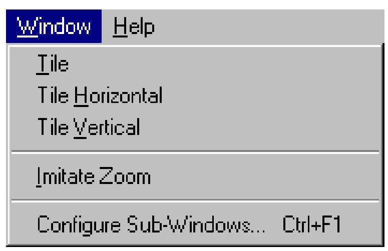

Fig.3-16. Window menu.

3.8.a. Tiling Windows

If you have more than one image open, the Tile commands on the Window menu allow you to arrange your images neatly on the screen.

Tile will resize your windows and arrange them on the screen left to right and top to bottom.

Tile Vertical will resize your windows and arrange them side-by-side on the screen.

Tile Horizontal will resize your windows and stack them top-to-bottom on the screen.

3.8.b. Imitate Zoom

Imitate Zoom applies the magnification and positioning of the active window or subwindow to all the displayed windows or subwindows.

First make sure that the window you want to imitate is active, then select Imitate Zoom from the Window menu. The other windows will change their view to match that of the active window

Note: Imitate Zoom will only work on comparable images with similar dimensions.

3.8.c. Configure Subwindows

If you have more than one image loaded in a window (e.g., in the case of matchsets), you can control how the window is partitioned into subwindows using the Configure Subwindows command.

Select Configure Subwindows from the Window menu to open the Configure Subwindows dialog box.

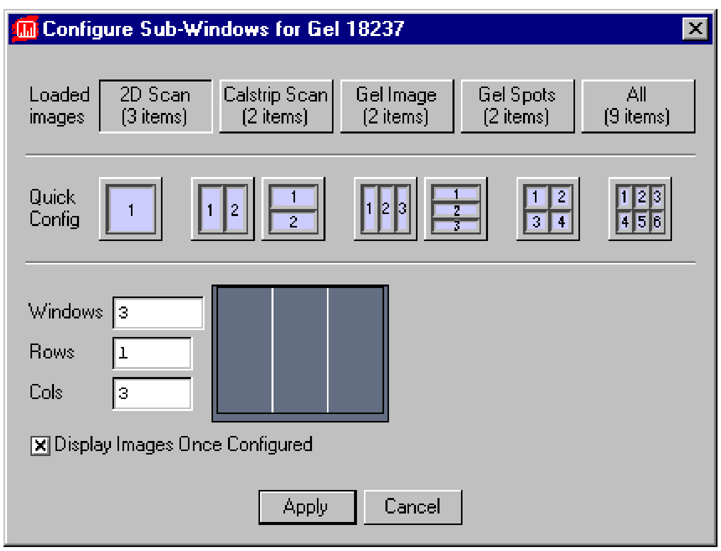

Fig.3-17. Configure Subwindows dialog box.

Loaded Images

The buttons next to the Loaded images prompt indicate the types of images that are loaded (2D scans, Gel Spots images, Gel Images, etc.) and indicate how many of each type are loaded. Click on a button to restrict your display to all the images of that type. (Click on All to not restrict by type.)

Quick Config

The buttons next to Quick Config will immediately partition the window into the subwindow configuration depicted on the button. You do not need to click on the Apply button. (Note that if you have elected to display images of only one type using the Loaded Images button, only those types of images will be displayed in the subwindows.)

Custom Configuration

Alternatively, you can create the configuration of your choice by typing values in the Windows, Rows, and Cols fields. Windows specifies how many subwindows to create. Rows and Cols specify how to layer the subwindows.

Display Images Once Configured

Note: For most users, this checkbox should remain selected. Deselect this checkbox only if you have a slow computer and are displaying a lot of images.

If you have a matchset with many members and do not want to take time to refresh your subwindows after reconfiguring, deselect the Display Images Once Configured checkbox. Then, when you reconfigure your subwindows, the images will not immediately appear in the windows.

Click in a subwindow to make the image in that subwindow appear. Now perform a command in that window such as Zoom In. Select View > Advanced View > All Windows > All Same to reproduce that command in all the other subwindows and refresh the images in them with the appropriate zoom factor.

Apply

Click on the Apply button to implement your configuration selections.

3.9. Cropping Images

If you are merging multiple exposures and / or matching spots between images, you will need to trim your scans to make them all the same size and shape. The tools for doing this are located on the Image menu.

The crop tools can also be used to reduce the memory size of a scan.

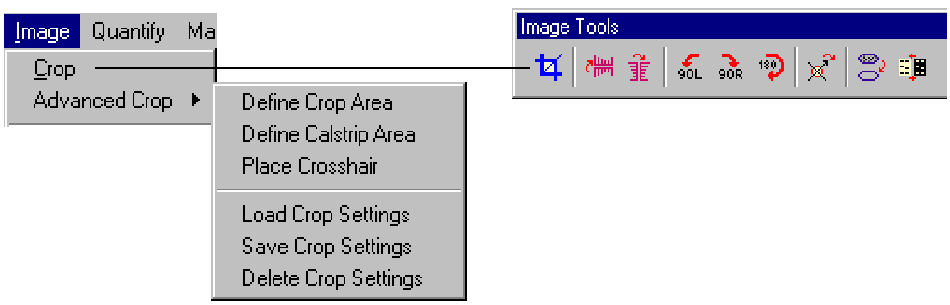

Fig.3-18. Cropping commands.

The Crop command is automatically assigned after you complete a scan. Alternatively, you can click on the Crop button on the Image toolbar or select Image > Crop. Your cursor appearance will change to a Crop symbol.

3.9.a. Drawing a Crop Box

Define the region to be cropped by dragging the cursor across the image, creating a box. Everything outside the box will be deleted.

Information about the physical dimensions of the crop area (given in millimeters and number of pixels) is listed at the bottom of the crop box, as is information about the memory size of the image inside the crop area.

1. To *reposition* the crop box, position your cursor at the center of the box. The cursor appearance will change to a multidirectional arrow symbol. You can then drag the box to a new position.

2. To *resize* the box, position your cursor on a box border or corner. The cursor appearance will change to a bidirectional arrow. You can then drag that border or corner in or out, resizing the box.

3. To *redraw* the box, position your cursor outside the box. The cursor appearance will change back to the Crop tool, and you can draw another box, replacing the one you just drew.

3.9.b. Advanced Crop Commands

If you are merging multiple exposures of the same gel (see Chapter 4) and / or matching spots between different gels, the images must all have the same dimensions, which means that they must be cropped to exactly the same size. The advanced crop commands allow you to save the crop box size and position from one scan and apply it to the others. They also allow you to crop images of gels and images of calstrips out of the same scan.

These commands are located on the Image > Advanced Crop submenu.

Define Crop Area/Define Calstrip Area

Define Crop Area allows you to draw a crop box as described in the previous section.

Define Calstrip Area allows you to draw a second crop box on the same scan. This is useful for scans the contain both an image of the gel and an image of the calstrip.

First draw your crop box on the image, them select Define Calstrip Area and draw a second crop box around the calstrip portion of the image. When you crop the calstrip crop box (see next section), the resulting image will be labeled and saved as a calstrip scan.

Note: Be careful to define your regular crop box before using Define Calstrip Area. Also, use the Define Calstrip Area command to crop only images of calstrips, as the resulting cropped image will be defined as a calstrip image (i.e., with a .csc extension in Windows).

Place Crosshairs

Place Crosshair allows you to mark a reference point within the crop box. Select Place Crosshair, then click inside the box on an easily recognizable spot that is present on each exposure you scan. The crosshair enables you to correctly align the crop box to enclose the same region on all your images.

Crop Settings

With a crop box and crosshair displayed, select Save Crop Settings. Enter a name for your crop settings in the pop-up dialog box. Click on the Apply button to save your settings.

You can recall saved crop settings in other scan windows using the Load Crop Settings function. To delete crop settings, select Delete Crop Settings and choose the settings to be deleted from the list.

3.9.c. Completing the Crop

Once you are satisfied with your crop box and / or calstrip box, position your cursor within the box slightly off-center. The cursor appearance will change to a scissors symbol. You can then click on the mouse to perform the crop.

A pop-up box will ask you whether you want to: (1) crop the original image, (2) save a copy of the area inside the crop box as a separate image, keeping the original image intact, or (3) cancel out of the cropping operation.

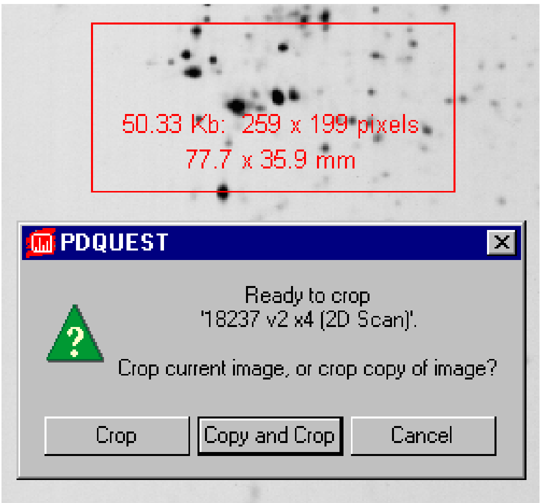

Fig.3-19. Crop box and pop-up Crop dialog.

If you select Crop, your cropped image will be displayed in the image window. If you select Crop and you have defined both a regular crop box and a calstrip crop box, both will be displayed as subwindows within the image window.

If you select the Copy and Crop button, a dialog box will open in which you can enter the name of the cropped image, its version number, exposure number, and / or file type. Click on OK, and the image window will display both the original image and the cropped image in subwindows.

3.10. Flipping and Rotating Images

The matching functions in PDQUEST will tolerate minor distortions in the shape and orientation of images. However, if your images are grossly misaligned, you may have to flip and / or rotate them. These commands are located on the Image menu and toolbar.

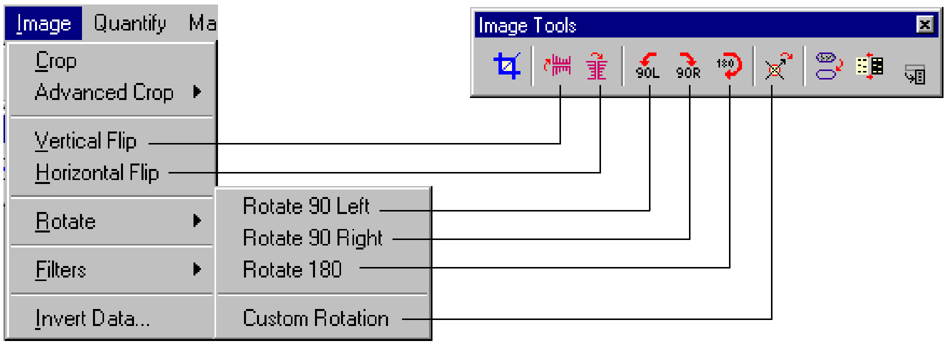

Fig.3-20. Flipping and rotating commands.

Flipping

To flip the image right-to-left, select Horizontal Flip from the Image menu or toolbar. To flip the image top-to-bottom, select Vertical Flip.

90° Rotations

Select Rotate 90 Left, Rotate 90 Right, or Rotate 180 from the Image > Rotate menu or Image toolbar to perform the specified rotation. You will be asked to confirm your choice before the command is executed.

Custom Rotation

If you need to rotate your image in increments other than 90°, you can use the Custom Rotation command.

Select Custom Rotation from the Image > Rotate menu or Image toolbar. A green “plus” sign will appear next to your cursor. Click on the image you want to rotate and a circular overlay with an orange arrow will appear. A small dialog box also will open, indicating the angle of rotation in degrees and radians.

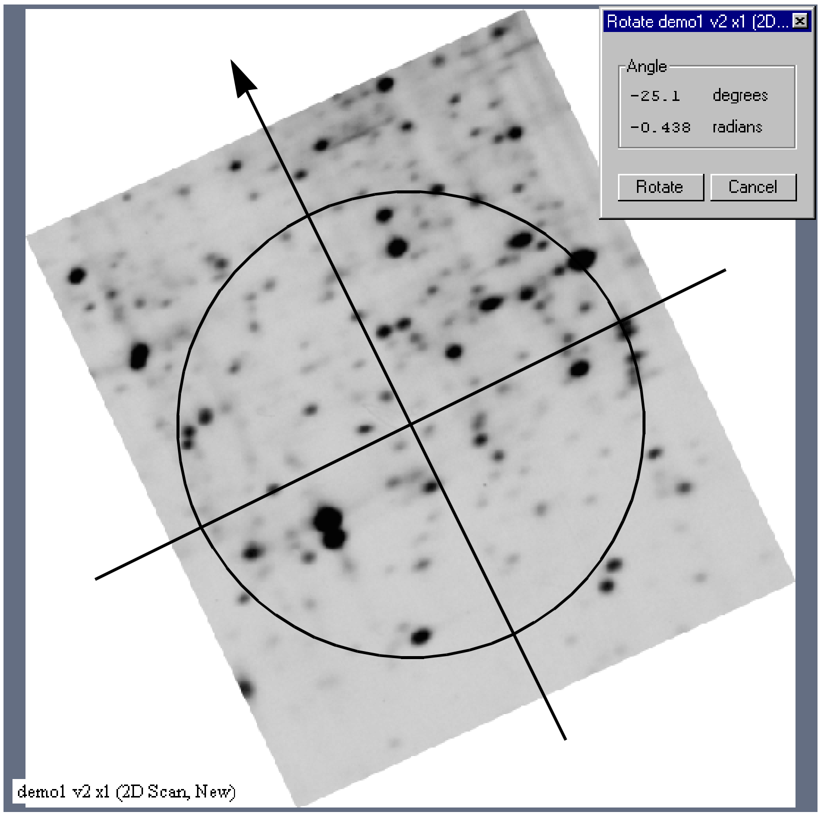

Fig.3-21. Custom rotation; the arrow points in the direction of the new top of the image.

To perform the rotation, position your cursor on the arrowhead and drag. As you drag, the arrow will rotate and the angle in the dialog box will change. Position the arrow so that it points in the direction of the new top of the image. You can “fine-tune” your rotation as much as you like.

To complete the rotation, click on the Rotate button in the small dialog box. A subwindow will open containing the rotated image. You will then have the option of renaming your new image and changing the version number.

If you are not satisfied with your rotated image, simply delete it and start over.

Note: Because an image is composed of square or rectangular pixels, Custom Rotation has to perform some minor smoothing on the image to turn it at a non-90^o angle. In addition, any analysis performed on the image cannot be rotated and will be lost.

3.11. Filtering Images

Filtering is a process that removes small noise features on an image while leaving larger features (like data) relatively unaffected. PDQUEST has a wide range of filters for removing different types of noise from images. Depending on the nature of your data, you will probably need to use only one or two of the available filters. However, you should experiment with several different filters before selecting the ones that work best for your images.

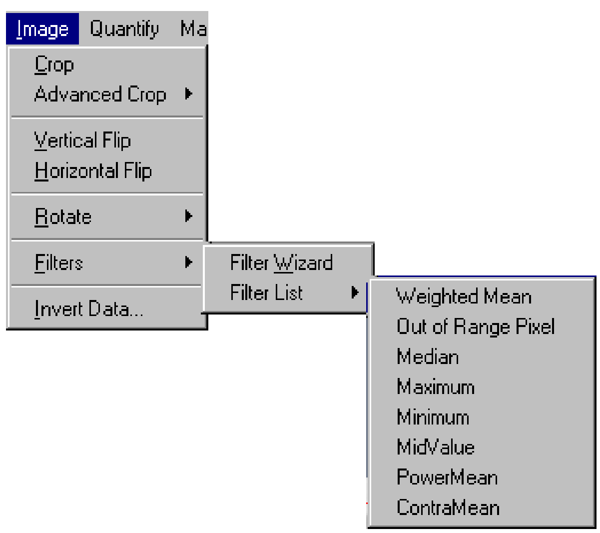

Fig.3-22. Filtering commands.

Note: Since filtering is an irreversible process, you will be asked if you want to create a copy of the original image before you filter. If you are experimenting with various filters, you should create copies of your image and compare them side-by-side. If you filter the original image and save it, *you cannot return to the original, unfiltered state.*

3.11.a. Using the Filter Wizard

The Filter Wizard is designed to guide you through the filter selection process. First, you will identify the type of noise in your image. Next, you will select the size of the filter to use on that noise. Finally, you will filter the image.

To open the Wizard, select Image > Filters > Filter Wizard or click on the Filter button on the Image Tools toolbar.

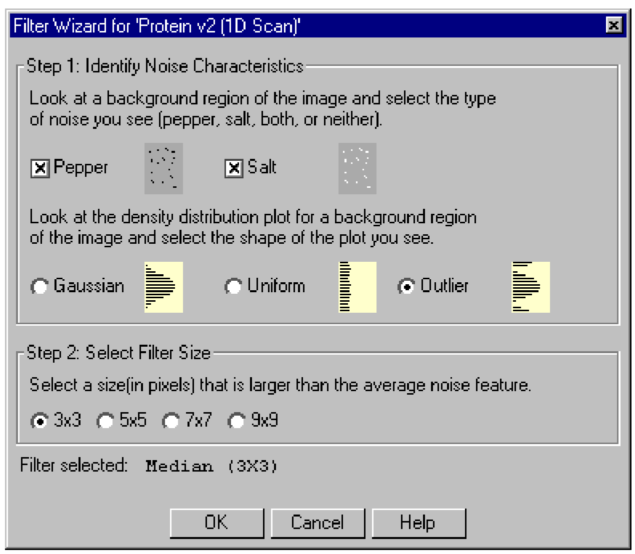

Fig.3-23. Filter Wizard dialog box.

When the Wizard dialog box opens, a vertical plot of the density distribution will pop up next to the image, showing the data range in the image and the amount of signal at each point in the range. (This is the View > Plot Density > Plot Vertical Trace histogram.)

This histogram can be used in conjunction with the Wizard to identify the type of noise in your image.

Step 1: Identify Noise Characteristics

The first step in the Wizard is to identify the type of noise in your image. The Wizard allows you to select from a variety of different noise types and combinations.

To determine the type of noise in your image, it is a good idea to first magnify a background region of the image (i.e., a region where there is no data) using the Zoom Box command on the Main toolbar. Note that as you magnify a region, the density distribution histogram changes to reflect only the signal within the region you have magnified.

After examining the magnified region, select one, both, or neither of the check boxes described below.

- **Salt.** This type of noise appears as specks that are lighter than the surrounding background. The density distribution histogram of this type of noise displays noise peaks at the high end of the range. This type of noise is common in electronic cameras with malfunctioning pixels. It can also be caused by dust or lint in the imaging optics or scratches on photographic film. Salt is a type of outlier noise (see below).
- **Pepper.** This type of noise appears as specks that are darker than the surrounding background. The distribution histogram of this type of noise displays noise peaks at the low end of the range. Its causes are similar to those of salt noise. Pepper is a type of outlier noise (see below).

Next, select one of the following option buttons to describe additional features of your noise.

- **Gaussian.** The distribution histogram of this type of noise has a Gaussian profile, usually at the bottom of the data range. This type of noise is usually an electronic artifact created by cameras and sensors, or by a combination of independent unknown noise sources.

- **Uniform noise.** This type of noise appears in the histogram as a uniform layer of noise across the data range of the image.
- **Outlier noise.** This category of noise includes salt and pepper noise (see above). The distribution histogram of this type of noise displays noise peaks at the high and low ends of the range.

After you have identified the type of noise, go to Step 2.

Step 2: Select Filter Size

Image noise is filtered by means of a filtering window (or kernel), which is measured in pixels. This filtering window slides across the image, processing the pixels within it.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. To select an appropriate size, magnify a background region of your image so that you can see the individual pixels. The filter size you select should be larger than the average noise feature but smaller than your data features.

Note: A smaller filter will alter your image less than a larger filter. Large filters can result in better suppression of noise, but can also blur desirable features in the image.

Step 3: Begin Filtering

After you have completed your selections, the filter name and size will be displayed at the bottom of the Filter Wizard dialog box.

To being filtering, click on the OK button. Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to Copy and Filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.11.b. Selecting a Filter Directly

If you already know the type and size of filter you want, you can select it directly by selecting Image > Filters > Filter List. The pulldown menu includes all the filters available in the Discovery Series.

The types of filters in the Discovery Series are:

- **Weighted Mean.** This filter is useful for reducing Gaussian noise. It calculates the weighted mean of the pixels within the filtering window and uses it to replace the value of the pixel being processed.
- **Out of Range Pixel.** This filter is useful for suppressing salt-and-pepper noise; its effect on Gaussian noise is minimal. This filter calculates the mean of the pixel values in the filtering window, including the pixel being processed. If the difference between the mean and the individual pixel value is above a certain threshold, then the individual value is replaced by the mean.
- **Median.** Also useful for suppressing salt-and-pepper noise, this filter calculates the median value of the pixels within the filtering window and uses it to replace the value of the pixel being processed. The median filter produces very little blurring if a small-sized window is selected.
- **Maximum.** This filter is useful for eliminating pepper noise in an image (it worsens the effect of salt noise). It replaces the value of the pixel being processed with the maximum value of the pixels within the filtering window.
- **Minimum.** This filter replaces the value of the pixel being processed with the minimum pixel value within the filtering window. This filter is useful for eliminating salt noise in an image (it worsens the effect of pepper).
- **MidValue.** This filter is useful for suppressing uniform noise within an image; however, it worsens the effect of pepper and salt. This filter replaces the value of the pixel being processed with the mean of the maximum and minimum pixel values within the filtering window.
- **Power Mean.** This filter is useful for suppressing salt and Gaussian noise within an image (it worsens the effect of pepper noise). It replaces the value of the pixel being processed with the power mean of the pixel values within the filtering window.

- **ContraMean.** This filter is useful for suppressing pepper and Gaussian noise within an image (it worsens the effect of salt). It replaces the value of the pixel being processed with the contra-harmonic mean of the pixel values within the filtering window.

To begin filtering, select a filter type from the pull-down list. A pop-up box will ask you to select a filter size.

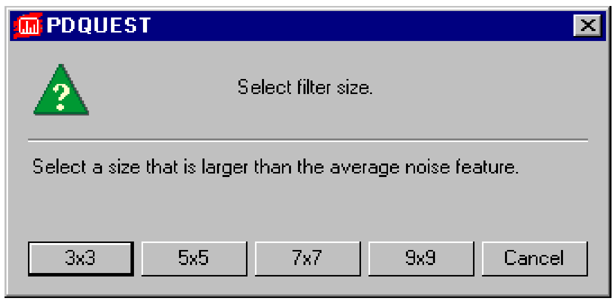

Fig.3-24. Selecting a filter size.

The available filter dimensions range from 3 x 3 pixels to 9 x 9 pixels. (See the previous section for guidance on selecting a size.)

Because filtering is an irreversible process, a pop-up box will give you the option of filtering the original image, creating a copy of the image to filter, or cancelling out of this operation.

If you choose to copy and filter, you will be asked to enter a new name and / or version number for the new copy before the operation is performed. Once this information has been entered, the filtering operation will be performed.

3.12. Invert Data

The Invert checkbox in the Transform dialog box (section 3.7.c.) inverts only the appearance of the image. In some cases, however, you may want to invert the actual image data.

If your image has light spots on a dark background (i.e., the signal intensity of the background is greater than the signal intensity of the sample), you will need to use the Invert Data function before you can analyze the image.

This function is reversible, so if you change your mind, you can always switch it back.

To invert your image data, select Image > Invert Data. You may need to use the Transform function to adjust the appearance of your inverted image.

4. Calibrating 2-D Gels

You can use calibration strips (calstrips) to calibrate your 8-bit scans for accurate quantitation. The commands for this are located on the Quantify > Calibrate Gels submenu.

Note: Calstrips can only be used with 8-bit images. If you are working with 16-bit images, calstrips will have no utility.

If you aren't working with calstrips, please proceed to the next chapter.

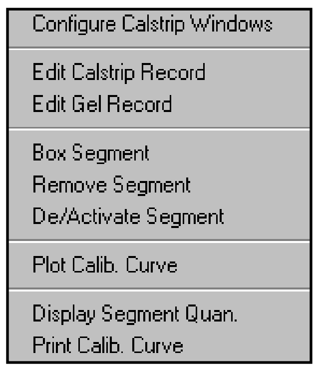

Fig.4-1. Quantify > Calibrate Gels submenu.

Creating a Calstrip File

If calstrip images are currently part of your gel images, you will have to crop them out and save them as separate files for use in PDQUEST. To do so, select Advanced Crop > Define Calstrip Area from the Image menu, draw a box around your calstrip, then crop it and save it as a calstrip file. See section 3.9.b. for instructions.

Configuring Calstrip Windows

If you are displaying multiple calstrips linked to multiple exposures (see section 5.1., Merging Multiple Exposures), Configure Calstrip Windows will automatically arrange the calstrip images in subwindows.

4.1. Calibration Strip Data

Values and units for each segment of your calstrip must be entered into a file called a Calstrip Record prior to calibration.

To do this, select Edit Calstrip Record from the Quantify > Calibrate Gels submenu. The Calstrip Record dialog box will open.

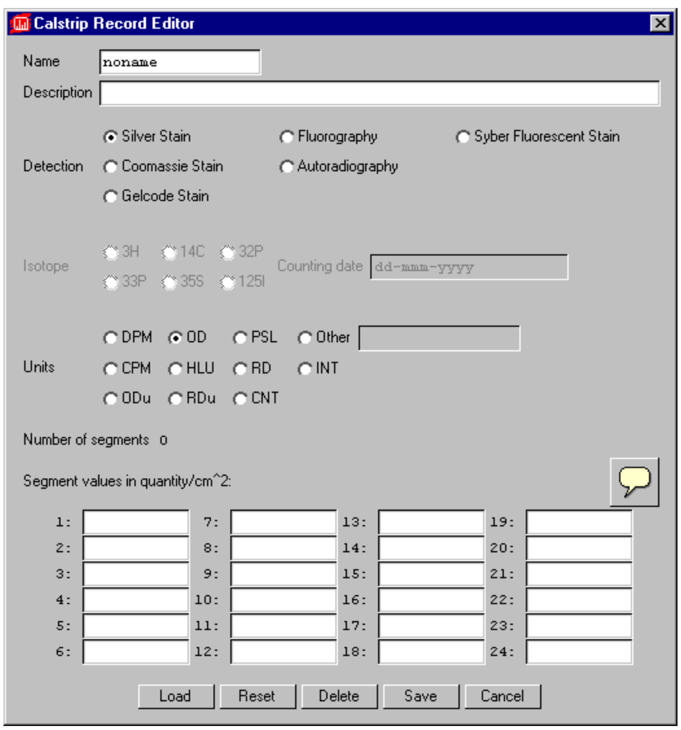

Fig.4-2. Calstrip Record Editor dialog box.

Enter a title for the data record next to the Name prompt. You can enter descriptive information about the calstrip in the Description box.

Next, select the Detection method / stain that was used to make the calstrip.

For Fluorography and Autoradiography detection methods, select the Isotope used. Also enter the date the calstrip was made next to the Counting date prompt. The format in which the date should be entered is dd-mmm-yyyy (e.g., 01-Nov-1993 or 01-11-1993). The first three letters of the name of the month or the number of the month may be entered.

Specify the units for the quantitative values by clicking on one of the buttons next to the Units prompt or typing the units in the text box next to the Other button.

Finally, enter the Segment values for each segment of the calstrip. These values should correspond to the calstrip segments as they appear from left to right (i.e., if the darkest segments are on the left, the values will decrease).

Range of Segment Values and Spot Quantities

The range of segment values should be approximately 2.0 to 2 x 10^4. The minimum value the Calstrip Record will accept is 2.0. A typical maximum is 2 x 10^4, but this value may vary. The range should span four orders of magnitude.

Typical spot quantities will range from 10^{-2} to 5 x 10^3, depending on the size of the spot.

An example of segment values (DPM / cm^2) for an ^{35}S calstrip are:

Segment #:	**1.**	278579.0	**7.**	14892.0	**13.**	827.0
	2.	159376.0	**8.**	9207.0	**14.**	516.0
	3.	100299.0	**9.**	5791.0	**15.**	315.0
	4.	67078.0	**10.**	3670.0	**16.**	134.0
	5.	40895.0	**11.**	2220.0	**17.**	93.0
	6.	24686.0	**12.**	1396.0	**18.**	44.0

Such a calstrip will produce spot quantitation values that typically range from 2.4 DPM to 502.6 DPM.

If you are working with Coomassie gels and you want to make a calstrip, you should begin with a stock solution and make approximately 10 serial dilutions, each of which will be used to make one segment of the calstrip.

For example, you might begin with a 50 µg / ml stock solution of BSA and make segments of the calstrip starting at 50 µg and progressing down to 5.0 ng in constant ratio steps.

In order to enter numerical values in the Calstrip Record that will produce interpretable spot quantitation values, convert all the numbers to ng (i.e., multiply all µg values by 10^3).

For the example mentioned above, the values would range from 50,000 ng to 5.0 ng.

Saving/loading Calstrip Records

When you have completed the information in the dialog box, click on the Save button at the bottom of the box.

To access the information in an existing Calstrip Record, click on the Load button. A list of saved Calstrip Records will be displayed. Select the one whose data you wish to see by clicking on it. The data associated with that record will be displayed on the form.

To delete a saved Calstrip Record, click on the Delete button.

Click on the Cancel button to close the Calstrip Record Editor form.

4.2. Scan Information

After the calibration data has been entered, you will also need to enter some information about the gel that goes with it.

Start by loading the 2-D and calstrip scans. Display the calstrip scans by using the Configure Calstrip Windows function (Quantify > Calibrate Gels > Configure Calstrip Windows).

Click on Gel Record in the Calibrate Gels submenu and click on the calstrip scan. The Gel Record dialog box will open, in which you can enter information about the calstrip.

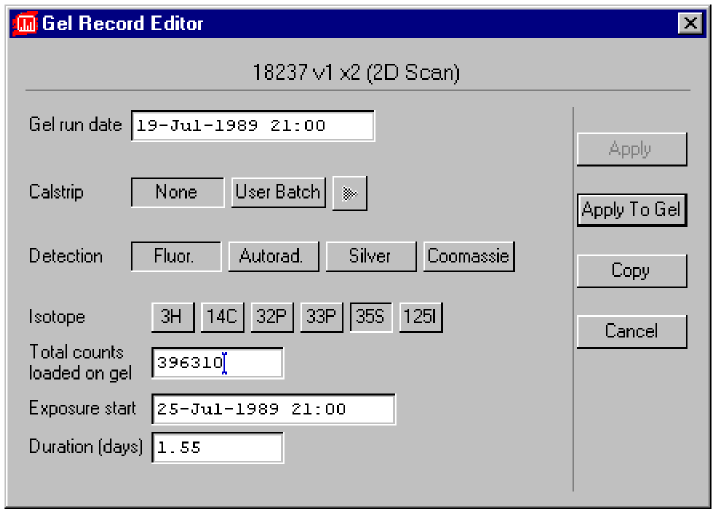

Fig.4-3. Gel Record Editor.

Enter the date on which the gel was run in the text box next to the Gel run date prompt.

Next to the Calstrip prompt, click on the button that corresponds to the kind of calibration associated with this gel. In most cases, you or someone in your lab has made the calstrips, so the button labeled User Batch is the appropriate choice.

Next to the User Batch prompt, click on the arrow. From the list displayed, select the calibration record name for the calstrip. The Detection and Isotope fields are automatically obtained from the specified Calstrip Record.

Click on the appropriate button to indicate the type of detection used for your gel, if not already defined from the batch file.

Click on the Isotope button to indicate the isotope used in the gel. This is usually, though not necessarily, the same isotope as the one used in the calstrip.

Next, enter the number of counts loaded on the gel.

Enter the day on which you began exposing the gel to film in the text box next to the Exposure start prompt.

Note: Most calibration problems arise from incorrectly entering the dates used to calculate radioactive decay factors for the isotopes. Wrong values can make the calstrip appear to have decayed to nothing.

Next to Duration (days), enter the number of days that the film was exposed.

Once this data has been entered, you will want to apply it to the 2-D scan and the calstrip scan. To do so, click on the Apply button.

If you want to apply different calibration data to your 2-D scan than to your calstrip scan (if, for example, your 2-D scan and calstrip scans use different isotopes), use the Apply Here button to apply the gel record to the currently displayed image only (2-D or calstrip scan).

Copying Gel Record Information

If you have scanned several exposures of the same gel, most of the calibration data will be the same for all of them (except for the exposure start date and length). Instead of entering all the information for each exposure, you can enter it once, and then copy it between exposures using the Copy button.

Clicking on Copy will open a list of available gel records. Select the record that you would like to apply to the current image and click the mouse.

Click on the Cancel button to exit the operation.

4.3. Sampling the Calibration Strip Segments

The next step in calibrating your gels involves sampling the densities in the segments of the calstrip to establish a correlation between the sampled intensities recorded by the scanner and the counts entered into the Calstrip Record.

If you have not already done so, display the calstrips. If you have gels with multiple exposures, use the Configure Calstrip Windows function to automatically divide the screen into enough windows to display all the calstrips for a given gel.

4.3.a. Flip Calstrip (if necessary)

If the orientation of the scanned calstrip does not correspond to the order in which the segment activities are entered in the Calstrip Record, a message will be displayed, indicating that the calibration density decreases while the counts increase (or vice versa).

To change the orientation of the calstrip so it corresponds to the Calstrip Record segment activities, use the Horizontal Flip function (under the Image menu). If the dark segments were on the left, they will now be on the right, and vice versa. Again, we recommend that you enter your data from highest to lowest activity and display the calibration strip with the darkest segment on the left.

4.3.b. Box Segments in a Calstrip

Once the calstrip is in the correct orientation, go back to the Quantify > Calibrate Gels submenu and select Box Segment.

Go to the first (leftmost) segment of your calstrip. Sampling boxes must be drawn beginning with the leftmost segment of the calstrip and moving right.

With Box Segment selected, drag a box on the calstrip. When you release the mouse button, the area will be sampled.

Note: When defining your box, be sure to avoid sampling regions that are very close to the borders of the segment or any irregular areas (i.e., areas that are overly dark or light).

The number of each segment will be displayed in the upper left corner of each sampling box.

Above the box will be a number between 0.00 and 2.00. This represents the O.D. value for the segment.

Beneath each box is the number of counts associated with that segment from the Calstrip Record.

If you make a mistake, simply redraw a new box over the old one. Repeat this procedure for all the segments of the calstrip.

Note: All segments entered in the calstrip record must be boxed starting from the left. The leftmost segment corresponds to Segment 1 on the Calstrip Record. Your calstrips and calstrip data should be designed so that either (1) the darkest segments are on the left and the count values decrease, or (2) the lightest values are on the left and the count values increase.

You can draw more boxes than the number of segments defined in the Calstrip Record. However, they are automatically and permanently deactivated.

4.4. Plot the Calibration Curve

Once you have created the sampling boxes, select Plot Calib. Curve to display the calibration curve.

A graph will be displayed that correlates the optical density (usually between 0.00 and 2.00 O.D.) with DPMs or other specified units.

The Y axis represents the optical density (0.00–2.00) in each sampled segment, while the X axis represents the corresponding DPM or other unit per image unit area calculated from the calibration record data.

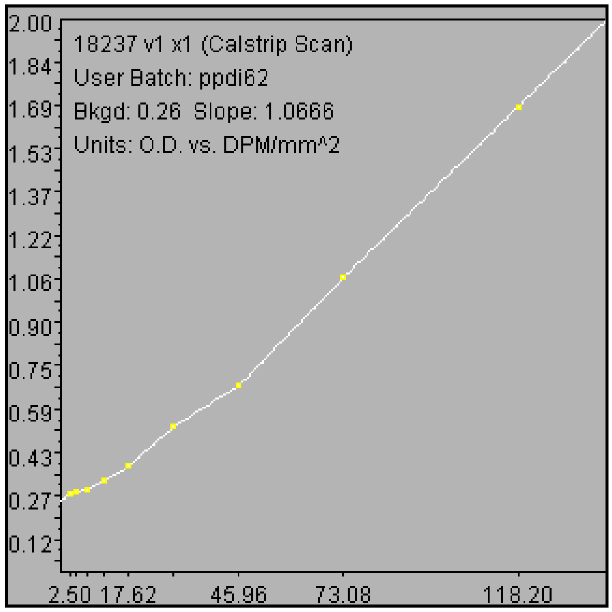

Fig.4-4. Example of a calibration curve.

Calibration Curve Problems

Find errant data points and their corresponding segments using the function Display Segment Quan.

Above each sampling box will be displayed the average O.D. value within the box, and just below the box will be displayed the corresponding DPM (or other specified unit). This should tell you which sampling boxes correspond to errant data points on the graph.

Usually, problems arise when a box straddles the boundary between two adjacent segments or the density is very uneven within the segment due to bleeding from an adjacent segment. These sorts of problems should be remedied by redoing the sampling boxes.

Use the function Remove Segment to remove an errant sampling box.

Draw a new sampling box on the segment using Box Segment.

Place the new box carefully to avoid repeating the problem. For example, keep the box away from segment boundaries. If the segment shows bleeding, place the smallest box possible in an area of the segment with representative O.D. Try not to sample blotchy areas.

Replot the calibration curve using the function Plot Calib. Curve.

If there is still a problem, you can opt not to include that particular segment in the calculation of the calibration curve. Select De / Activate Segment and click on the problem segment. The segment will still exist, so its counts in the corresponding Calstrip Record do not need to be changed or deleted. However, the segment will not be included in the calculation of the calibration curve.

If you change your mind and want to reinclude the segment, select De / Activate Segment again and click on the segment. The segment will be reactivated and included in calibration curve calculations.

4.5. Repeat for Each Calstrip

If you have multiple film exposures of the same gel, repeat the steps outlined above for each exposure. The calibration strip batch number for each film exposure of a single gel should be the same since the same physical calibration strip was used to make each exposure.

5. Spot Detection

This chapter describes how to merge multiple exposures (if any) and detect spots.

5.1. Merging Multiple Exposures

If you are working with 8-bit images and are using calstrips on your scans, PDQUEST gives you the capability to merge multiple exposures of the same gel into a single image with a greater dynamic range.

Note: You can only merge 8-bit images with calstrips. If you are working with 8-bit images or are not using calstrips, please proceed to the next section.

Dark spots will saturate on long exposures, while faint spots will be missed on short exposures, and either extreme can be missed on medium duration exposures. Therefore, it is frequently necessary to combine the quantitative data from a number of exposures of the same gel exposed for different lengths of time into a “merged” synthetic image. Merging creates such an image automatically as you spot detect.

Merging occurs automatically when you auto-detect spots, provided all your exposures and calstrips are loaded into the same image window.

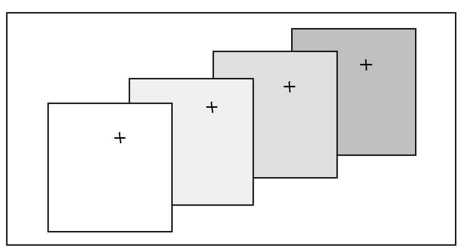

Fig.5-1. Diagrammatic representation of merging.

Cropping and Aligning Exposures of the Same Gel for Merging

Before you can merge multiple exposures, you must crop and align them so that they are exactly the same size and position. To do this, you will use the Advanced Crop commands on the Image menu. See section 3.9.b. for instructions.

5.2. Auto-Detecting Spots

This section describes how to use PDQUEST to detect and quantify spots in 2-D protein gels that have been scanned and prepared according to the instructions in the previous sections of this guide. If you are satisfied with your calibration data, or if you are working with uncalibrated gels, you are now ready to detect and quantitate the protein spots on your gel.

Note: If you detecting spots on a calibrated gel with multiple exposures, all of the exposures and calstrip scans must be loaded.

5.2.a. The Auto-Detect Form

Select Auto-Detect from the Quantify menu to display the Auto-Detect form.

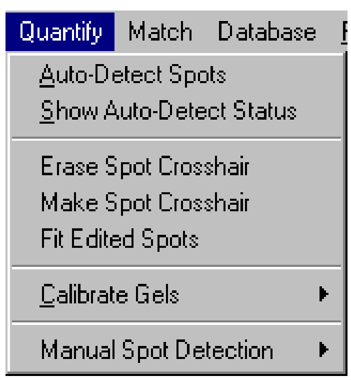

Fig.5-2. Quantify menu.

The Auto-Detect form will ask you to select the gels that you want to spot detect. You can choose all loaded scans or a subset by selecting names from the list of loaded scans.

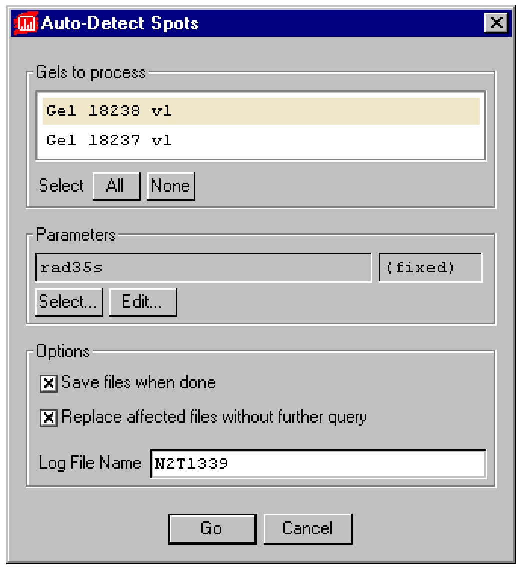

Fig.5-3. Auto-Detect dialog box.

Click on the Select button to see a list of available autodetection parameters. Select the set of parameters that should be used for autodetecting the spots.

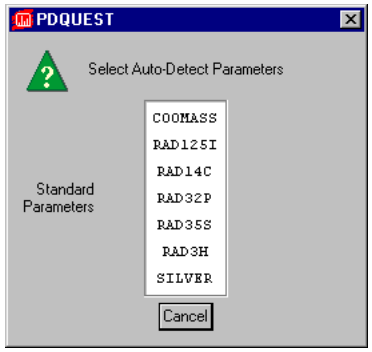

Fig.5-4. Choose a set of parameters for spot detection.

A number of predefined Standard Parameters are available. Parameters such as Silver and Coomass are optimized to detect spots on those types of gels.

Note: The spot detection parameters include various smoothing and background subtraction techniques which can reduce the effects of background density and gel artifacts and improve the quality of spot detection. For more information on these parameters, refer to Appendix I.

Besides the set of Standard Parameters, you can also create user-defined parameters. To create a set of user-defined parameters, see section 5.4.

Select the set of parameters you want to use. The list will disappear and the name of the parameter file you chose will appear next to the Parameters button.

Select "Save files when done" to auto-save the results of autodetection. If these gels have been previously spot-detected, specify whether you want to replace existing files by clicking on Yes or No next to "Replace affected files without further query?"

Near the bottom of the Auto-Detect dialog box is a text box displaying the default Log File Name. You will need to know this file name if you check the results of spot detection using the Show Status function (see below). You can change this name to a file name of your choice.

Once you have selected your autodetection parameters, click on the Go button. If you decide not to proceed with spot detection, select Cancel.

As spot detection proceeds, the Auto-Detect Status form will appear on-screen.

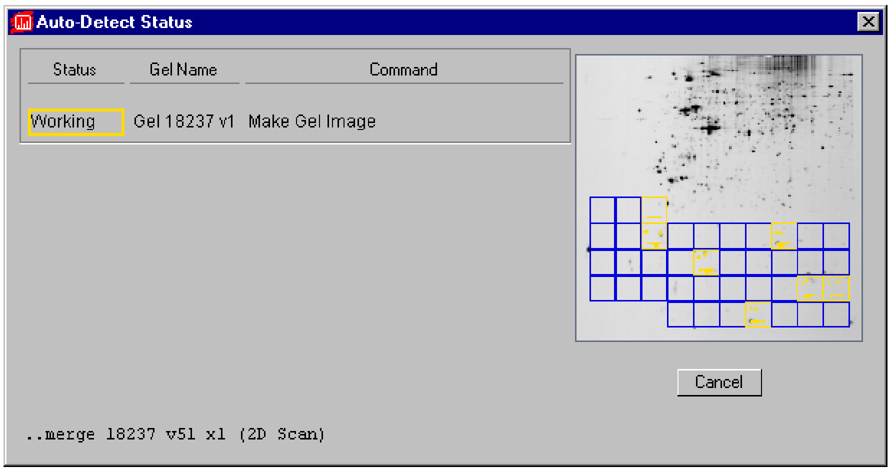

Fig.5-5. Auto-Detect Status form.

Spot Detection Status

The Auto-Detect Status form lists the status of each gel that you selected for detection while the detection is occurring. Messages appearing at the bottom of the form display information relating to the individual steps in spot detection as described in Appendix I.

A preview image in the upper-right corner of the form displays a graphical depiction of each step as it occurs.

Click on Cancel to interrupt spot detection before it is finished.

Completing Spot Detection

When autodetection is complete, a pop-up box will ask if you want to proceed with creating a matchset.

If you are confident about the results of your detection, choose Yes, and the Create Matchset form will appear (skip ahead to Chapter 6 for more information on matchsets and the Create Matchset form). If you want to review the results of your detection before creating a matchset, choose No.

If you did not select “Save files when done” in the Auto-Detect dialog, save the results of your autodetection by clicking on the Save button on the main toolbar.

Review the Spot Detection Procedure

Once you have autodetected spots, you can check the procedure using the Show Status function in the Quantify menu. This tells you when spot detection was started, which gels were detected (or failed to detect), which steps were performed, and when spot detection was completed.

5.3. Reviewing Spot Detection

When you autodetect spots in a gel scan, you create two new files with the same root file name as the scan: a Gel Image file and a Gel Spots file.

The **Gel Image fi le** contains the merged and processed image. The **Gel Spots file** contains a synthetic image with Gaussian representations of the spots in your Gel Image file. The Gaussian fitting is essential for the purposes of higher-level spot analysis. Therefore, it is essential that the “synthetic” spots in the Gel Spots file accurately match the “real” spots in the Gel Image file.

When autodetection is complete, both the Gel Spots and Gel Image files will be loaded into the same image window with the scan that was used to create them.

Use the View > Assign tool and / or Window > Configure Subwindows command to arrange your images in the window.

View > Advanced Assign > Interchange Images (F11) will toggle between the Gel Image and Gel Spots file.

To display all spot crosshairs, use either Spot Crosshairs or All Spot Crosshairs (in the View > Spot Overlays submenu).

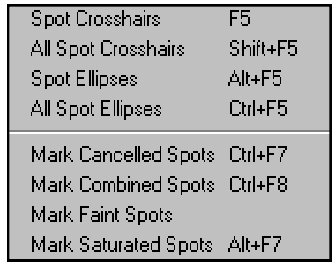

Fig.5-6. Spot Overlays submenu.

The crosshairs mark the exact centers of spots in the Gel Spots file. The crosshairs in the Gel Image are exact duplicates of those in the Gel Spots file. Each spot in the Gel Image should have a crosshair on top of it to indicate that this intensity has been detected as a spot.

Automatic spot detection should accurately detect a high percentage of the spots you can see on the gel. Due to gel artifacts, etc., some spots may not be properly detected and fitted by Auto-Detect. If this is the case, you can manually add / delete spots.

Note: Usually, problems with spot detection can be corrected on a spot-by-spot basis during matching, If you prefer, however, you can correct problems before matching.

Viewing Tips

When studying spots on an image, you can magnify selected regions of the image using the Zoom Box tool on the main toolbar. Also, you can magnify and review the image section by section by using the View > Advanced View > 3x3 Sections function.

These and other viewing tools are described in detail in Chapter 3.

5.3.a. Identifying and Correcting Spot Detection Errors

Spot detection errors include undetected spots, too many spots detected, saturated regions, and spots that have not been properly fitted (crosshairs off-center, wrong spot geometry in the Gel Spots, etc.).

When checking the shape and position of spots, use Interchange Images (F11) to toggle between the Gel Image and the Gel Spots files.

Adding and Erasing Spots

To manually add a new spot center on your Gel Image, select Make Spot Crosshair from the Quantify menu, then click on the unmarked spot. The spot will be added to the Gel Spots file.

Note: Make Spot Crosshair will invoke the Gaussian modeling algorithm for quantifying the density around each new spot.

To remove an incorrect spot center, select Erase Spot Crosshair, then click on the wrongly marked spot on the Gel Image. The spot will be removed from the Gel Spots file.

Use Plot Cross-section (View > Plot Density > Plot Cross-section) to aid in accurately positioning new spots.

Saturated Spots

Because Auto-Detect may or may not have correctly fit spots in saturated regions, saturated spots are not quantified. The best you can do is to use spot crosshairs as place holders for saturated spots.

Locate saturated regions in the Gel Image using the function Highlight Saturated Pixels in the Transform window (View > Transform). To highlight specific spots that have been saturated, select View > Spot Overlays > Mark Saturated Spots (ALT+F7).

After you have identified saturated spots, redisplay the spot crosshairs.

Look for crosshairs in the saturated spots that appear to be off-center. Correct the problem by erasing the off-center crosshairs (Erase Spot Crosshair) and adding new crosshairs (Make Spot Crosshair) in the correct locations.

Remember that saturated spots are not quantified and centers in Gel Spot files are merely place holders for these spots.

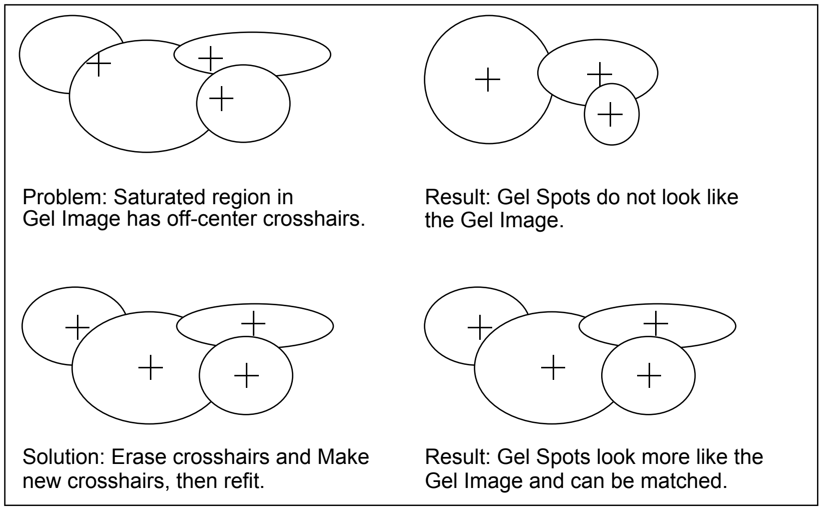

Fig.5-7. How to solve the problem of misplaced spot centers in saturated regions.

Isolated Undetected Spots

You may find areas in the Gel Image and in the gel that have some density above background, but that have not been detected as spots. Also, some areas with substantial density may have escaped spot detection. These areas are not marked with crosshairs in the Gel Image and there are no corresponding spots in the Gel Spots.

If necessary, redisplay the crosshairs, then look for spots in the Gel Image that are not marked with crosshairs and for which there are no corresponding spots in the Gel Spot file. Select Make Spot Crosshair from the Quantify menu, then click on the spot.

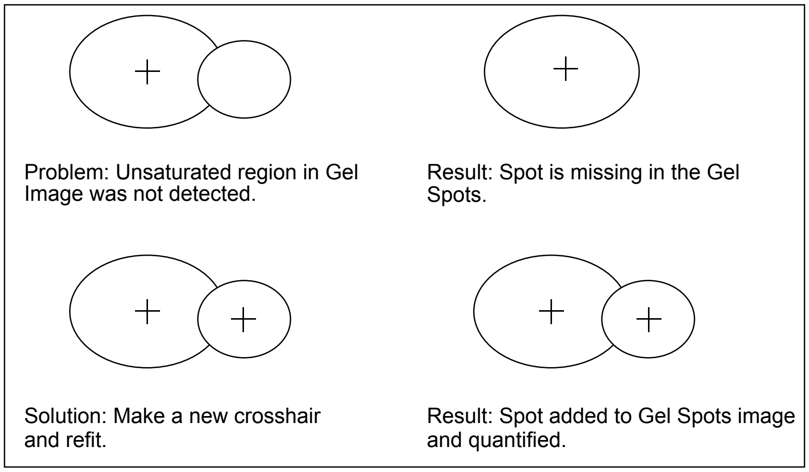

Fig.5-8. How to solve the problem of too few spot centers in isolated unsaturated regions.

Note: Often, the problem involves fuzzy or faint density regions that are not easily modeled by ideal Gaussian distributions.

Undetected Spots in Clusters

Most detection errors occur in regions of spot clusters. A common problem is undetected spots in these regions.

If necessary, redisplay crosshairs, then look for unsaturated clusters in which there are fewer spot crosshairs than you believe there should be, based upon your examination of the gels. Correct the problem by marking the undetected spots manually.

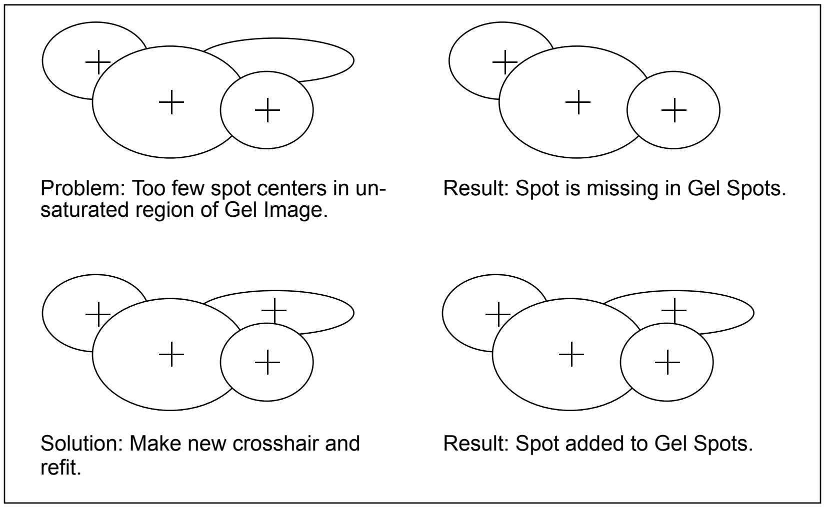

Fig.5-9. This illustration depicts a situation in which undetected spots are clustered.

Note: Sometimes, even to a trained eye, it is not obvious how many spots are in a cluster. If you can, refer to the lighter film exposures to resolve clusters.

Too Many Spots in Clusters

Another possible error that occurs in regions of spot clusters is too many spot centers. In this case, Auto-Detect is dealing with an overload of information and fits too many ideal distributions to the overlapping density.

If necessary, redisplay crosshairs, then look for unsaturated clusters in which there are more spot crosshairs than you believe there should be, based upon your examination of the gels. Delete the incorrectly identified spots by using the Erase Spot Crosshair tool.

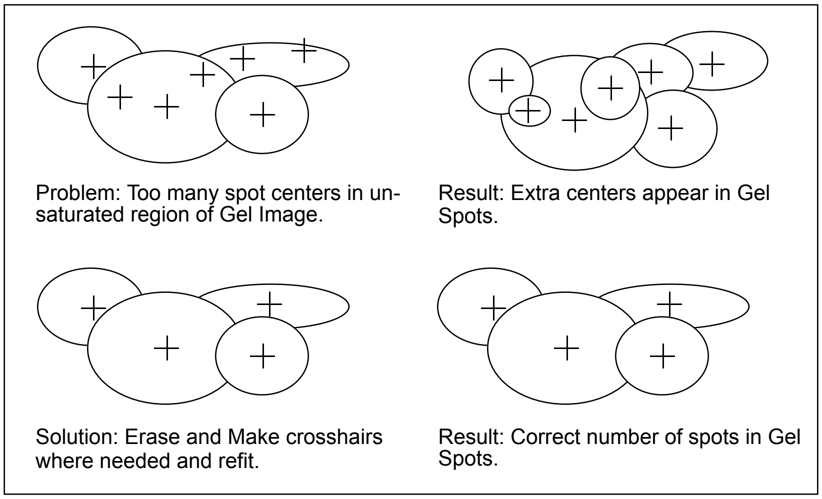

Fig.5-10. Illustration depicts a situation in which there are too many detected spots in a cluster.

Note: It is not always obvious how many spots comprise a cluster. If you can, refer to the lighter film exposures to resolve clusters.

5.4. Customizing Autodetection Parameters

If you have problems autodetecting, you can specify your own autodetect parameters.

Click on the Edit... button below Parameters in the Auto-Detect dialog box to open the Edit Auto-Detect Parameters dialog box.

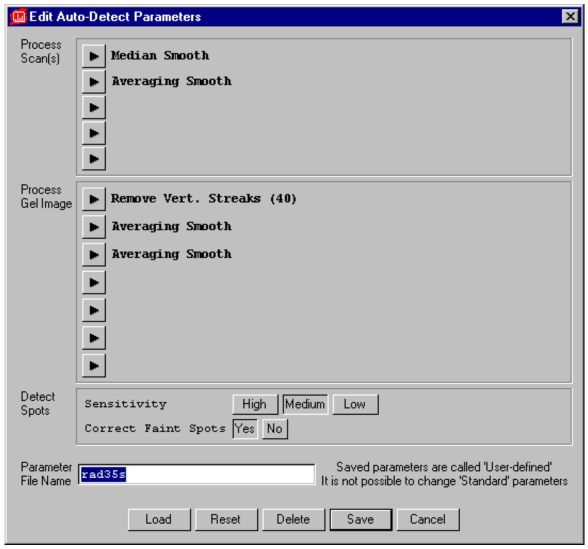

Fig.5-11. Auto-Detect Parameters dialog box.

Here, you can specify the smoothing and background subtraction operations that are applied to the 2-D scan and the Gel Image as well as the sensitivity level at which spots are detected.

Refer to Appendix I. for details on the different parameters.

The following is a summary of the procedures that are applied during spot detection.

Processing the Scan

In the first section of the Auto-Detect Parameters form, you can specify the operations to be applied to the scanned image.

There are five buttons with arrows in them under Process Scan(s). To see a list of the available operations, click on the first arrow button. A list of options will appear.

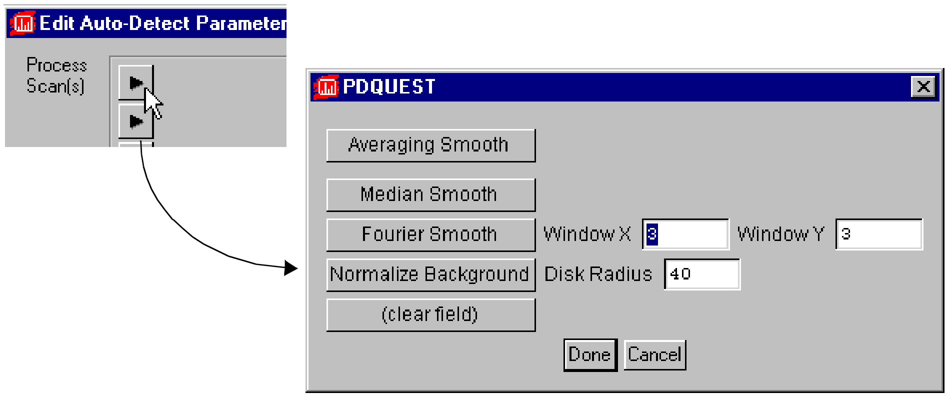

Fig.5-12. Selecting new Process Scan(s) auto-detect parameters.

To select the operation to be applied to the scan, click on it with the cursor and then select the Done button. The list will disappear and you will see the operation that you selected next to the first arrow button.

You can repeat this procedure for each of the arrow buttons, thereby processing the scan in up to five different ways.

For more information about these parameters, see Appendix I.

Processing the Gel Image

The next step in the autodetection process is to create a Gel Image from the 2-D scan. The Gel Image is an intermediate image on which smoothing and background subtraction are performed while preserving the original scan.

The next section of the dialog box controls the processing of the Gel Image. As with processing a 2-D scan, click on one of the arrows to see a list of the smoothing and background subtraction options available.

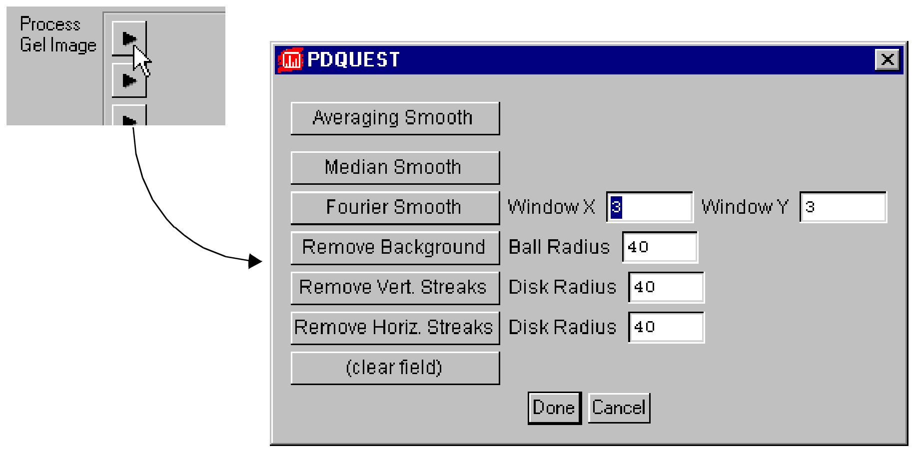

Fig.5-13. Selecting new Process Gel Image auto-detect parameters.

To select one of these operations, click on it and then click on the Done button. The list will vanish and the selected option will appear next to the arrow button.

For more information about these parameters, see Appendix I.

Detect Spots

After the Gel Image has been processed, spots can be detected. In this section of the Auto-Detect Parameters form, you can specify whether the spot detection should be performed at high, medium, or low sensitivity.

At higher sensitivity, more spots will be detected, but there is a greater chance of noise and gel artifacts being mistaken for actual protein spots.

At low sensitivity, you are less likely to erroneously detect noise as real spots, but you increase the possibility of missing faint or diffuse spots.

You can also indicate whether a correction should be applied to faint spots, since the quantitation of faint spots tends to be underestimated.

Enter a File Name and Save the Parameters

The parameters that you set can be saved by entering a file name in the text box next to the Parameter File Name prompt near the bottom of the dialog box and clicking the Save button. The parameter file name that you enter will appear in the User-defined section of the Auto-Detect Parameters form.

If you want to review a particular set of parameters, click on the Load button to display a list of the available files. Click on a file to display all the operations associated with that parameter file.

The Reset button will restore the dialog to its default values.

The Cancel button closes the dialog box. Any unsaved changes will be lost.

6. Creating and Editing Matchsets

After the protein spots in your gels have been detected, you are ready to compare one gel to another. To do so, you must create a matchset.

Matchsets are sets of gels grouped together for the purpose of qualitative and quantitative comparison. Individual spots on one gel can be compared to comparable spots on every other gel in the matchset.

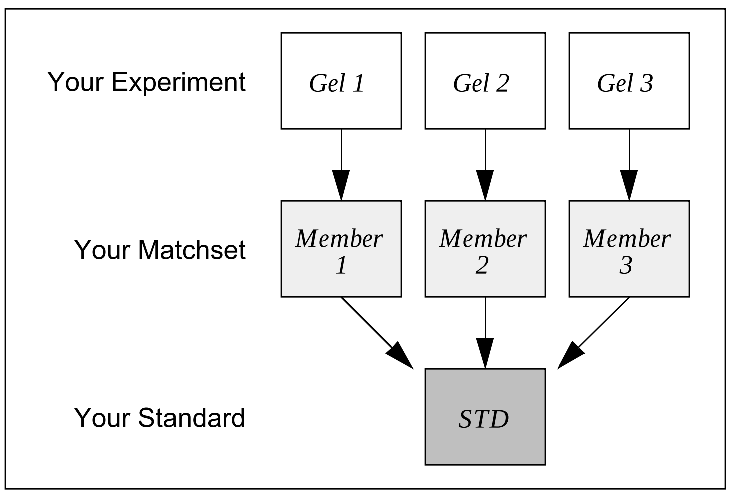

Fig.6-1. How experiments are organized into matchsets.

Matchsets can be created from any number of gels. Matching is coordinated through an image called the "**Standard** ," which is created from the members of your matchset. Ultimately, your Standard image will contain copies of every protein spot from all the member gels.

Matchsets can be composed of members with different quantitation units. (Gel Images calibrated with calstrips have units of DPMs, while uncalibrated images use ODs.) It is important to remember that it is *not* appropriate to make quantitative comparisons between gels with different units, although qualitative comparisons are still valid.

6.1. Creating a Matchset

Matchsets are created from Gel Spots files and can also contain Gel Image files.

Note: Gel Image files can be edited (spots may be added and erased), whereas Gel Spots files cannot. If you have not already edited the results of spot autodetection (see section 5.3.), you should load your Gel Images when creating a matchset. However, note that Gel Image files are much larger than Gel Spots files. If computer memory or disk space is an issue, you should perform spot editing before creating the matchset, then not include Gel Image files in the matchset.

The commands for creating and editing matchsets are located on the Match menu.

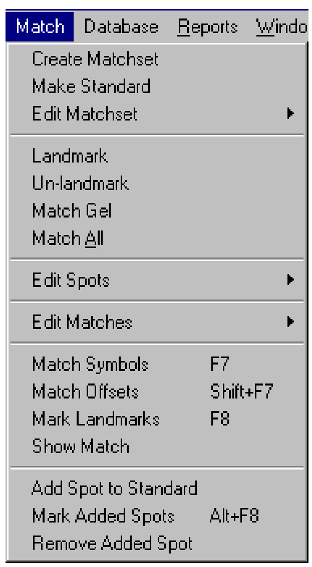

Fig.6-2. Match menu.

Select Match > Create Matchset to open the Create Matchset dialog.

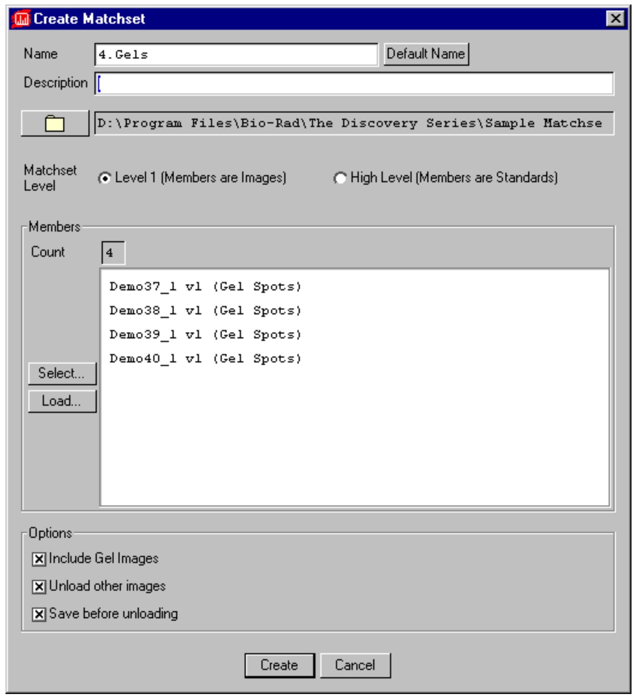

Fig.6-3. Dialog box for creating a matchset.

At the top of the dialog box, you are prompted to enter a name for your new matchset. To assign your matchset a default name generated by the program, click on the Default Name button.

You can also enter a description of the matchset in the text box next to the Description prompt.

Select the directory in which the matchset should be stored. The currently selected directory is displayed next to the Directory button. To save the matchset in a different directory, click on the Directory button; the Select Directory form will appear. Click on the Matchsets button in the Select Directory dialog to select the working Matchsets directory (specified in Edit > File Paths).

Next, specify whether you are creating a “lower level” matchset or a “higher level” matchset by clicking on the appropriate button next to the Matchset Level prompt. (Lower level matchsets are created from original gel images; higher level matchsets are created from the Standards of lower level matchsets. For more information on higher level matchsets, see section 9.2.)

Selecting Currently Loaded Files

If you are creating a lower level matchset, select the Gel Spots files that should be included. If the files to be included are already loaded, click on the button labelled Select.

A list of currently loaded Gel Spots will be displayed. Select the files that you want include in your matchset and click on Done. The files you chose will be listed next to the Members prompt.

Selecting Unloaded Files

To include Gel Spots that are not currently open, click on the Load button. The Select New Member dialog box will be displayed. Select the files that you want to include in the matchset and click on Add Selected.

The names of the files you selected will appear in the Members list in the Create Matchset dialog box.

Creating the Matchset

Specify whether you want to include Gel Images in the matchset by choosing Yes or No next to the Include Gel Images? prompt.

Indicate if any other loaded images should be saved and unloaded before this matchset is created by selecting Yes or No next to the Unload other images? prompt. Also select whether you want to save these other images before unloading them (Save before unloading).

Once you have finished entering information on the Create Matchset form, click on the Create button.

You will be prompted to select an image to serve as the template for the Standard. Generally, this will be an image with the most spots, or an image that is most representative of the entire matchset. (See the following section for more information.) If you are unsure which file you want to select for your Standard, click on Cancel—you can select the Standard template later. Otherwise, select the appropriate file name.

A new window will open displaying all the members of the newly created matchset.

6.2. Matchset Standard

For each matchset, you must choose one gel that will serve as the template for the matchset Standard. A copy of that image will be created and designated the Standard.

The goal is to match all gels in the matchset to this Standard. The Standard represents all the data in the matchset and serves as the basis for comparing different experiments in higher level matchsets (see Chapter 8).

The Standard does not contain any image data, although many calculations, such as those that yield relative molecular weights and isoelectric points for protein spots, are performed in the Standard. Also, spot annotation occurs in the Standard.

6.2.a. Criteria for Selecting the Standard Gel

Before choosing a gel to serve as the template for your Standard, you may want to review how many spots were detected in each Gel Spots file you are adding to the matchset. With the Gel Spots file open and its window selected, select File Info from the File menu. The info box will tell you how many spots were detected in the gel, as well as file size, total data in the spots, etc.

This information can help you decide which gel should serve as the Standard template.

The criteria for selecting the template for the Standard are listed below. Ideally, you want the Standard to satisfy all of the following criteria. If you intend to build a database, or analyze data from many different experiments, we recommend that you select a gel that is representative of your system and / or is a gel run from a representative control sample. Otherwise, choose the gel with the most spots that comes closest to satisfying the other criteria.

Number of Spots

In creating a Standard, you start with the spots in the template, then you manually add spots from the other gels in the matchset. Choosing a template with the most spots minimizes number of spots that you will need to add later.

Representative Gel

The Standard is a representation of all the data in your matchset and is often used in reports. Therefore, you may not want to use an atypical gel as a template, even if it has the most spots. We suggest that you choose the most representative Gel Spots file (i.e., the one that looks like a typical gel run from samples in your biological system).

Control Gels

Choose a control gel as the Standard so that it can be included in every experiment you plan to analyze and used as the Standard in every matchset. This makes it easier to compare results between different experiments.

Spot Quality

Editing the Standard and matching other gels to the Standard are easier when the Standard consists of high quality spots.

6.2.b. Creating a Standard

To create a Standard, select Make Standard from the Match menu. A pop-up box will appear. Select the member image to serve as the Standard template from the list.

The Standard image will appear in the upper left subwindow of the matchset and will have the name “REF <matchset name>-std.”

At this point, we suggest that you save the matchset by clicking on Save.

6.3. Editing a Matchset

The commands for editing the structure of your matchset are located on the Match > Edit Matchset submenu.

You can add members to and remove members from an existing matchset. Selecting Add Member from the Match > Edit Matchset submenu will open a version of the Create Matchset dialog box (see above). You can add a new Gel Spots file to the matchset using this dialog as if you were creating the matchset for the first time.

To remove a matchset member, select Remove Member and select the member to be removed from the pop-up list. You will receive a warning that this command cannot be undone.

Matchset Info opens a dialog box in which you can enter a description and comments about your matchset.

Reorder Matchset opens a dialog box in which you can select a member name and move it up or down in the matchset hierarchy. This hierarchy affects the order in which the matchset members are displayed on the screen (top to bottom, left to right).

Making/Erasing Spots in the Matchset Members

If you have not already done so, review the spots in the Gel Images of your matchset members and, if necessary, add or remove spots using the Make Crosshair and Erase Crosshair tools. See section 5.3., Reviewing Spot Detection.

Note: Spots can only be added to and erased from Gel Images, not Gel Spots files. Therefore, these commands will not work on the Standard, which does not have a Gel Image file.

As always, when editing spots, it is useful to magnify selected portions of the images using the zoom tools and Imitate Zoom command described in Chapter 3.

Editing Spots in the Matchset Standard and Members

PDQUEST has tools for editing the spots in the Gel Spots files of the matchset members, as well as the Standard. These functions are located on the Edit Spots submenu.

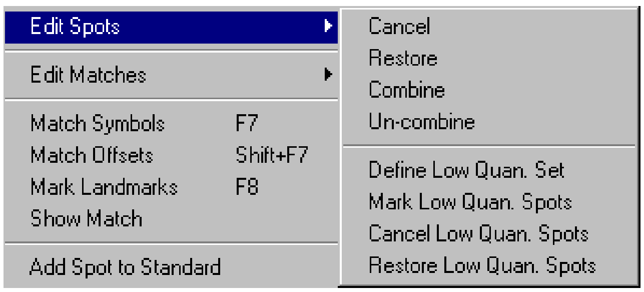

Fig.6-4. Edit Spots submenu.

Cancel will push unwanted spots into the background (i.e., the spot remains, but is not included in matching or other calculations). Select Cancel from the Match > Edit Spots submenu and click on the spot in the Standard or member image that you want to cancel. To highlight the spots you have cancelled, select Mark Cancelled Spots from the View > Spot Overlays submenu.

If you change your mind, select Restore from the Edit Spots submenu and click on the point of the cancelled spot. The spot will reappear.

Combine will combine the quantitation of a cluster of spots into one spot. Select Combine from the Match > Edit Spots submenu and draw a box around the cluster of spots in the Standard or member images that you want to combine. When you finish drawing the box, click on the point in the box where you want the combined spot to be placed. To highlight the spots you have combined, select Mark Combined Spots from the View > Spot Overlays submenu.

If you change your mind, select Un-combine from the Edit Spots submenu and draw a box around the combined spot. The spot will be uncombined into the original multiple spots.

Note: Spots edited in the Standard or any matchset member are not automatically updated in the other members or Standard. You must make the same changes to the same spots/clusters in all the images of the matchset.

6.4. Landmarking

Landmarks are reference spots used by PDQUEST to align and position matchset members for matching. They are used to compensate for slight differences and distortions in the member gels.

When choosing landmark spots in your member images, you should look for spots that are:

- Present in all members of the matchset.
- Well-resolved.
- Near a corner of the image.
- Isolated—i.e., very few spots in its immediate vicinity.
- Located in or near regions of high spot concentration.

Entering Landmarks

To begin landmarking spots, select Landmark from the Match menu and click on a spot on the Standard.

Note: Always initiate landmarking in the Standard.

The cursor will jump to the same area in the next member.

If the landmarking routine can identify the same spot in that member, it will appear marked.

If the routine is unsure about placing the landmark in that member, it will appear unmarked, and the area in which the routine expected to find the spot will be circled. Click on the appropriate spot manually.

Continue for all the members in the matchset.

To highlight all the landmarks in a matchset, select Mark Landmarks (F8) from the Match menu.

Unlandmarking

If you accidentally landmark the wrong spot, select Unlandmark from the Match menu and click on the incorrect landmark.

Note: Unlandmarking a member image will only remove the landmark from that member. Removing a landmark from the Standard unlandmarks that landmark from all the members of the matchset.

Tips for Landmarking

If a spot is missing in one of the gels, skip that gel and continue landmarking the other gels.

We suggest that you landmark approximately 5 spots per section of the gel. As a rule of thumb, landmark approximately 10% of the spots on your gel (depending upon the number of spots).

6.5. Auto-Matching

Now you are ready to match the members of your matchset to the Standard.

First, display each member of your matchset, including the Standard. Make sure that you entered sufficient landmarks (≥ 3) in the gels you want to match to the Standard.

Click on the function Match All (in the Match menu) to match all the members in one step. On-screen messages will indicate the progress of the matching.

Spots that have been matched in all the images will be marked on the Standard with a letter.

6.6. Checking and Editing Matches

The accuracy of the PDQUEST matching algorithm depends upon several factors, not the least of which is the number of landmarks you have entered. When the first round of matching is complete, we suggest that you review the results and make any corrections necessary to the matches called by PDQUEST.

The general procedure involves making as few corrections as possible, adding new landmarks where necessary, and rematching the entire gel to the Standard.

The commands for checking and editing matches are located on the Match > Edit Matches submenu.

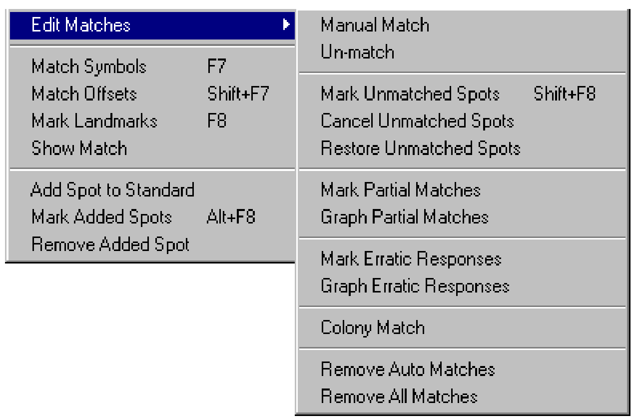

Fig.6-5. Edit Matches submenu.

Configure the Screen for Editing Matches

To edit matches, configure two vertical subwindows (using the Window > Configure Subwindows command). Display the Standard image in the left window and the member image you want to edit in the right window. When you are done editing one member image, replace it with the next member.

6.6.a. Identifying Unmatched Spots

PDQUEST has a number of tools for displaying and identifying your unmatched spots.

Match Symbols

To label all the spots that were not matched in the automatic matching step, select Match Symbols (F7) from the Match menu.

- Spots matched in all the images will be labeled with letters in both the Standard and member images.
- Spots that are in the Standard but not in all the member images will be circled in red in the Standard.
- Spots that are in a member image but not in the Standard will be circled in red in the member image.

Show Match

To look at individual matches, select Show Match (F9) and click on a spot in the Standard. The matched spots in the member images will be highlighted. If a spot is not matched in the member image, a circle will be drawn in its estimated location.

Match Offsets

Match Offsets displays the vector offsets from spot center locations in the member gels to the equivalent (matched) spot center locations in the Standard. Select Match Offsets (SHIFT+F7) from the Match menu to display the offset lines.

Look for the following conditions that indicate possible matching problems:

- Offset lines that cross each other (this is a key problem).
- Grossly nonparallel offset lines in local regions.
- Offset lines of unequal lengths.
- Areas of the gel with few offsets; this indicates unmatched regions.

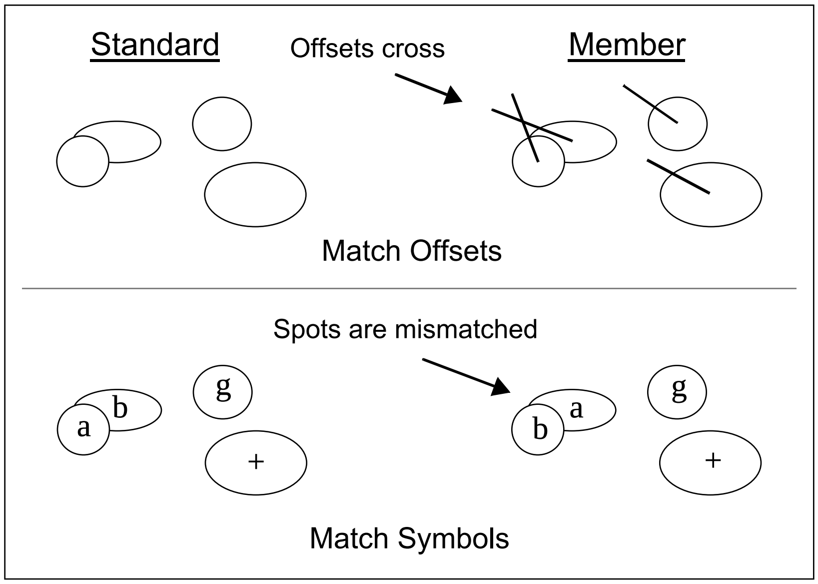

Fig.6-6. Example of incorrect matches.

Partial Matches

Partial matches are spots in the Standard that are not matched in one or more member images. This may indicate spots that either are not present in the gel or gels in question, in which case there is nothing to match, or spots that were missed during matching for one reason or another (e.g., the spot may have been cancelled in the gel).

Note: These commands only work on the Standard. In addition, Graph Partial Matches will only work in single-window mode.

Mark Partial Matches (Match > Edit Matches) highlights spots that are partially matched in the Standard.

Graph Partial Matches displays histograms connected to each partially matched spot in the Standard. These histograms show whether a spot is matched or unmatched in each member image. (For more information on graphs, see section 7.4.)

Each bar in a histogram represents the quantitation of the spot in each member of the matchset. The bars are ordered from left to the right based on the order in which the members were loaded. If a spot is not matched in a member, there will be no bar for that member in the histogram.

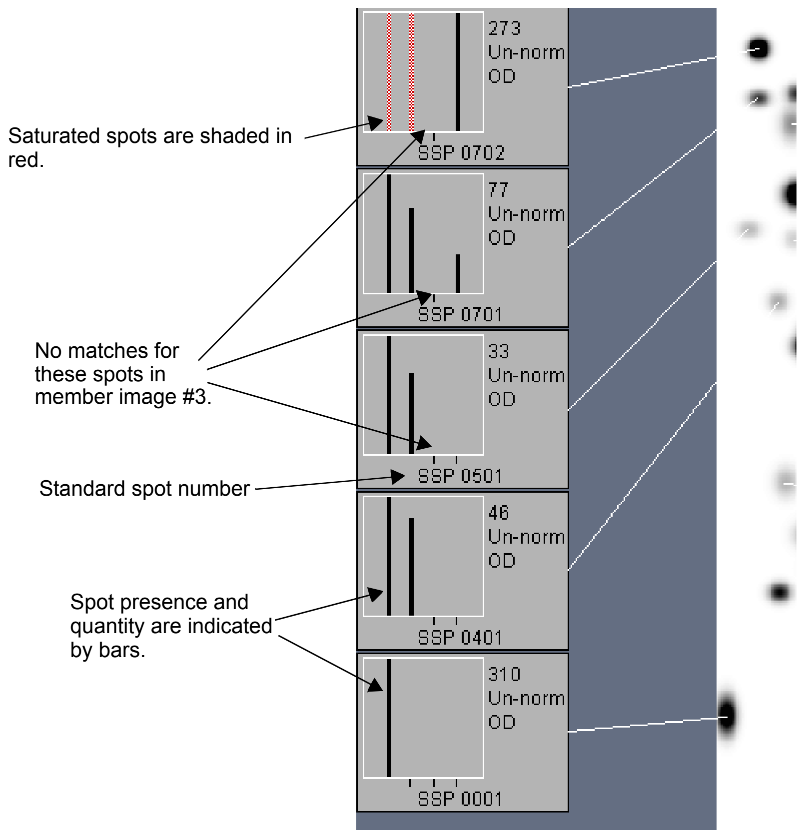

Fig.6-7. Examples of histograms of partially matched spots.

Histograms can only be displayed for one region of an image at a time. To shift regions, select Reports > Graph > Graph Advance (ALT++) or Graph Retreat (ALT+-). Graphs will be displayed for spots in the next region of the image.

The histogram also displays the intensity and Standard Spot (SSP) number for each spot. (For more information on SSP numbers, see section 8.1.) Decide which spots need to be checked and make a list of SSP numbers and gel numbers in which to reexamine the matching.

Be sure to check all the spots in one gel before moving on to the next.

To correct partial matches on a member gel, first configure two vertical windows, then display the Standard in the left and the member image in the right.

Use the Database > Spot Information > Find Spot (SHIFT+F9) function on the Standard to locate a spot whose SSP number is included on the list of spots needing to be re-examined. Then use the Zoom Box function followed by Imitate Zoom to get an enlarged view of the area around the spot in both images.

Check Erratic Spots

Mark Erratic Responses highlights all matched spots in the Standard whose quantity in the member images varies by a significant factor. Select Mark Erratic Responses from the Match > Edit Matches submenu, and enter a fold change factor in the pop-up box. (For example, a change factor of 3 will highlight spots whose quantity varies by a factor of three or more among member images.)

Graph Erratic Responses (in the Match > Edit Matches submenu) displays histograms of those matched spots whose quantity in the member images varies by a significant factor. Once again, enter a fold change factor in the pop-up box and click on Apply.

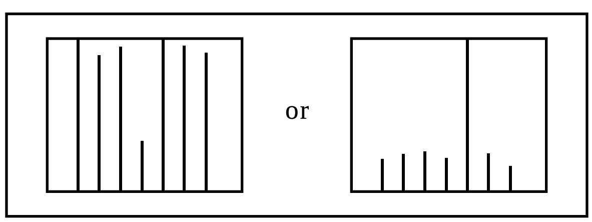

Fig.6-8. Example histograms of spots showing erratic responses.

6.6.b. Correcting Mismatches

Additional Landmarking

Landmarking spots that have been mismatched correctly matches the spots in question and also improves matching in the region when you rematch the gel to the Standard. Select Landmark from the Match menu, click on the spot in the Standard, then click on the spot in the member to landmark it.

Note: Landmarking will improve the modeling of the Standard, facilitating additional matching.

Enter additional landmarks in regions that have few matched spots.

Manual Matching

If a spot is present in all the member images but is not matched in the Standard, select Manual Match from the Match > Edit Matches submenu, click on the spot with the red circle in the Standard, then click on the same spot in all the member images. The circled spot in the Standard will changed to matched.

To unmatch a manually matched spot, select Un-match from the Match > Edit Matches submenu and click on the manually matched spot.

Note: Manual matching will <u>not</u> change the modeling of the Standard, and therefore will not facilitate additional matching.

Adding Spots to Member Images

It is possible that a spot is in a Gel Image but was not detected and therefore was not included in the matchset member. If this is the case, use Make Crosshair to add it to the member (see preceding chapter). After that, you will want to match it to the Standard using Manual Match.

If you cannot match a spot because it is not in the Standard, then you may need to add the spot to the Standard (see below).

Rematching Your Gel

After you have added several new landmarks, rematch the entire gel with the Match All command.

When matching is complete, Save your matchset.

6.6.c. Adding Spots to the Standard

In this step you look for spots in each member that are not included in the Standard. These may be spots induced by the experimental protocol for a particular gel.

It is important to locate these spots and add them to the Standard. Only then will you be able to match them to equivalent spots in other members.

Once again, configure your matchset into two vertical windows. Assign the Standard to the left window and the desired member to the right window.

Select Match Symbols (F7) and look for spots with red circles in the member image. These spots are unmatched. You can also use the function Unmatched Spots (SHIFT+F8) to find these spots.

There are usually four situations that result in unmatched spots; only the last requires that you add spots to the Standard.

1. A spot near the unmatched spot may be mismatched to the Standard.

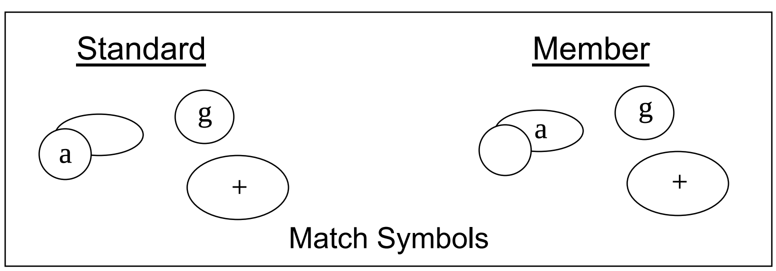

Fig.6-9. Mismatched spots.

2. The spot may be cancelled in the Standard. If so, restore the spot and match.

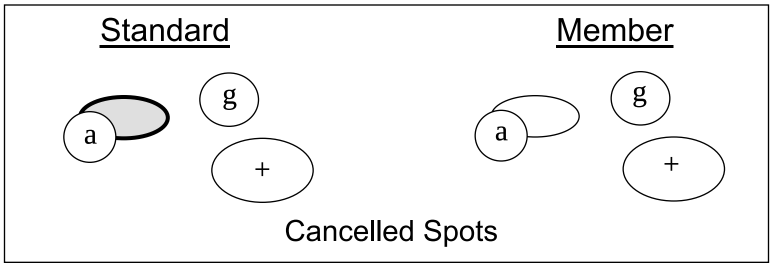

Fig.6-10. Restoring a spot that was cancelled in the Standard.

3. If the spot is in the Standard, but not matched in the member, match the spot.

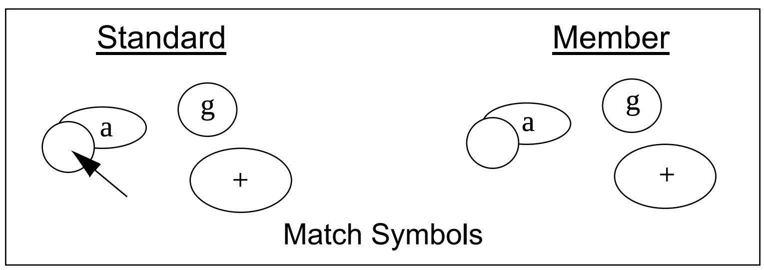

Fig.6-11. Matching unmatched spots.

4. The spot is not in the gel from which the Standard was made. This spot may be added to the Standard as described below.

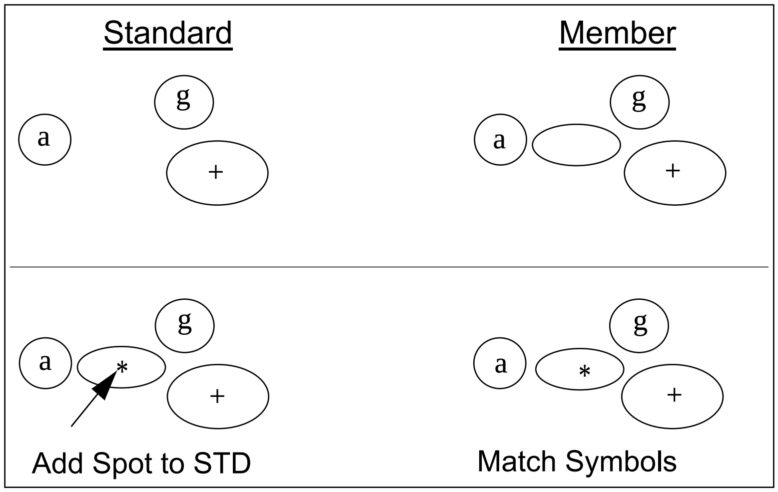

Fig.6-12. Adding a spot to the Standard.

Select Add Spot to Standard from the Match menu. Click on the center of the unmatched spot in the member. The cursor jumps into the Standard window to the position the spot is expected to be, based on the surrounding matches.

When you click in the Standard, the spot will appear and a match is automatically made to the spot in the member.

Repeat this procedure for the entire member image, then for all the members of the matchset.

To remove a spot that you have added, select Remove Added Spot from the Match menu and click on the added spot in the Standard.

Mark Added Spots highlights the spots that you've added to the Standard. They are marked by blue squares.

After adding a spot from one member to the Standard you may find that an unmatched spot in another member can be matched to the added spot. If so, make the appropriate matches.

Unless your system induces and/or represses many proteins, you may need to add no more than a dozen or so spots from all members to the Standard.

7. MrpI Values/Spot Quantitation

PDQUEST has various tools for calculating and displaying the molecular weights / isoelectric points and quantities of the spots in your matchsets.

Known molecular weights and isoelectric points can be entered for proteins using the MrpI Standards. Using the values that you provide, PDQUEST calculates the molecular weight (Mr) and isoelectric point (pI) values for all the other proteins in your Standard.

Since most experiments are carried out in duplicate, data analysis can be performed on each gel individually or on sets of duplicate gels grouped together. Functions in the Database > Replicate Groups submenu allow you to group sets of duplicate gels and determine the average quantitation of their protein spots.

The Normalization function will normalize spot quantitation among gels in a matchset.

Graphs (histograms) of spot quantitation and MrpI data can be displayed using the functions in the Reports > Graph submenu. Examining the histograms can give you a general sense of the trends in protein expression.

7.1. MrpI Standards

You can enter the molecular weight and isoelectric point values for your known protein spots. With this data, PDQUEST can calculate the Mr and pI values for all the spots in your Standard.

The functions needed to enter and display Mr and pI data are found in the Database > MrpI Standards submenu.

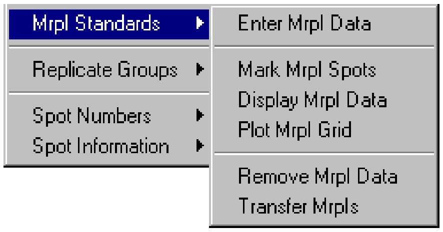

Fig.7-1. MrpI Standards submenu.

Note: PDQUEST needs to know the orientation of the gel in order to calculate pI values. To enter the correct orientation, go to the Preferences dialog box (Edit > Preferences) and click on either Acid-Base or Base-Acid next to the pH Gradient prompt.

Entering MrpI Values for Known Spots

To determine Mr and pI values for all the spots in your Standard, you first need to enter Mr and pI data for a few known spots.

Select Enter MrpI Data from the Database > MrpI Standards submenu, then click on a spot in the Standard whose Mr and/or pI values are known. A dialog box will ask you to enter the Mr and pI data for that spot.

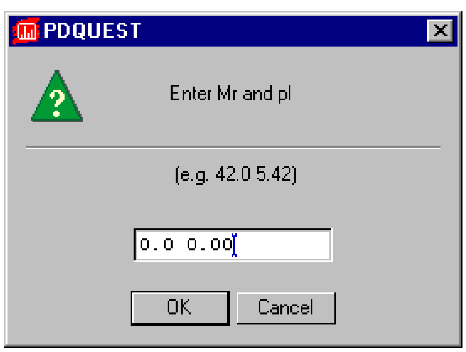

Fig.7-2. Pop-up box in which Mr and pI data are entered.

Type the Mr values first, then the pI data, and be sure to separate them with a space. Click OK when you are done.

If you make a mistake entering the data, you can select Enter MrpI Data again and click on the spot. The values you entered will appear in the dialog box and you can make the necessary corrections.

If you want to remove the Mr and pI data for a spot, select Remove MrpI Data and click on the spot.

Displaying Mr and pI Data

To see the proteins for which you have entered Mr and pI data, you can use either Mark MrpI Spots or Display MrpI Data. The first will highlight the spots for which you entered data. The second will display Mr and pI values next to the spots.

For a graphical display of the overall trend of Mr and pI values across the entire gel, first make sure the Standard subwindow is active, then select Plot MrpI Grid from the Database > MrpI Standards submenu.

A grid of Mr and pI axes will appear over the Standard. You can use the grid to verify that the values that you entered make sense; you can also use it to estimate the Mr and pI values of proteins of interest.

To see the exact Mr and pI values of your favorite spot, select Spot Parameters from the Database > Spot Information submenu and click on the spot.

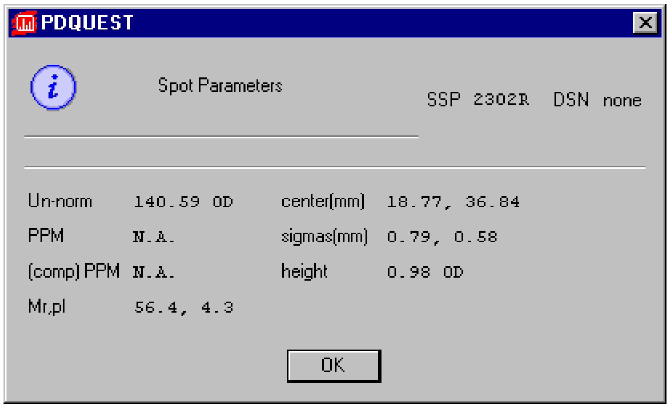

Fig.7-3. Spot Parameters pop-up box.

In the pop-up box, the Mr and pI values will be displayed. These values are calculated based upon the known Mr and pI data that you entered.

Note: If the pI of the spot of interest falls outside of the highest and lowest known pI values, a pI value for that spot will not be calculated. Empirical results have shown that extrapolated pI values are unreliable, due to gel distortions and other factors.

7.2. Replicate Groups

Because we recommend that you perform your experiments more than once, PDQUEST offers you the option of looking at each gel individually or looking at the average of the data of duplicate gels.

A replicate group is the name given to a set of gels that are duplicates of each other. For example, three separate gels of a control cell lysate theoretically should produce the same quantitation for all spots on a 2-D gel. In practice, however, slight variations in the quantitation will probably be seen. Instead of choosing the one that you think is best, you can take the average and use those values for spot quantitation.

Creating a Replicate Group

Note: All your duplicate gels must be in the same matchset before you can create a replicate group.

To make a replicate group, open the Replicate Groups submenu (Database > Replicate Groups).

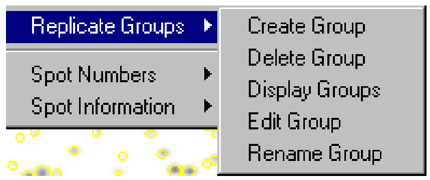

Fig.7-4. Replicate Groups submenu.

Select Create Group. In the dialog box, enter a name for the replicate group next to the Name prompt.

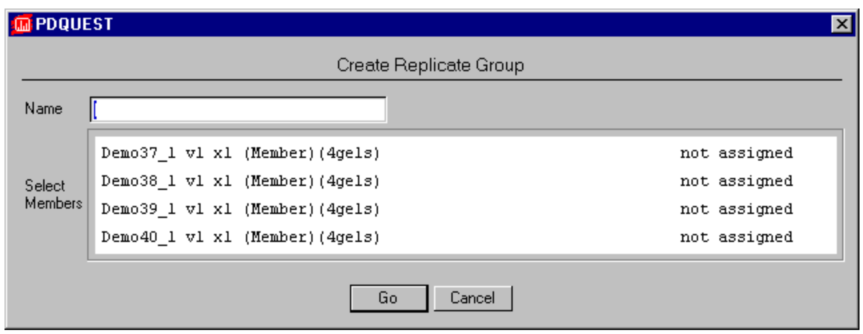

Fig.7-5. Enter a name for the replicate group in the dialog box.

Next you will be asked to select the gels that you want to include in your replicate group. Click on each gel that should be included.

Click on Go when you are finished. In the upper left message bar, you will see the total number of gels in this group.

Replicate groups can be altered using the commands Rename Group, Delete Group, and Edit Group.

- Rename Group allows you to enter a new name for the replicate group.
- Delete Group deletes the selected replicate group.
- Edit Group lets you add gels to or remove gels from a replicate group.

Quantitating Your Replicate Group

After you have created a replicate group, you can calculate the average quantity of the members of the group by selecting Quantitation Mode under the Edit menu (CTRL+F6) and clicking on the Replicate Group Quantitation button in the pop-up form.

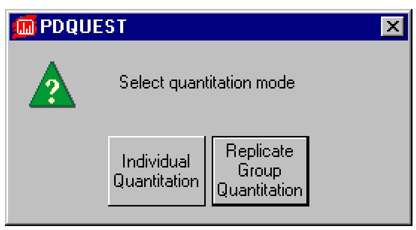

Fig.7-6. Quantitation Mode pop-up.

7.2.a. Displaying Replicate Group Quantitation

Replicate group quantitation can be displayed using the Quantity Table function under Database > Spot Information.

Click on the spot whose quantitation you want to see. Depending upon the quantitation mode you selected, one of two kinds of pop-up boxes will appear.

If you chose individual quantitation, the box will display the quantitation values found for that spot in each of the gels.

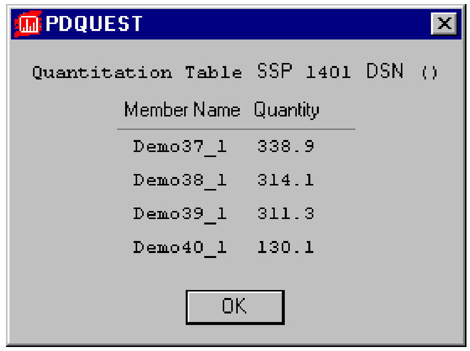

Fig.7-7. Quantity table displaying individual spot quantitation.

If you selected Replicate Group Quantitation, the Average value of that spot's quantitation will be displayed, as will the coefficient of variance (CV) and the number of gels in the replicate group (Count).

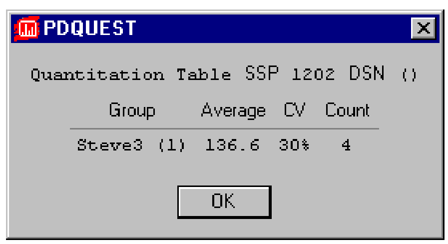

Fig.7-8. Quantity table displaying replicate group quantitation.

7.3. Normalization

Before you graph, export, or print protein spot data from a matchset, you will probably want to normalize the quantitation. Normalization is a process by which quantitative data from different gels are adjusted so that comparisons between samples can be made. (Normalization is not appropriate with higher level matchsets.)

Normalization affects histograms of spots and the quantitative data printed in reports.

With either the Standard or a member image selected, select Normalize from the Edit menu to open the Quantity Normalization dialog box.

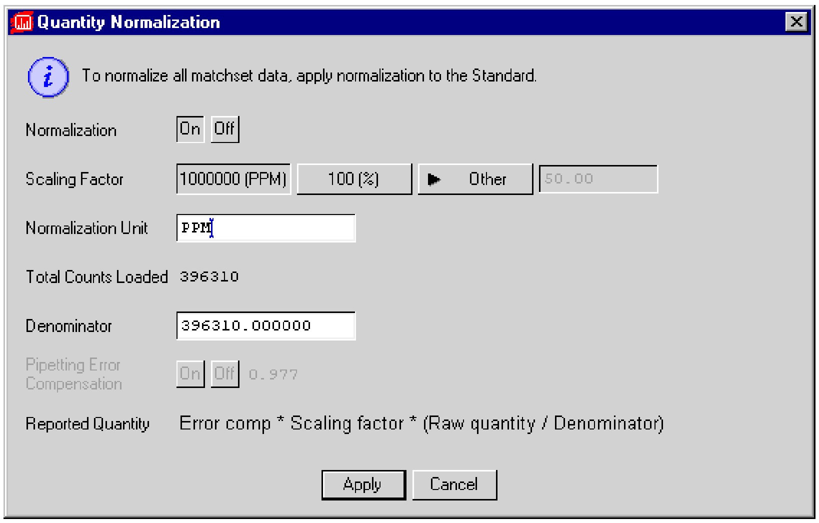

Fig.7-9. Quantity Normalization dialog box for member images.

At the top of the form, indicate that you want to apply normalization by clicking the On button. Choosing this option in the Standard turns normalization on for the entire matchset. Clicking Normalize on a member image turns on normalization for that member only.

At the bottom of the form, next to the Reported Quantity prompt, is the equation for calculating normalization. The Scaling Factor is a constant by which the quantity is multiplied to give a more meaningful value. A common scaling factor is 10^6, giving units of parts per million (PPM).

Next to the Scaling Factor prompt, specify the scaling factor that you would like to use by clicking on the 1000000 (PPM) or 100 (%) buttons. Alternatively, click on the Other button and enter your own scaling factor.

If you choose 1000000 or 100, the Normalization Unit will be automatically set to PPM or %, respectively. If you entered your own scaling factor, enter the normalization units in the text box next to the prompt.

The Denominator is the number by which a spot's raw quantity is divided. Normalization is often calculated by taking the counts in a spot and dividing by the number of counts loaded onto the gel. In the text box next to the Denominator prompt, the total number of counts loaded is used as the default.

You can change the denominator by typing in a new value.

Note: The Denominator prompt will not be active if you clicked Normalize in the matchset Standard. Since the Standard is a composite image containing data from all the members of the matchset, there are no counts associated with it.

Pipetting Error Compensation can correct for minor systematic errors inherent in scintillation counting, pipetting, and exposing films. The normalization process is based on the assumption that these minor errors effect all proteins equally. It also assumes that few protein spots change within the experiment or that when changes do occur, they do not alter the random distribution from one sample to the next.

Pipetting Error Compensation can be applied to all the members of a matchset by using the Normalize function in the Standard window.

When you are satisfied with the information entered in the Quantity Normalization dialog box, click on the Apply button at the bottom (Apply To All if you are in the Standard).

If you decide not to normalize spot quantitation, click on Cancel.

7.4. Graphs

Histogram graphs can be used to quickly compare spot quantitation in each member of a matchset. A brief look at a histogram can give you a sense of the general trends in spot quantitation.

Histograms are also useful as tools for detecting problems such as spots that are not matched and spots whose quantitation is suspiciously off the average. Reviewing the histograms of spots in a User Set is a quick and easy way to confirm that the members of the User Set meet the criteria that was specified when the set was created.

Histograms of Matched Spots

Histograms are graphs consisting of bars whose lengths represent the data being plotted. Each bar represents the quantitation of the spot in each member of the matchset.

The bars represent the gels from the left to the right in the order in which they were loaded. To check the order, use Assign (ALT+A). If you would like to change the order, use the function Reorder Members (Match > Matchset > Reorder Members).

As long as the spot is matched to the Standard, you can display a histogram for it.

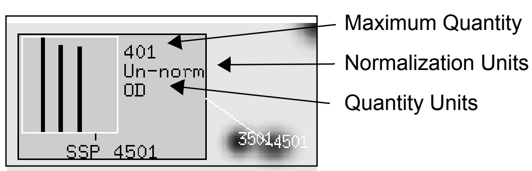

Fig.7-10. Sample histogram with individual quantitation.

The Standard Spot (SSP) number is displayed beneath the histogram.

The number at the upper right of the histogram is the quantitation of the maximum bar in the graph. The other bars are drawn proportional to the highest bar.

Beneath the maximum quantitation, the normalization units are displayed. Normalization is a process by which quantitative data from different gels are adjusted so that you can compare different samples to one another (for more information on normalization, see section 7.3.).

If a spot is saturated, the bar representing it on the histogram will be shaded red instead of black.

If replicate groups are defined for the matchset, you can look at the quantitation of individual spots or the average data of duplicate gels. You can specify which way you want to review spot quantitation by selecting Quantitation Mode from the Edit menu.

If you specify Replicate Group Quantitation, the bars on the histograms will display not only the maximum spot quantitation for each replicate group, but also the average spot quantitation and the standard deviation. Each bar represents one replicate group. The order of the bars corresponds to the order in which the replicate groups were made.

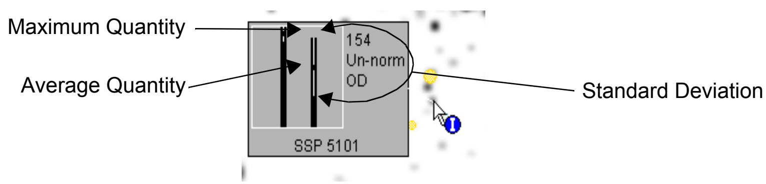

Fig.7-11. Sample histogram with Replicate Group Quantitation.

7.4.a. Different Histogram Displays

Once a User Set has been selected, use the Graph Set function to display the histograms of the spots in the set (Database > Analysis Sets > Graph Set).

All other functions for displaying histograms on-screen are found in the Graph submenu of the Reports menu. The majority of these functions will only work on the Standard image.

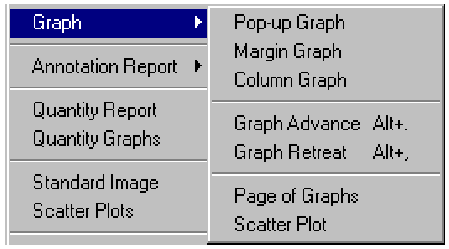

Fig.7-12. Reports > Graph submenu.

Pop-up Graph

Pop-up Graph shows a histogram next to a selected spot. Select Pop-up Graph from the menu, then click on a spot.

With Pop-up Graph, only one histogram can be displayed in a window at a time. If you select another spot, the first histogram will disappear and a new one will be displayed.

Margin Graph

Margin Graph will display histograms for several spots along the left and the right margins of the screen. Select Margin Graph, then click on several spots. Their histograms will be displayed in the left and right margins.

Depending on the size of your window, you will be able to display up to 10 margin graphs at once. If you keep clicking on spots after you've displayed 10 graphs, you will begin replacing graphs with your more recent selections.

Column Graphs

Column Graph will fill the right and left margins with histograms of spots in the Standard. Select Column Graph from the submenu. The spots whose graphs are displayed are all located in a narrow column of the image, centered around the location of the cursor.

To change the column of spots, use the Graph Advance / Graph Retreat tools.

Graph Advance/Graph Retreat

If you are displaying Column Graphs or graphing spots using the Graph Partial Matches or Graph User Set commands, you will need to shift the display to graph spots in different regions of the image.

Graph Advance will shift the column of spots being graphed to the right; Graph Retreat will shift the column of spots being graphed to the left.

Page of Graphs

Page of Graphs displays the histograms of all the spots in your Standard as they would appear in a printed report. Select Page of Graphs from the Reports > Graph submenu to open the window.

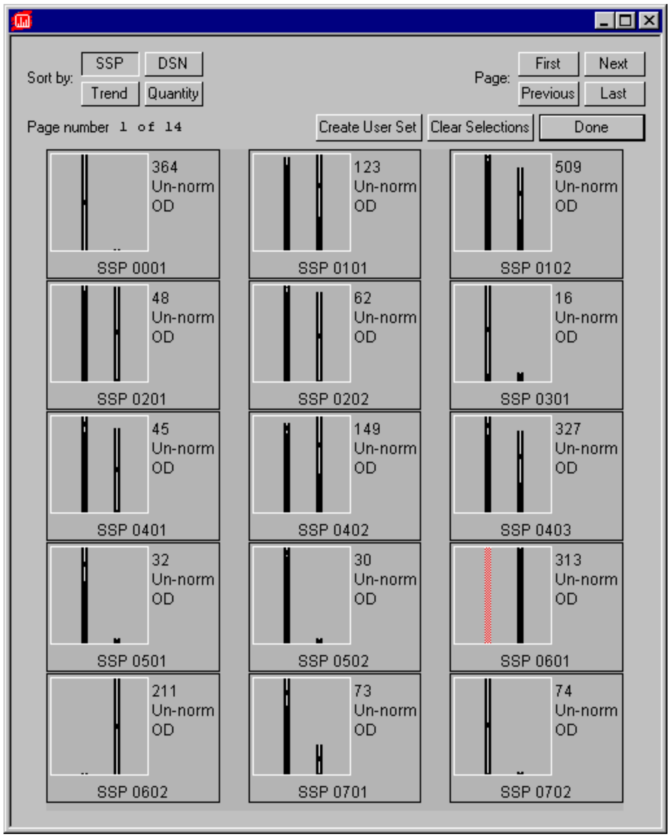

Fig.7-13. Page of Graphs window.

You can peruse your page of graphs by clicking on the Page buttons in the top right corner of the dialog. You can rearrange your page of graphs by click on the "Sort by" buttons in the top left corner of the dialog.

You can create a User Set containing some or all of the spots whose graphs are displayed. Point and click on each of the graphs that you want to include in the User Set. As you click on each, the border around the graph will become highlighted, indicating that it is selected.

If you make a mistake, click again on a graph to deselect it or click on the Clear Selections button to deselect all the graphs.

After selecting all the graphs of interest, click on the Create User Set button. A dialog box will be displayed in which you are asked to enter a name and a description of the User Set. Choose Create if you want to proceed.

See section 9.1. for more information on User Sets.

To close the Page of Graphs window, click on Done.

Mrpl Values/Spot Quantitation

8. Annotations and Numbering

PDQUEST has a number of features for numbering and annotating your spots.

Every matched spot is assigned a Standard Spot (SSP) number. SSP numbers are increasingly important when you begin comparing spots between different matchsets using higher level matchsets (see the next chapter).

Besides the standard SSP numbering system, an alternative database spot number (DSN) system is also available. The advantage of the DSN system is that it indicates the general location of each spot, based on the MrpI grid.

Notes, comments, and other pertinent information can be kept in a file associated with each protein spot. The functions needed to enter and edit this data are found in the Annotations menu.

8.1. Standard Spot Numbers

Standard spot (SSP) numbers are unique numbers assigned to each spot in the matchset Standard. In general, spots in the lower left region of the image have low SSP numbers, while spots in the upper right region have high SSP numbers.

8.1.a. Assigning SSP Numbers

SSP numbers are automatically assigned when you create a matchset Standard. To create SSP numbers, PDQUEST first divides the Standard into rows and columns, placed so that there are approximately the same number of spots in each section and that no section has over 100 spots. The first two digits of a spot's SSP number correspond to its section's X and Y coordinates. Spots within a section are numbered sequentially.

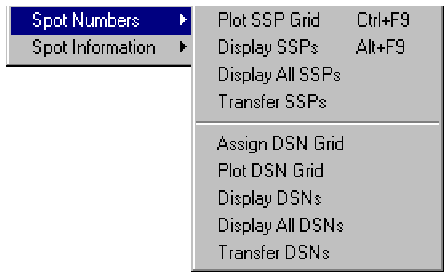

Fig.8-1. Spot Numbers submenu.

8.1.b. Displaying SSP Numbers

Select Plot SSP Grid from the Database > Spot Numbers submenu to display the SSP grid on the Standard. Numbered vertical and horizontal grid lines will appear on the Standard.

The grid is helpful for locating spots by SSP number. However, it is an approximation and should only be used as a guide. If enough spots are added to a single grid block to use up the 99 available numbers, numbers will be assigned from other blocks. SSP numbers are always four digits.

Display SSPs displays SSP numbers next to every spot in the active window. You may need to zoom in on the image to be able to read them.

Display All SSPs displays SSP numbers next to every spot in every open window.

Transfer SSPs is used to copy the SSP numbers from one higher level matchset to another.

8.2. Database Spot Numbers

Database spot numbers (DSNs) are an alternative to the SSP numbers. The DSN system provides a more flexible way to number protein spots in a Standard. A coordinate grid can be established for each axis: Mr and pI. The grid coordinates are identified with letters (A–R, excluding I and O because they look like numbers). The letters are intended to represent specific Mr and pI ranges so that the numbering between different gels can be comparable. These ranges are as follows:

Mr values:	0-10=A	87-117=J
	11-14=B	118-158=K
	15-19=C	159-215=L
	20-25=D	216-293=M
	26-34=E	294-398=N
	35-46=F	399-541=P
	47-63=G	542-736=Q
	64-86=H	737-1000=R
pI values:	0-2=A	7.0-7.5=J
	2-3=B	7.5-8.0=K
	3-4=C	8.0-8.5=L
	4-5=D	8.5-9.0=M
	5.0-5.5=E	9-10=N
	5.5-6.0=F	10-11=P
	6.0-6.5=G	11-12=Q
	6.5-7.0=H	12-14=R

DSNs have the format ABnn. “A” is the Mr grid letter, “B” is the pI grid letter, and “nn” is the number of the spot within that section of the grid.

8.2.a. Assigning DSNs

If you have already entered Mr and pI values for known spots, you can create a DSN grid using this data by selecting Assign DSN Grid from the Database > Spot Numbers submenu. This automatically assigns a DSN number to each spot in the Standard.

To display the grid (this does not occur automatically when the grid is created), click in the Standard window and select Plot DSN Grid. A grid of vertical and horizontal axes will appear, with the letter assignments of each axis displayed in the window margins.

To see the spot assignments, select Display DSNs. To read the numbers, you will probably need to zoom in on a smaller region of image.

Transfer DSNs is used to copy the DSN numbers from one higher level matchset to another (see Chapter 9).

8.3. Annotations

You can add notes, comments, and other pertinent information to a file associated with each protein spot using the functions in the Annotations submenu of the Database menu.

As the size and complexity of your database increase, annotations can be an essential tool for keeping track of experimental data.

8.3.a. Entering Annotation Data

Before you can begin creating annotations for individual protein spots, you first need to create the category of information to be entered.

Some examples of annotation categories are: Protein Name, Amino Acid Composition, Binding Properties, Translational Regulation, etc.

Creating a Category

To create a category, select Create Category from the Database > Annotations submenu. A dialog box will ask you to enter a name and a description for the category.

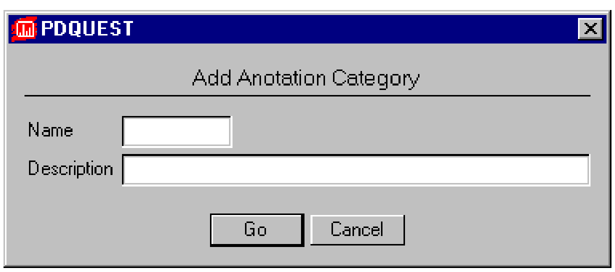

Fig.8-2. Enter a name for the new annotation category.

A description of the category is optional. Click on the Go button when you are done.

Once a category has been created, it is automatically active. You can enter spot annotations for the category without having to select it first.

Selecting an Existing Category

To enter information in an existing category, first open the category by choosing Select Category from the Database > Annotations submenu and selecting the category name.

If annotations have been entered for spots in this category, the spots will be highlighted when the category is selected.

Entering Annotations

Once a category is active, you can enter annotations for individual spots. Select Edit Annotation and click on a spot to open the annotations form.

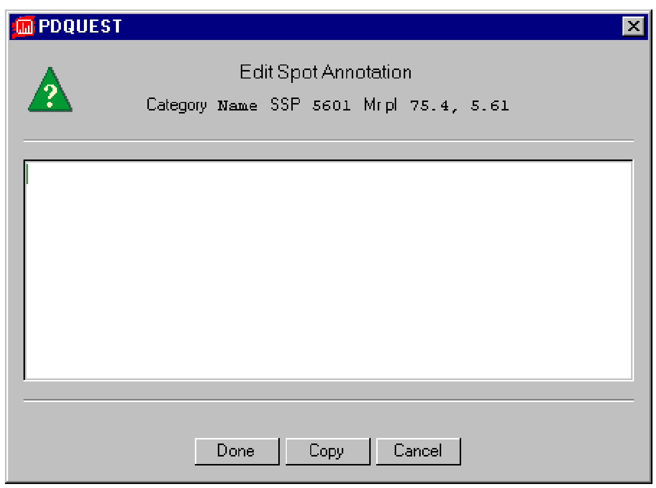

Fig.8-3. Edit Annotation dialog box.

When you have finished entering the information for that spot, click on the Done button at the bottom of the box. The box will disappear and the spot will become highlighted.

Repeat these steps for all the spots you want to annotate in the selected category.

Deleting/Renaming a Category

To delete a category, select Delete Category from the Database > Annotations submenu and select the category to delete from the list.

To rename a category, select Rename Category and select the category name from the list. Enter a new category name in the pop-up box.

Creating Standard Categories

Create Standard Categories is a function that adds 38 predefined categories to your annotation database. These categories are then available for you to enter and edit annotation information about spots in your matchset.

Select Create Standard Categories from the Database > Annotations submenu. The categories will be added to the matchset. To view the categories, use the Select Category command.

Saving Annotations and Categories

Annotations and categories for a matchset are saved when you save the matchset. No separate operation is necessary.

8.3.b. Displaying Annotation Data

Several functions are available for displaying annotation data.

Select Mark Category to highlight all the spots that have information entered in the selected category.

Select Mark All Categories to highlight all the spots in the Standard that have data entered in any category, regardless of whether the category is selected.

Show Spot Annotation will display the annotation information in the selected category for an individual spot that you click on. Select Show Spot Annotation, then click on a spot.

Show All Annotations will display the annotation information in a selected category for every spot in the Standard.

Peruse Annotation displays a list of the annotation categories that have information for a particular spot. Select Peruse Annotation, then click on a spot of interest. A list of annotation categories for that spot will be displayed. Select a category from the list to display the annotation information in a pop-up box next to the spot.

To conceal an annotation, select Hide Annotation and click on the spot displaying the annotation.

9. Databasing Spots

As you accumulate more information about your proteins, it becomes increasingly important to store the data in an organized and accessible format. PDQUEST provides several functions to build and query such protein spot databases.

Use the functions in the Database > Analysis Sets menu to create user sets containing statistically and biologically related spots. The spots in a user set meet quantitative or qualitative criteria that you specify. There are six different types of user sets that you can create: Quantitative, Qualitative, Arbitrary, Boolean, Statistic, and Annotation.

An important part of databasing is the ability to compare data between experiments. PDQUEST allows you to create a database from different experiments through the use of higher level matchsets (See section 9.2.).

9.1. Analysis Sets

The analysis sets functions in PDQUEST are designed to help you create and study sets of proteins that are statistically and scientifically meaningful.

It is virtually impossible to analyze the gel or experiment as a whole, so PDQUEST provides a way to break down all the information on the gel into smaller, more manageable pieces. This is done by creating User Sets (i.e., groups of spots that satisfy criteria that you specify).

There a six different kinds of User Sets that you can create:

Quantitative User Sets: Include spots whose quantitation has increased or decreased “x” fold (where x is a user-determined value) or whose quantitation has changed beyond or within the fold change bounds that you specify.

Qualitative User Sets: Include spots whose quantitation is “x” times above the minimum level of detection in one gel but below the minimum level of detection in

	another gel—i.e., spots that have been turned “on” or “off.”
Arbitrary User Sets:	Are composed of spots that you choose. They begin as empty sets and you select the proteins that are to be members of them.
Boolean User Sets:	Contain a group of spots formed by the combination of two other sets. For example, User Set C could be made up of the spots that are present in both User Set A and User Set B.
Statistical User Sets:	Include spots that are found to be significant according to the statistical test that you indicate.
Annotation User Sets:	Are made up of spots whose annotations include a particular key word that you specify.

9.1.a. Creating a User Set

With the matchset open, select Create Set from the Database > Analysis Sets submenu.

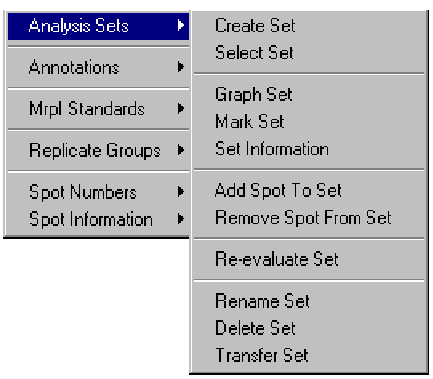

Fig.9-1. Analysis Sets submenu.

The User Analysis Set dialog box will open.

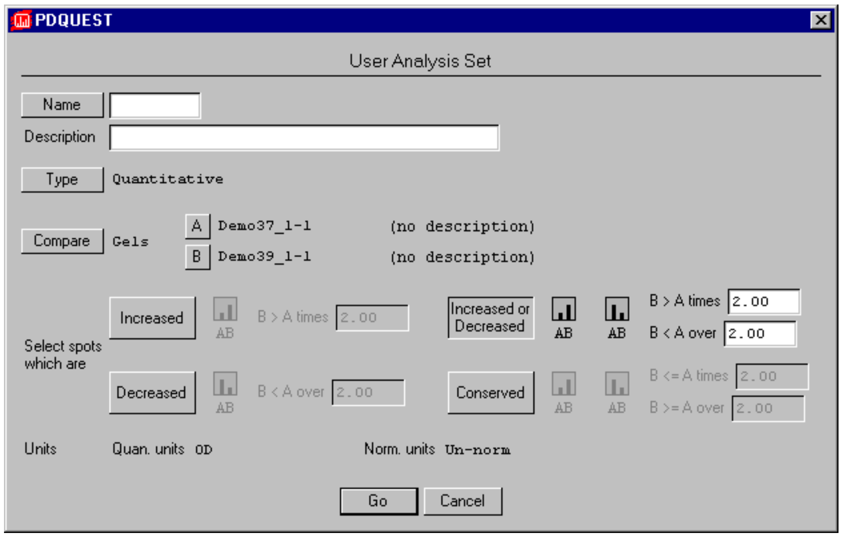

Fig.9-2. Example of a User Set dialog.

If you want to update a User Set that already exists, click on the Name button at the top of the User Analysis Set dialog box. A list of existing sets will be displayed. Select the name of the set you want to update by clicking on it. The dialog box will be reconfigured and the information about that User Set will be displayed.

Selecting the Type of User Set

The first step in creating the User Set is selecting the type you want to create. In the User Analysis Set dialog box, click on the Type button.

A list of User Set types will appear. Click on a type to select it. (See the following sections for descriptions of the different types of User Sets.)

The list will disappear and the User Analysis Set dialog box will reconfigure to display the test you chose. You will see the name of your selection displayed next to the Type button.

Selecting the Operands to Compare

All User Sets (except Arbitrary and Annotation Sets) involve comparisons: How does a protein or group of proteins (in a gel or replicate group) behave under one set of conditions versus another? To answer this question, you need to specify which sample should be compared with which other sample.

First, indicate whether you want to look at the quantitation of individual gels or the average quantitation of groups of gels. Click on the Compare button and select either Gels or Groups from the list displayed. Your choice will appear to the right of the Compare button.

Next, click on the A button and choose a sample from the list displayed. (If you selected Gels to compare, a list of all the gels will be displayed. If you selected Groups, a list of replicate groups will be displayed.)

After clicking on your choice, you will see it displayed next to the A button.

Repeat this procedure with the B button to enter the second sample.

Displaying the Spots that Are Included in a Set

After you have entered the settings for a User Set, click on the Go button to create the set. The dialog box will disappear. On the Standard, the spots that are included in the User Set will be highlighted.

Once a User Set is active, use the function Mark Set (in the Analysis Sets submenu) to highlight all the spots contained in that User Set.

Another way to display the proteins in a User Set is with the Graph Set function (in the Database > Analysis Sets submenu). This function will display histograms of the spot quantitation along the left and right margins of the screen. (For more information on histograms, see section 7.4.)

To clear the User Set display, click on Hide Overlays on the main toolbar.

Opening/Deleting/Editing an Existing User Set

To open an existing set, choose Select Set from the Database > Analysis Sets submenu and select the User Set from the list.

You can also open an existing User Set from within the User Set dialog box. Click on the Name button at the top of the dialog and select the name of the set you want to load from the list. The settings and information for that set will be displayed in the dialog box.

To delete a saved User Set, select Delete Set from the Database > Analysis Sets submenu and select the set to delete from the list of names.

To rename a User Set, select Rename Set from the Database > Analysis Sets submenu and select the set to rename from the list. Enter a new name for the set in the pop-up box.

To recalculate the members of a User Set (e.g., if you have added or matched more spots), select Re-evaluate Set from the Database > Analysis Sets submenu. The set will be recalculated and the members will be displayed on the Standard.

9.1.b. Quantitative User Sets

Quantitative User Sets allow you to perform fold change analyses. To create a Quantitative User Set, select Database > Analysis Sets > Create Set. Quantitative User Set is the default configuration of the User Set dialog box.

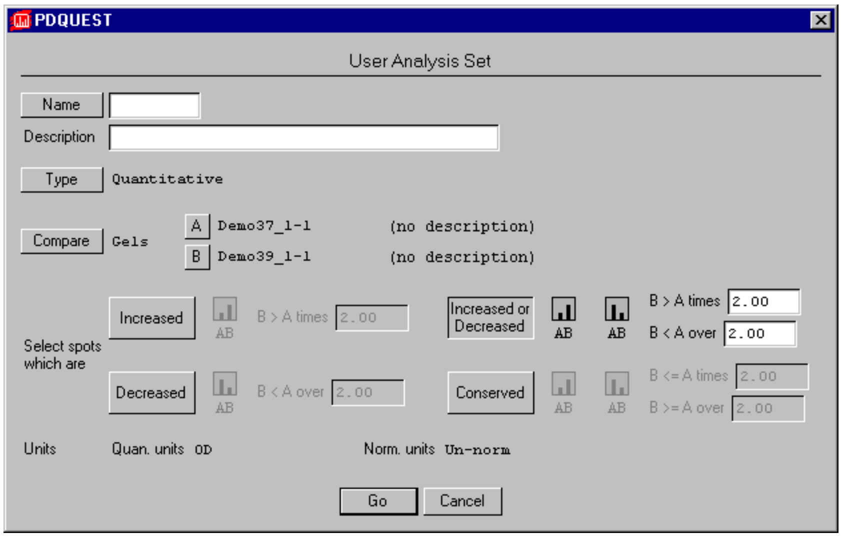

Fig.9-3. Quantitative User Set dialog.

After entering your selections for comparison (e.g., control gels versus experimental gels), you need to decide what kind of quantitative changes you want to examine. You can select Increased, Decreased, Increased or Decreased, or Conserved.

Next to each choice is a diagrammatic representation of the type of change the test will find and an explanation of the type of quantitative criteria a spot must meet to be included in the User Set.

Do you want to look for a protein whose quantitation went up under the experimental conditions as compared to the control? If so, click on the Increased button next to the “Select spots which are” prompt.

“A” represents all matched spots in sample A and “B” represents the corresponding matched spots in sample B.

Comparisons are made using spot quantity only. Positional comparisons, etc., are not possible.

The comparison is performed as follows: A threshold value is established by multiplying and/or dividing A by the factor you specify. B is then compared to the threshold.

For example:

1. If you ask to create a User Set by clicking on the Increased button and enter a factor of 2.0, the set will include spots whose quantity in B is at least twice that of the corresponding spot in A.

2. If you click on the Increased button and enter a factor of 0.5, the user set will include spots whose quantity in B is greater than half the quantity of the corresponding spot in A.

3. Clicking on the Decreased button and entering a factor of 2.0 will find spots whose quantity in B is less than half the quantity of the corresponding spot in A.

4. Creating a user set by clicking on Decreased and entering a factor of 3.0 finds all the spots whose quantity in B is less than one-third the quantity of the same spot in A.

5. If you are interested in finding the spots whose quantitation has significantly changed either way (either increased or decreased), select the Increased or Decreased option. The proteins found in this test have increased in quantitation above the upper bound or decreased below the lower bound. If you enter an upper bound of 2.0 and a lower bound of 3.0 you will find spots whose quantity in B is more than twice the quantity in A or less than one-third the quantity in A.

6. To find spots whose quantitation in B has not changed significantly from A, click on the Conserved button. Enter factors to define the upper and the lower bounds. If, for example, you enter an upper bound factor of 3.0 and a lower bound factor of 2.0, the user set will include spots whose quantity in B is between three times and one-half times the quantity of those spots in A.

After entering the fold-change factor(s), the corresponding diagram will automatically update to show you the quantitative relationship these criteria define.

Click on the Go button at the bottom of the User Analysis Set dialog box to create the set you have specified.

Be sure to enter a name for the User Set. If you forget, a pop-up box will remind you to do so.

9.1.c. Qualitative User Sets

Qualitative User Sets contain spots that were not detected in one gel but were detected in the other. This test would detect a protein that was expressed under experimental conditions but was not expressed in the control sample, or visa versa.

Note: We strongly recommended that all the spots in such a User Set be double-checked for accuracy. It is possible that a protein might actually be present in both gels but was missed during matching.

Creating a Qualitative User Set

To create a Qualitative User Set, select Database > Analysis Sets > Create Set, click on the Type button in the dialog box, and select Qualitative from the list displayed. The dialog box will change to a Qualitative User Set configuration.

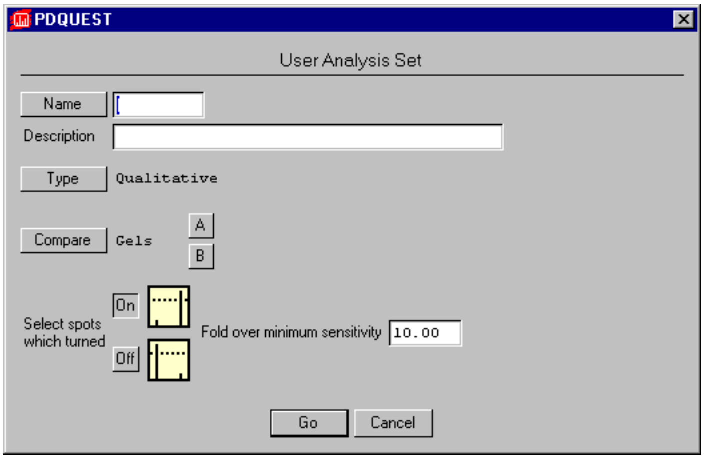

Fig.9-4. Qualitative User Set dialog.

As described above, choose your control and experimental gels.

To find proteins that were not present in the control sample (Gel A) but were expressed under experimental conditions (Gel B), click the On button next to the “Select spots which turned” prompt.

For spots present only in the controls (Gel A) and not in the experimental samples (Gel B), click on the Off button.

Finally, you need to tell PDQUEST what “turned on” or “turned off” means. How many times greater than the minimum detectable spot should the quantitation be for a spot to be included in this User Set? Should it be 5-fold greater than the minimum? If so, enter 5.00 next to the Fold over minimum sensitivity prompt.

Click on the Go button to find all the spots that fit the criteria that you specified. They will be highlighted on the Standard.

9.1.d. Arbitrary User Sets

Arbitrary User Sets are composed of any group of spots that you choose. To create an Arbitrary User Set, select Database > Analysis Sets > Create Set, click on the Type button in the dialog box, and select Arbitrary from the list displayed.

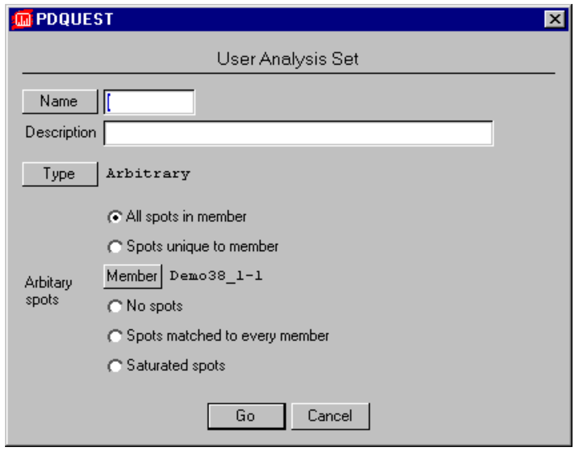

Fig.9-5. Arbitrary User Set dialog.

The first option ("All spots in member") will create a User Set containing all the spots in a selected member that are matched to the Standard. Click on the Member button to display a list of the gels from which to choose. Click on your choice and it will be displayed next to the Member button. The User Set will include all the spots in the member you selected that are matched to the Standard.

"Spots unique to member" will create a user set of those spots found only in the member that you specify and not in any of the other members of the matchset. Click on the Member button to display a list of gels from which to choose. Click on your selection and it will be displayed on the form next to the Member button.

"Spots matched to every member" will create a User Set of spots that are matched in all the members.

"Saturated spots" will create a user set of all spots that are saturated in any member.

The "No spots" option creates an empty User Set to which you can add any spots you choose. See the following section for detailed instructions.

Adding Spots to/Removing Spots from an Arbitrary User Set

To add spots to an empty User Set, you first select the Arbitrary User Set you want to add to, then you click on the spots you want to add to it.

Note: Spots may be added on a spot-by-spot basis only to Arbitrary User Sets.

Choose Select Set from the Database > Analysis Sets submenu to display a list of User Sets. Select the Arbitrary User Set you want to add spots to.

With the set selected, select Add Spot To Set from the Database > Analysis Sets submenu and click on a spot. The spot will be highlighted, indicating that it has been added to the User Set. The name of the User Set to which it has been added is displayed in the upper left corner of the window.

If you want to remove a spot from the User Set, select the set, then select Remove Spot From Set from the Database > Analysis Sets submenu and click on an added spot. The spot will be unhighlighted, indicating that it has been removed.

9.1.e. Boolean User Sets

Boolean User Sets are formed by combining two previously defined User Sets using Boolean operators. You can create Boolean Sets that include:

1. The intersection of two sets of spots (proteins present in both User Set A *and* User Set B).
2. The union of two User Sets (spots in set A *or* in set B).
3. Spots that are unique to set A plus spots that are unique to set B.
4. Spots that are found only in set A.
5. Spots that are found only in set B.

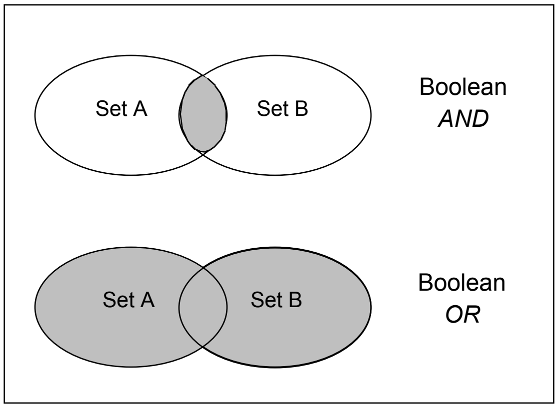

Fig.9-6. Venn diagrams depicting the AND and OR Boolean operations.

To create a Boolean User Set, select Database > Analysis Sets > Create Set, click on the Type button in the dialog box, and select Boolean from the list displayed.

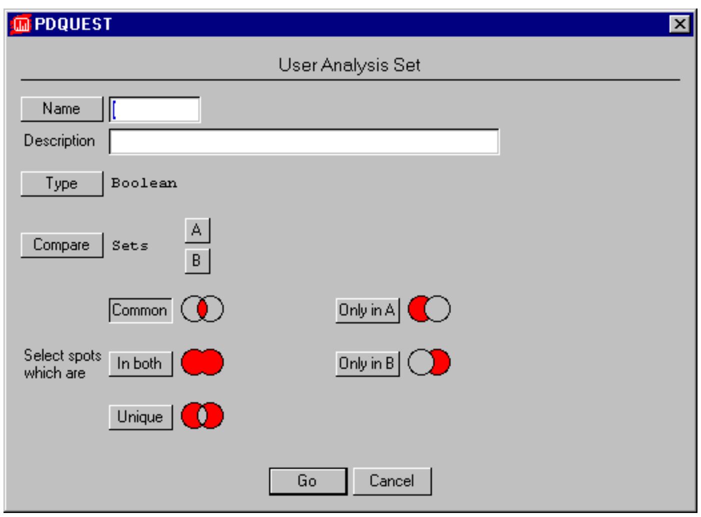

Fig.9-7. Boolean User Set dialog.

To make a Boolean User Set, select the two User Sets you want to compare to one another from the lists displayed when you click the A and B buttons.

Enter the name and description to be given to the User Set.

Select the Boolean operator to be used to create the set by clicking on one of the buttons next to the “Select spots which are” prompt.

When you click on the Go button, the dialog box will disappear and the spots that meet the criteria that you specified will be highlighted in the Standard image.

9.1.f. Statistical User Sets

Statistical User Sets are composed of spots from replicate groups found to be significant according to the statistical test that you specify.

To create a Statistical User Set, select Database > Analysis Sets > Create Set, click on the Type button in the dialog box, and select Statistic from the list displayed. The dialog box will be reconfigured to let you select the replicate groups to be compared and the statistical test to be performed.

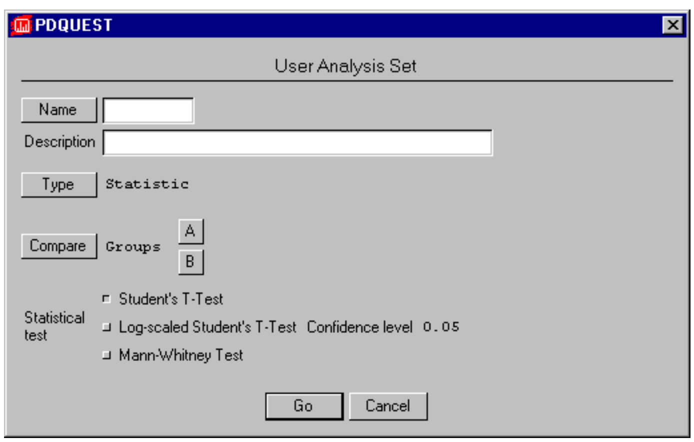

Fig.9-8. Statistic User Set dialog.

Statistical User Sets only work with groups, because the tests require multiple gels. The Student's T Test and Log Student's T Test require at least two gels per group. The Mann-Whitney Test requires at least four gels per group.

Choose the groups by clicking on the A and B buttons as described in previous sections.

Indicate the statistical test you want to perform by clicking on the button next to the name of the test.

Enter a name for the User Set at the top of the dialog box and click the Go button.

Significant protein spots will be highlighted on the Standard.

9.1.g. Annotation User Sets

Another type of User Set that you can create is based on entries made in annotation categories. (For more information about annotations, see section 8.3) Annotation User Sets are composed of spots whose annotations include a particular key word or words that you specify.

To create an Annotation User Set, select Database > Analysis Sets > Create Set, click on the Type button in the dialog box, and select Annotation from the list displayed. The dialog box will reconfigure so you can enter keywords and select the annotation category to be searched.

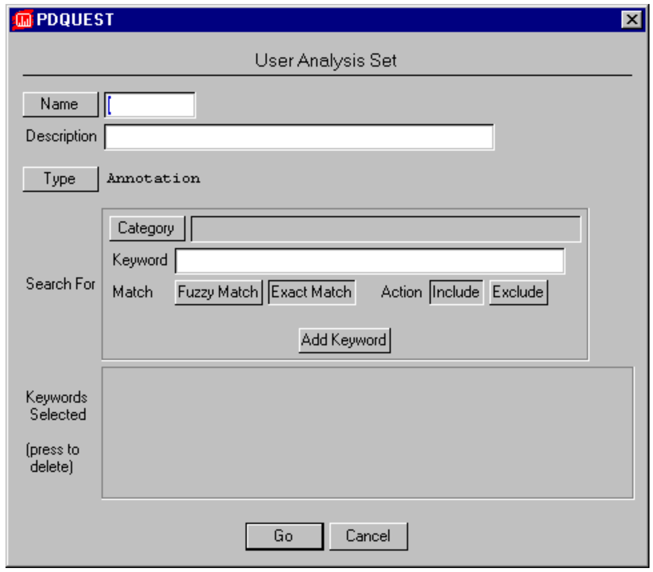

Fig.9-9. Annotation User Set.

Enter a name for the User Set in the text box next to the Name button. Enter more detailed information about the User Set in the text box next to the Description prompt.

Click on the Category button to see a list of all the annotation categories associated with this matchset. Click on the category that you want to search for the keyword that you are about to enter. When you do this, the list will disappear and the name of the category you selected will appear next to the Category button.

In the text box next to the Keyword prompt, enter the word you want to search for. The word can be a root and not a complete word. It need not be spelled correctly. For example, "polymrph" will find "polymorph," "polymorphic," and "polymorphism."

If you are not entirely certain of your spelling, or if you have entered a "word root" and want close, but not identical, matches, click on the Fuzzy Match button next to the Match prompt. The search program will then look for similar words in the annotation but will overlook some discrepancies. Fuzzy matching does require that the first letter of the keyword and the annotation be the same.

If you are sure of your spelling and are only interested in finding perfect matches to the keyword, click on the Exact Match button next to the Match prompt. If you select Exact Match, the search will not tolerate any discrepancies between the keyword and the annotation.

Selecting Include or Exclude next to the Action prompt determines whether the protein spots added to the User Set all have annotations that include the keyword (Include) or do not include the keyword (Exclude).

For example, suppose you are searching an annotation category called “Cell location,” which contains the locations of proteins in a cell, and you want the User Set to include all proteins except those found in the nucleus. You select the category enter “nuclear” as the keyword and click the Exclude button. The resulting User Set would include all spots in that category except those whose annotation includes the word “nuclear.”

After entering the keyword and specifying Fuzzy Match / Exact Match and Include / Exclude, click on the Add Keyword button. In the box below, the parameters for the keyword search are displayed. You can enter as many keywords as you want to be included in the search. Each time you click on the Add Keyword button, its parameters are added to the list in the box.

When you have entered all the keywords, click on the Go button at the bottom of the dialog box. The form will disappear and all the spots that are included in the User Set will be highlighted in the Standard.

9.2. Higher Level Matchsets

PDQUEST allows you to create matchsets from matchsets, so you can build a database by comparing and summarizing the results of a series of experiments.

“Higher level” matchsets are matchsets created from the Standard images of other (“lower level”) matchsets. A lower level matchset might be a matchset created from your original gels (a “level-1” matchset), or it could be another higher level matchset (that is nevertheless “lower” than the one you are creating). Many experimental protocols can be designed to take advantage of higher level matching.

PDQUEST allows you to edit spots in the member images of lower level matchsets and have those changes take immediate effect in the higher level matchset. Having lower level members accessible from the higher level Standard helps you to visualize the original gels that contain the data analyzed in the higher level Standard.

For example, when working with a higher level matchset, you may notice that a spot is present in one member and not the other. Your next step would be to go back to the level-1 matchset from which that member came, to see if any of the gels in the level-1 matchset contain the spot of interest. If you found the spot, you would add it to the Standard of the level-1 matchset so that it could be subsequently added to the Standard of the higher level matchset. Being able to access and edit the members of the level-1 matchset from the higher level matchset eliminates the need to unload the higher level matchset, load the level-1 matchset to edit the spot, unload the level-1 matchset and then reload the higher level matchset.

One experiment may find all the proteins that change during cell growth in culture, while another may identify nuclear proteins by cell fractionation. These two experiments can be analyzed individually in level-1 matchsets, each with their own Standard images. These two Standards can then be combined into a higher level matchset with its own Standard. Data from the two experiments are now linked through a third Standard. This higher level matchset may shed light on growth-related proteins found in the nucleus.

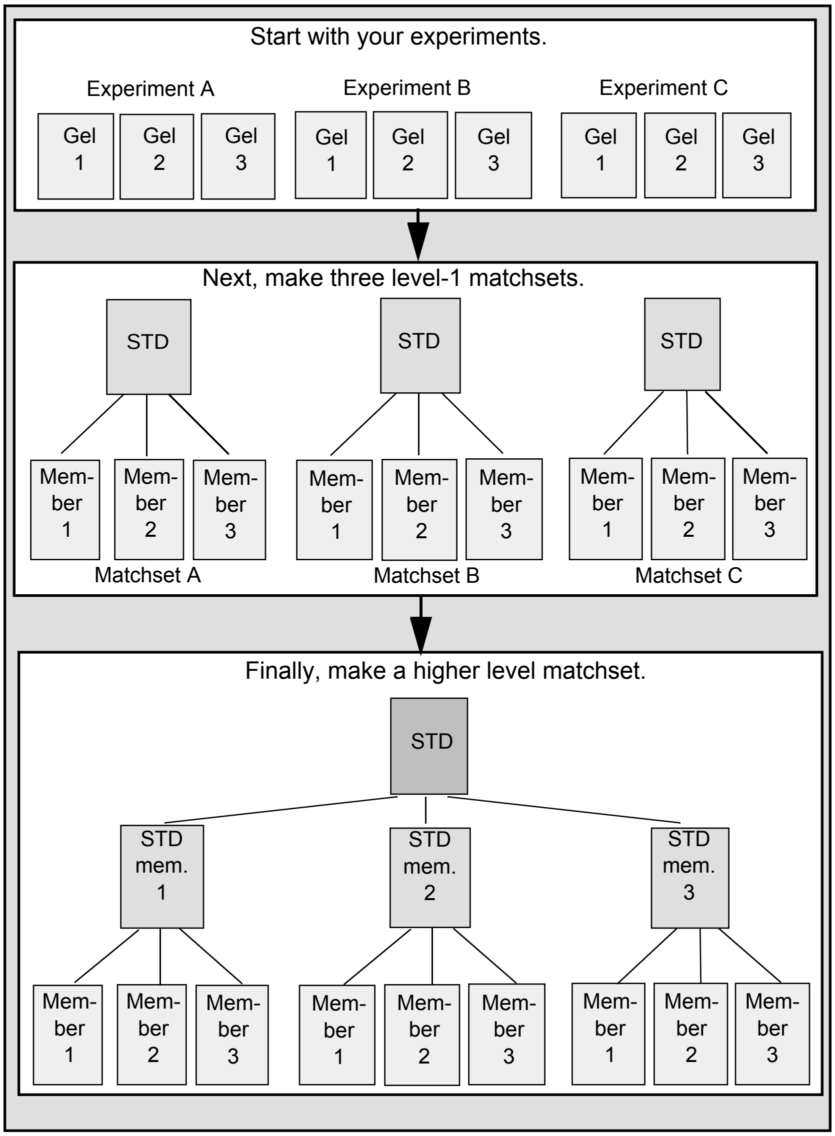

Fig.9-10. Creating a higher level matchset: In this example, the higher level matchset is made from 3 lower level matchsets, each containing 3 members.

9.2.a. Creating a Higher Level Matchset

As mentioned previously, higher level matchsets consist of lower level matchsets. To make a higher level matchset, select Create Matchset from the Match menu. The Create Matchset dialog box will open.

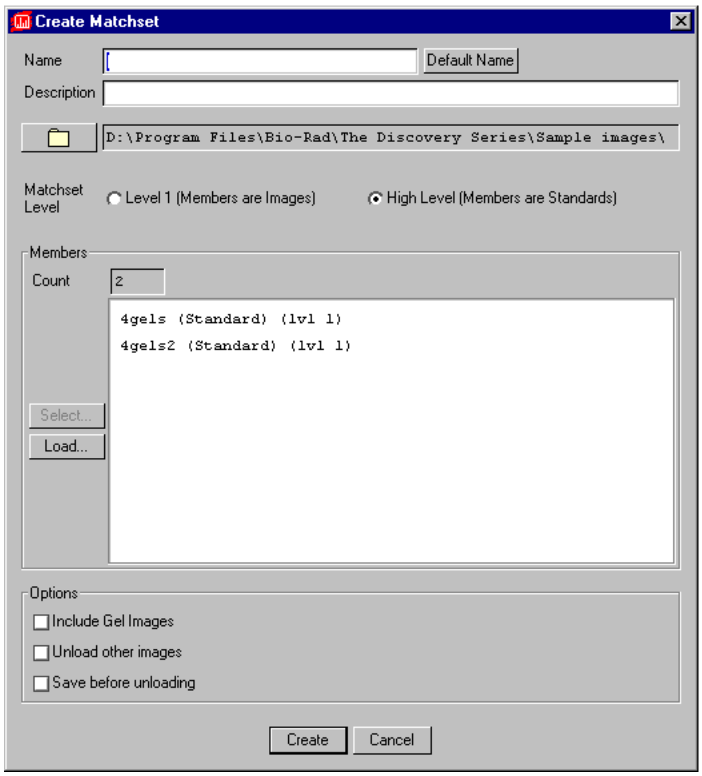

Fig.9-11. Dialog box for creating a higher level matchset.

Higher level matchsets are created in the same way as lower level matchsets (see Chapter 6), with the following exceptions:

- Next, to the Matchset Type prompt, select High Level instead of Low Level.
- You cannot include Gel Images in a higher level matchset, because higher

level matchsets are made up of Standards, which do not have Gel Images.

- The members that you select for a higher level matchset are lower level matchsets rather than Gel Spots files.

To select the matchsets for inclusion in your higher level matchset, click on the Load button. This will open the Select New Member dialog.

Select the matchsets you want to include by clicking on them. Once you have chosen all the matchsets that you want to include, click on the Add Selected button.

The names of those matchsets will appear in the Members list in the Create Matchset dialog box. When you are satisfied with your choices, click on the Create button.

You will asked to select a Standard for the higher level matchset (see below). This is similar to the Standard in a lower level matchset.

The screen will be configured into enough windows to display each member of the matchset. Each member will be assigned to a window.

9.2.b. Editing Higher Level Matchsets

You can add members to, remove members from, and reorder members in higher level matchsets just like lower level matchsets. These commands are described in section 6.3.

9.2.c. Creating a Standard for a Higher Level Matchset

As with lower level matchsets, each higher level matchset has a Standard. You can select the member to use as the template for the Standard when you create the matchset, or you can select it later using the Make Standard function.

Select Make Standard from the Match menu, then click on the matchset member that will be the template for the Standard.

To assign the same SSP numbers to the higher level matchset Standard as exist in the Standard template, click on Yes in the pop-up box. For more information about retaining and transferring SSP numbers, see below.

9.2.d. Matching and Editing Higher Level Matchsets

Once your higher level matchset has been created, you need to edit and match its members. The same protocols used with lower level matchsets can be applied to higher level matchsets.

Landmarks need to be entered before member matchsets can be matched to each other. Members should be carefully reviewed for spots that need to be added to the higher level matchset.

For more information about matching and editing, see sections 6.4.–6.6..

9.2.e. Transferring DSNs or SSP Numbers

Each matchset has unique DSN or SSP numbers assigned to proteins that may not correspond to those in other matchsets. So how can you correlate data between two or more matchsets?

Begin by creating a higher level matchset consisting of the two lower level matchsets and creating its Standard.

Apply the numbering scheme from one of the members to all other members of the matchset (SSPs or DSNs).

Depending on the numbering system, select Transfer SSPs or Transfer DSNs from the Database > Spot Numbers submenu.

When you use either of these functions, you will first be asked to select the image from which the numbers should be copied. Select the image, then indicate the matchset member to which the numbers should be applied.

Confirm the transfer of numbers by clicking on the Yes button.

If you transfer numbers between all the members, the spot numbers for the matched spots in the lower level matchsets will correspond to each other and to the spots in the higher level matchset as well. This makes it easier to correlate data from the reports of the member matchsets.

9.2.f. Transferring Mr and pI Data

The function Transfer MrpIs (Database > MrpI Standards > Transfer MrpIs) allows you to apply Mr and pI values from one Standard to the other members of a higher level matchset.

When you select this function, you will first be asked to select the matchset member the MrpI data should be transferred from.

Next, choose the matchset members to which the data should be applied.

Confirm the transfer of the MrpI data by clicking on the Yes button.

9.2.g. Transferring Annotations

If annotations are entered in the member matchsets of a higher level matchset, you can transfer them all to the higher level matchset Standard. This makes them all available from a single file. You can do this with the Transfer Category function.

A selected annotation category can be transferred between members in a higher level matchset, but only for spots that are matched between both members. If no spots are matched between the two, only the category name is transferred (no data is transferred).

Before you can transfer annotations, you need to select the annotation category. From the Database > Annotations submenu, choose Select Category, then click on the Standard with the annotations you want to transfer. A list of the available annotation categories for that Standard will open.

Select the category you want to transfer. This category will become active and ready to be transferred.

From the Annotations submenu, select Transfer Category. From the pop-up list, select the Standard from which the annotations should be transferred. Next, select the member to which the annotations should be transferred.

Complete the annotation transfer by clicking on the Transfer OK button.

A message in the status bar will indicate the number of spots with annotations in that category and how many were transferred.

To see to which protein spots the annotations were transferred, select the category in the window displaying the “recipient” member and use any of the annotation display functions described in section 8.3.

10. Reports

The reports described in this section (Quantity Reports, Quantity Graphs, Scatter Plots, Standard Images, and Annotation Reports) are available from the Reports menu.

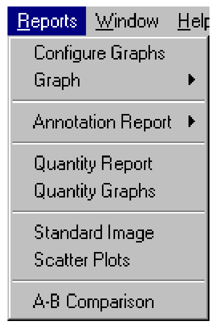

Fig.10-1. Reports menu.

Printing Reports

The print dialog for the various reports includes controls for choosing the printer and printer properties, selecting the orientation of the paper, and specifying the number of copies.

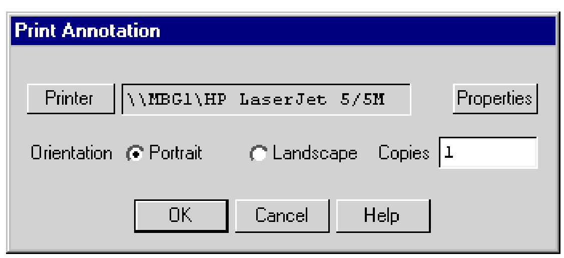

Fig.10-2. Print reports dialog box.

10.1. Quantity Reports

If you select Quantity Report, the general report information dialog box will open.

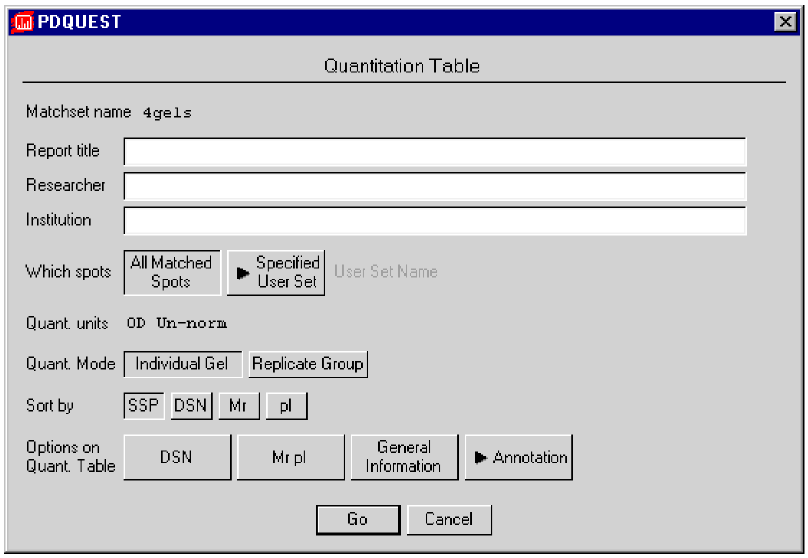

Fig.10-3. Quantitation Table dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified User Set. If you click on the Specified User Set button, a list of available User Sets will be displayed. The name of the set you choose will appear next to the User Set Name prompt.

If you want to see spot quantitation for each individual member of the matchset, select Individual Gel next to the Quant. Mode prompt. If you want to see the average quantitation for a Replicate Group, click on the Replicate Group button.

You can sort your spots by SSP number, DSN, molecular weight, or isoelectric point by clicking on the appropriate “Sort by” button.

Other Options

If DSN numbers are assigned to the protein spots and you would like to see them on the report, click on the DSN button.

Mr and pI data will be reported if you click on the Mr pI button.

Clicking on the General Information button will include a title page, a list of symbols and abbreviations, and information about the matchset, member gels, and their calibration data in the report.

You can print spot annotations by clicking on the Annotation button and selecting from the list of annotation categories. The report will include annotation information for the spots that have annotation data in the selected category.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

10.2. Quantity Graphs

If you select Quantity Graphs from the Reports menu, a dialog box will open in which you can specify the format of your quantitation histograms.

Fig.10-4. Quantitation Histograms dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified User Set. If you click on the Specified User Set button, a list of available User Sets will be displayed. The name of the set you choose will appear next to the User Set Name prompt.

If you want to see spot quantitation for each individual member of the matchset, select Individual Gel next to the Quant. Mode prompt. If you want to see the average quantitation for a Replicate Group, click on the Replicate Group button.

You can sort your spots by SSP number or DSN by clicking on the appropriate “Sort by” button.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

10.3. Standard Image Reports

Selecting Standard Image from the Reports menu allows you to print a map of up to 300 spots from the matchset Standard.

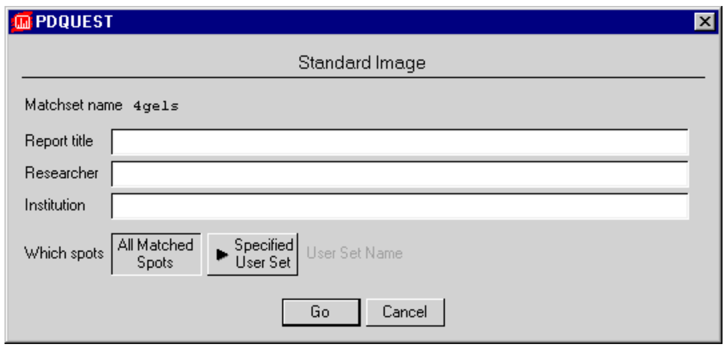

Fig.10-5. Standard Image dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified User Set. If you click on the Specified User Set button, a list of available User Sets will be displayed. The name of the set you choose will appear next to the User Set Name prompt.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

10.4. Scatter Plots

Scatter plots are graphs that show the relatedness of two gels. Each matched spot is plotted on a log scale as a point according to its quantitation in the first gel (X-axis) and its quantitation in the second gel (Y-axis). If the quantitation value of a spot is identical in both gels, the point representing that spot falls on the line with a slope of 1.00 (45° line). If the values are not equal, the point falls to the left or to the right of the line.

You can display a scatter plot for any two gels in a matchset, or print out the scatter plots for all the members of a matchset.

Displaying the Scatter Plot for Any Two Matchset Members

Select Scatter Plot from the Reports > Graph submenu. The Scatter Plot window will open.

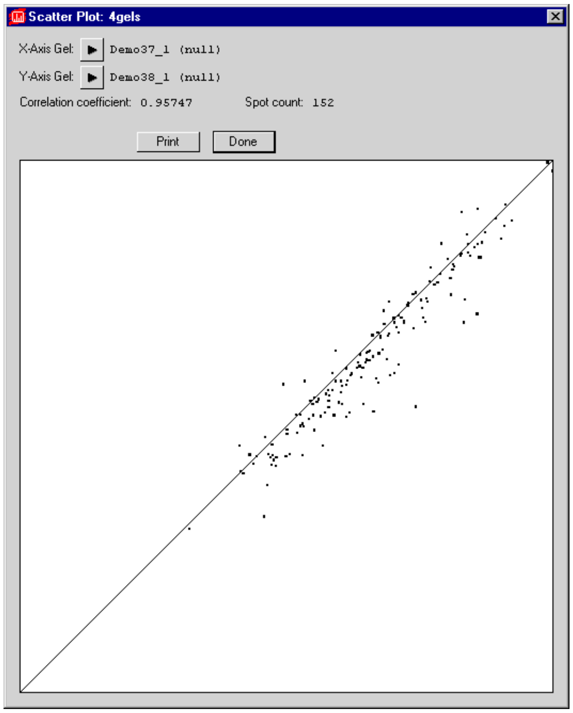

Fig.10-6. Scatter Plot window.

Select the members to compare by clicking on the arrows buttons next to X-Axis Gel and Y-Axis Gel.

The Spot Count is the number of matched spots in the two members, and the Correlation Coefficient is a measure of quantitative similarity between gels. A coefficient value of 1.00 indicates that the two gels are perfectly related (i.e., equivalent), while a low coefficient value (e.g., 0.40) indicates that the two gels are not very similar.

To print this scatter plot, click on the Print button in the window. The standard print report dialog will open, in which you can select the settings for your print and proceed to print.

Printing Scatter Plots of All Members of a Matchset

To print the scatter plots for all members of a matchset, select Scatter Plots from the Reports menu.

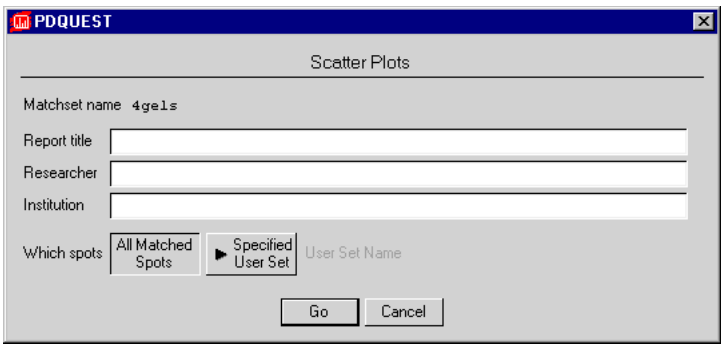

Fig.10-7. Scatter Plots dialog box.

You can type in a report title, the name of the researcher, and the name of the research institution in the appropriate fields.

You can specify the spots that will be included in your report: All Matched Spots or the spots in a Specified User Set. If you click on the Specified User Set button, a list of available User Sets will be displayed. The name of the set you choose will appear next to the User Set Name prompt.

Once you have selected your options, click on the Go button at the bottom of the form. This will open a smaller version of the standard print dialog, in which you can select your printer settings and proceed to print.

10.5. Annotation Reports

You can print your spot annotations using the functions in the Annotation Report submenu of the Reports menu.

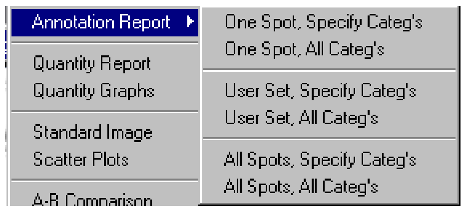

Fig.10-8. Annotation Report submenu.

Using the commands in this submenu, you can print annotations for individual spots, the protein spots in a User Set, or all the spots in the Standard. You can also print annotations for particular categories or all categories.

Individual Spots

To print the annotations for a particular spot, select One Spot, Specify Categ’s or One Spot, All Categ’s, then click on the spot in the Standard.

If you select One Spot, Specify Categ’s, a pop-up list will prompt you to select annotation categories to print. Once you have made your selections, click on the Done button below the list. The print dialog box will appear, in which you can select you printer settings.

If you select One Spot, All Categ’s, the print dialog box will appear immediately.

User Sets

To print annotations for spots in a specific User Set, select User Set, Specify Categ’s or User Set, All Categ’s from the Annotation Report submenu.

A list of User Sets will be displayed. Click on the name of the User Set whose annotations you want to print.

If you selected User Set, Specify Categ's, a pop-up list of categories will appear. Make your selection(s), then click on Done. The print dialog box will appear.

If you selected User Set, All Categ's, the print dialog box will appear as soon as you have chosen a User Set.

All Spots

All Spots, Specify Categ's prints selected annotation categories for all the spots in the Standard.

All Spots, All Categ's prints the annotations in all available categories for all annotated spots in the Standard.

11. Printing and Exporting

This chapter describes the commands for printing and exporting images and data. These commands are located on the File menu.

11.1. Printing

The commands for printing images, actual-size images, scan reports, and video printouts are located on the File > Print submenu.

1. Print Image prints a copy of the image and any overlays that are currently displayed.
2. Print Actual Size prints an actual-size copy of the image.
3. Scan Report prints the image and information about its scan history, number of pixels, data range, etc.
4. Video Print prints images and reports to a video printer.*

*Video printing requires installation of the video board and cable that come with the Gel Doc 1000 / 2000 gel documentation system. The video board and cable can also be ordered separately.

11.1.a. Printing Images

Print Image prints a copy of the active image window and any image overlays that are displayed.

File > Print > Print Image opens the Print Image dialog box.

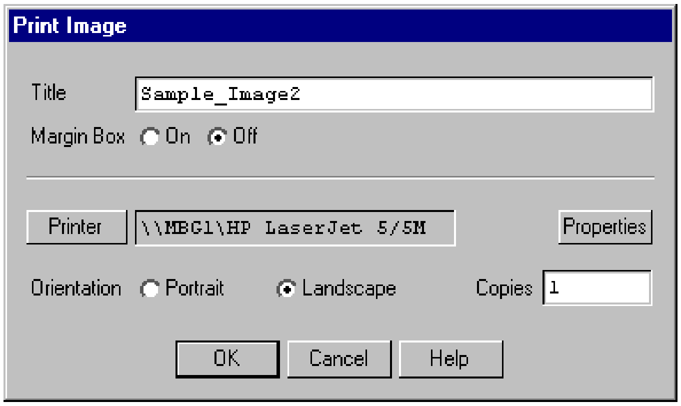

Fig.11-1. Print Image dialog box.

At the top of the dialog box, enter the title that you want to appear above the image when you print.

Choose whether or not a margin border should appear around the edge of the image by clicking either the On or Off button next to the Margin Box prompt.

Click on Printer to display a list of your available printers. After you have selected a printer, you can click on Properties to select the standard Windows print settings for that printer.

Note: If you are using Windows 95 with certain printer drivers, you may experience difficulty printing some overlays and overlay symbols on images. While there is no single solution to this problem, you should try adjusting some settings in the Printer Properties dialog box, under the Graphics and Advanced tabs—specifically, the Dithering, DPI, Rasterizing, and RET settings.

Select the paper orientation by clicking on Portrait or Landscape next to the Layout prompt. (This can also be set in the Properties dialog box.)

Indicate how many copies should be printed in the text box next to the Copies prompt.

When you are satisfied with all the print parameters, click on the OK button to send the image to the printer.

11.1.b. Print Actual Size

You can print your images at their actual size using the Print Actual Size command.

Note: If you are using the Gel Doc or the Fluor-S, you must specify the correct image area size when capturing your images to ensure accurate 1:1 printing. You can specify the image area size in the acquisition window for the instrument. See the chapter on each imaging device for more information.

Select Print Actual Size from the File > Print menu. The standard Print Image dialog box will open (see previous section).

11.1.c. Scan Report

Scan Report allows you to print out a single-page report of an image and its associated information. The format of the report is designed to provide a concise yet thorough summary of the most relevant features of an image for documentation purposes.

The report includes the following information:

- The image.
- The report date.
- The image title and a brief description.
- The directory location.
- The date the image was scanned.
- The type of imaging device.
- The imaging area and number of pixels.
- Pixel size.
- The intensity range, image color, and memory size.
- Image background information.
- Relevant lane and band information.

To print a scan report of a particular gel, select Scan Report from the File > Print submenu. This will open a smaller version of the standard print dialog box (see Print Image, above).

Note: TIFF images do not contain all the tagged information that would normally be included in an image file (for example, imaging device, scan date, image color, etc.). For this reason, the Scan Report may list this information as “Unknown” for imported TIFF files.

11.1.d. Video Printing

The Video Print command allows you to print images and reports on a video printer. To create a video printout of the active window, select Video Print from the File > Print submenu.

Note: Video printing requires installation of the video board and cable that come with the Gel Doc 1000 / 2000 gel documentation system. The video board and cable can also be ordered separately.

Settings for the Mitsubishi P90W/P91W Video Printer

There are three settings for the Mitsubishi P90W / P91W video printer. Set Contrast to 0, Brightness to 0, and Gamma to 5.

The dip switches should stay in the orientation in which they are shipped: Pin 1 is up (on), and Pins 2–10 are down (off).

11.2. Exporting

Matchset data can be exported for analysis by spreadsheet programs, and scans can be exported as TIFF images for analysis by other image analysis software.

11.2.a. Exporting Matchset Data

Data from a matchset can be exported in ASCII format to your hard disk, file server, or other storage medium for analysis by spreadsheet programs.

With the matchset loaded, select Export > Matchset Data from the File menu. This will open the Export Matchset Data dialog box.

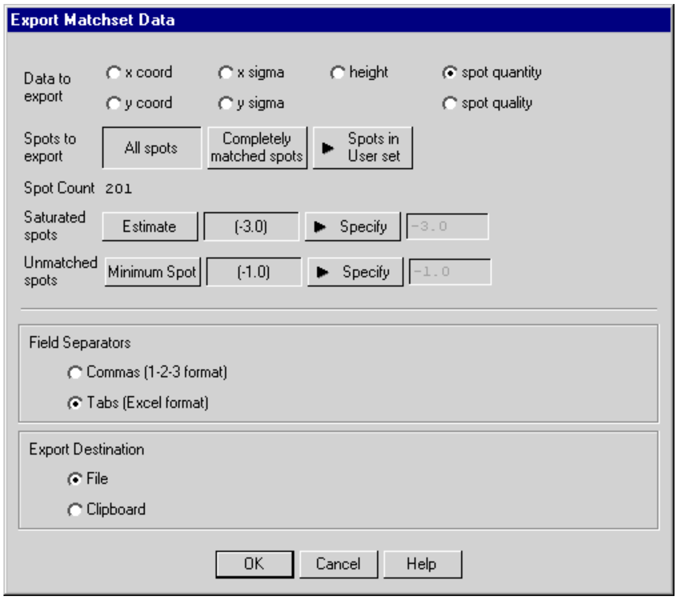

Fig.11-2. Export Matchset Data dialog box.

Next to the “Data to export” prompt, select the characteristic of the spot that you want to export.

Specify the spots whose data should be exported. You can choose either all the spots in the matchset, those spots that are matched in every member of the matchset or the spots in a particular User Set.

The Spot Count will automatically adjust to reflect the total number of spots selected.

If you are exporting spot quantity data, select the way in which saturated spots and unmatched spots are reported.

The quantitation of saturated spots can be estimated, reported as (-3.0), or reported in some other way that you specify. If you opt for the latter, click on the Specify button and enter the value that should be reported.

If a spot is unmatched in one or more members of a matchset, it can be reported as the minimum spot quantitation value (the minimum quantity needed for a spot to be detected for the gel). Alternatively, it can be reported as (-1.0) or as a value that you enter in the text box next to the Specify button.

Next, select the separators you want to use to separate columns of data in your spreadsheets: commas or tabs.

Your data can be exported as a file or to the clipboard.

If you choose the File option button, clicking on OK will open a version of the Save As dialog box, in which you can enter a name and export location for the file.

If you choose the Clipboard option button, clicking on OK will save your data to the clipboard.

11.2.b. Exporting an Image

To export a 2-D scan or calstrip scan as a TIFF image, select Export > Export to TIFF Image from the File menu. The Export Scan dialog will open.

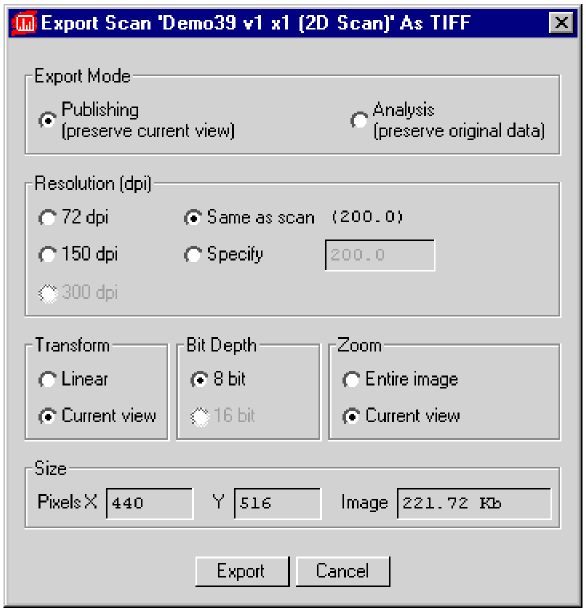

Fig.11-3. Exporting TIFF Images.

Note: Only 2-D scans and calstrip scans can be exported as TIFF images.

Analysis Export Mode

Analysis mode exports the scan data, unmodified by PDQUEST's viewing controls.

If you select Analysis mode, the other controls in this dialog box will become inactive.

Publishing Export Mode

Publishing mode exports a TIFF image that looks like the image as it is currently displayed in PDQUEST (with whatever Transform adjustments you may have made).

In Publishing mode, you can specify a resolution for the TIFF image by selecting 72 dpi (typical computer screen resolution), 150 or 300 dpi (standard printing resolutions), the same resolution as the scan file, or any resolution you want to specify (up to the resolution of the scan).

If you have log transformed the image, you can specify a linear transform, or preserve the current view.

If your image is 16 bits, you can compress it to 8 bits for export to TIFF.

Finally, if you are only displaying part of the image due to magnification or repositioning, you can preserve the current view or export the entire image.

Exporting the Image

The size of the pixels in the image and the file size of the image are listed at the bottom of the dialog. When you are ready to export, click on the Export button.

A Save As dialog box will open. The default file name will have a .tif extension, and the file type will indicate that this is a TIFF image. You can change the file name or select a different directory to save in.

PDQUEST will generate a grayscale TIFF image compatible with Melanie software.

Appendix A. Gel Doc 1000/2000

Fig.A-1. Gel Doc 1000/2000.

Before you can begin acquiring images using the Gel Doc® 1000 / 2000 gel documentation system, the system must be properly installed and connected with the host computer. See the Gel Doc hardware manual for installation, startup, and operating instructions.

To use the Gel Doc, you will need to have the Bio-Rad-supplied acquisition board installed in your PC or Macintosh. The drivers for this board will be installed when you install PDQUEST.

Make sure that your Gel Doc camera is turned on. If the camera is not turned on, the Gel Doc acquisition window will open but the screen will be black, and you will be unable to Video Print.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

A.1. Opening the Gel Doc Acquisition Window

To acquire images using the Gel Doc, go to the File menu and select Gel Doc.... The acquisition window for the instrument will open, displaying a control panel and a video display window.

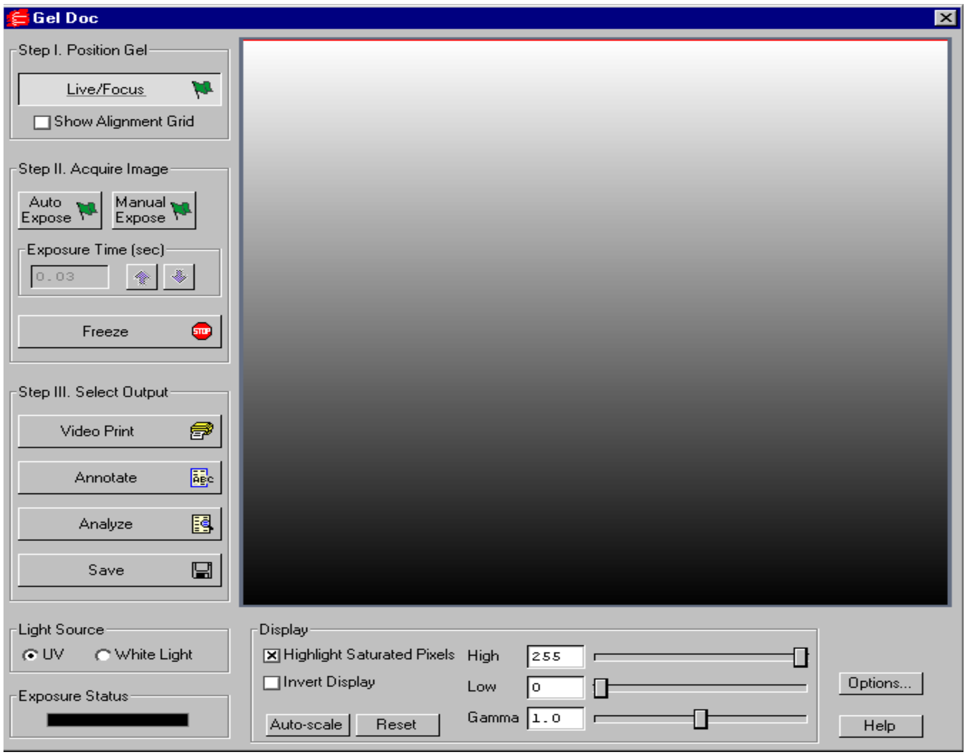

Fig.A-2. Gel Doc acquisition window

The Gel Doc video display window will open in "live" mode, giving you a live video display of your sample. If no image is visible, make sure the camera is on, check the cable connections, and make sure that the protective cap is off the camera lens. Also check to see that the transilluminator is on and working.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Gel Doc:

1. Position and focus the gel or other object to be imaged.

2. Acquire the image.

3. Select the output.

A.2. Step 1. Position Gel

The Gel Doc window will open in “live” mode, giving you a live video display of your sample. (Frames will be captured and displayed at about 10 frames per second, depending on the speed of your computer.) In this mode, the Live/Focus button will appear selected.

You can use live mode to zoom and focus the camera on your sample area and position your sample within the area. You can also select the Show Alignment Grid checkbox to facilitate positioning.

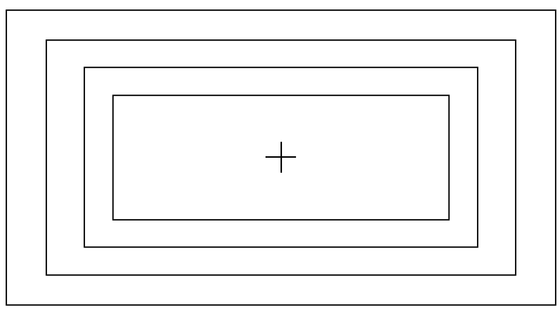

Fig. A-3. Gel Doc alignment grid.

The image should stay in focus while zooming. If focus is not maintained, refer to your Gel Doc hardware manual.

Note: After positioning your sample, you should check the Imaging Area dimensions under Options (see below) to make sure that they conform to the size of the area you are focusing on. To determine the size of the area you are focusing on, you can place a ruler in the Gel Doc box so that it is visible by the camera.

A.3. Step II. Acquire Image

For many white light applications, you can skip this step and save and print images directly from live mode.

For UV light or chemiluminescent applications, or if your aperture is closed down in white light applications, this mode will probably not give you the best possible image. To illuminate faint areas of your sample, you will need to select an exposure time for the image.

“Exposure” refers to the integration of the image on the camera CCD over a set period of time. The effect is analogous to exposing photographic film to light over a period of time. The Gel Doc has two exposure modes: Auto and Manual.

To acquire an image, you will typically perform the following steps:

1. Select Auto Expose to determine an approximate exposure time.
2. After Auto Expose is finished, “fine tune” that exposure time using Manual Expose. You can also change the camera aperture at this time.
3. Click on Freeze or any of the other buttons in the window to stop the exposure process.

Note: If you know the approximate exposure time you want (± 3 seconds), you can skip the Auto Expose step and go directly to Manual Expose.

Auto Expose

Clicking on the Auto Expose button will cancel live mode and begin an automatic exposure. (The Auto Expose button will appear selected throughout the exposure.) If you know the approximate exposure time you want (± 3 seconds), you can skip this step and go directly to Manual Expose.

In Auto Expose mode, the image is continuously integrated on the camera CCD until it has reached a certain percentage of saturated pixels. This percentage is set in the Options dialog box. (Default = 0.75 percent. See Options below.)

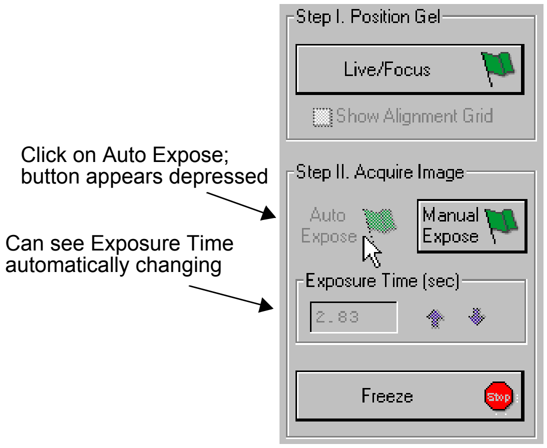

Fig. A-4. Acquiring an image, step 1: Select Auto Expose.

Once an image has reached the specified percent of saturated pixels, it is captured and displayed in the video display window, Auto Expose is automatically deactivated, the exposure time appears active in the Exposure Time field, and Manual Expose is activated.

Note: If you are having difficulty auto-exposing your sample, you can use Manual Expose to adjust your exposure time directly. Most applications only require an exposure time of a few seconds, which can be quickly adjusted using Manual Expose.

Manual Expose

Manual Expose is automatically activated after Auto Expose has deactivated. Or, if you know the approximate exposure time you want, you can simply click on the Manual Expose button.

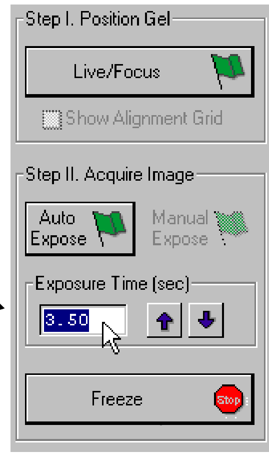

Fig. A-5. Acquiring an image, step 2: Adjust exposure time.

When Manual Expose is active (the button will appear selected), the last captured image will be displayed in the Gel Doc image window. The camera continues to integrate the image on the CCD, updating the display whenever the specified number of seconds is reached.

You can adjust the exposure time by changing the number of seconds in the Exposure Time field. To do so, enter a number directly or use the arrow buttons next to the field.

Freeze

Once you are satisfied with the quality of the displayed image, click on the Freeze button to stop the exposure process. The last full exposure will be displayed in the image window.

Note: Freeze is automatically activated if you adjust any of the subsequent controls (e.g., Video Print, Light Source, Display controls, etc.).

Click on Freeze to stop the exposure process

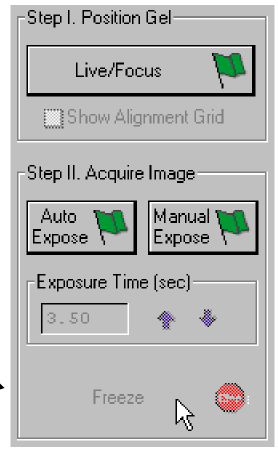

Fig. A-6. Acquiring an image, Step 3: Freeze the exposure.

A.4. Step III. Select Output

The Gel Doc window has several output options.

Video Print

Clicking on Video Print will automatically send the currently displayed frame (either live or integrated) to a video printer. You can add information about your image to the bottom of the printout by selecting the appropriate checkboxes in the Options dialog box. (See Options, below.)

Annotate

Clicking on Annotate will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

The Annotation Tools toolbar will also pop up, allowing you to annotate your image.

The image will not be saved until you select Save or Save As from the File menu.

Analyze

Clicking on Analyze will open a separate image window displaying the captured image. The default name for the image will include the date, time, and user (if known).

You can then analyze the image using the other features in PDQUEST.

The image will not be saved until you select Save or Save As from the File menu.

Save

Clicking on Save will open a separate image window displaying the captured image. A Save As dialog box will automatically open displaying the default file name for the image, which will include the date, time, and user (if known). You can then change the file name and storage directory.

You can also save your image as a TIFF image for export to other applications. (See Save As in section 2.5. for more information.)

A.5. Light Source

The Light Source option buttons allow you to set the type and scale of your Gel Doc data.

UV

Select this light source for fluorescent and chemiluminescent samples. The data will be measured in linear intensity units.

White Light

Select this light source for reflective and transmissive samples. The data will be measured in uncalibrated optical density (uOD) units.

A.6. Exposure Status

The Exposure Status bar shows the progress of your exposure. If your exposure time is greater than 1 second, the status bar display will give you a graphical representation of the remaining time before exposure is complete.

If the exposure time is less than 1 second, the status bar will not refresh itself for each exposure; it will remain at 100 percent.

A.7. Display

The Display controls are useful for quickly adjusting the appearance of your image for output to a video printer. Adjusting these controls will automatically freeze the video display and allow you to alter the image within the Gel Doc window.

These controls are similar to those in the Transform dialog box.

Note: The Display controls will only change the appearance of the image. They will not change the underlying data.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Invert Display

This checkbox will switch light spots on a dark background to dark spots on a light background, and visa versa.

Auto-scale

Clicking on Auto-scale will adjust your displayed image automatically. The lightest part of the image will be set to the minimum intensity (e.g., white), and the darkest will be set to the maximum intensity (e.g., black). You can then “fine tune” the display using the High, Low, and Gamma sliders described below.

High/Low Sliders

If Auto-scale doesn't give you the appearance you want, you can use the High and Low sliders to redraw the image yourself. Dragging the High slider handle to the left will make weak signals appear darker. Dragging the Low slider handle to the right will reduce background noise.

You can also type specific High and Low values in the text boxes next to the sliders. Clicking anywhere on the slider bars will move the sliders incrementally.

Gamma Slider

Some images may be more effectively visualized if their data are mapped to the computer screen in a nonlinear fashion. Adjusting the Gamma slider handle changes the light and dark contrast nonlinearly.

Reset

Reset will return the image to its original, unmodified appearance.

A.8. Options

Clicking on the Options button opens the Options dialog box. Here you can specify certain settings for your Gel Doc system.

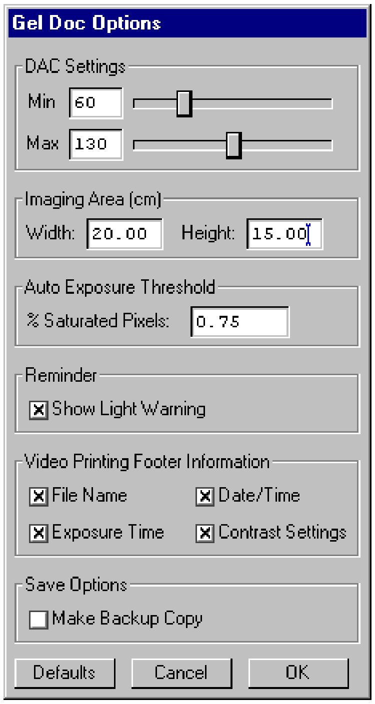

Fig. A-7. Available options in the Gel Doc acquisition window.

Clicking on OK implements any changes you make in this box. Clicking on Defaults restores the settings to the factory defaults.

DAC Settings

Note: The default DAC settings are highly recommended and should be changed with caution.

These sliders may be used to adjust the minimum and maximum voltage settings of your video capture board. The minimum slider defines the pixel value that will appear as white in the image, while the maximum slider defines the pixel value that will appear as black. The slider scale is 0–255, with the defaults set to 60 minimum and 130 maximum.

Imaging Area

These fields are used to specify the size of your imaging area in centimeters, which in turns determines the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. These are also important if you are comparing the size of objects (e.g., using the Volume Tools) between images.

Auto Exposure Threshold

When you click on Auto Expose, the exposure time is determined by the percentage of saturated pixels you want in your image. This field allows you to specify that percentage.

Typically, you will want less than 1 percent of the pixels in your image saturated. Consequently, the default value for this field is 0.75 percent.

Reminder

When this checkbox is selected, the software will warn you to turn off your transilluminator light when you exit the Gel Doc acquisition window or when your system is “idle” for more than 5 minutes.

Note: If you are performing experiments that are longer than 5 minutes (e.g., chemiluminescence), this should be deselected.

Video Printing Footer Information

The checkboxes in this group allow you to specify the information that will appear at the bottom of your video printer printouts.

Save Options

To automatically create a backup copy of any scan you create, select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Appendix B. GS-700 Imaging Densitometer

Fig. B-1. GS-700 Imaging Densitometer

Before you can begin scanning images with the GS-700 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-700 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-700, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

B.1. Opening the GS-700 Acquisition Window

To acquire images using the GS-700, go to the File menu and select GS-700.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

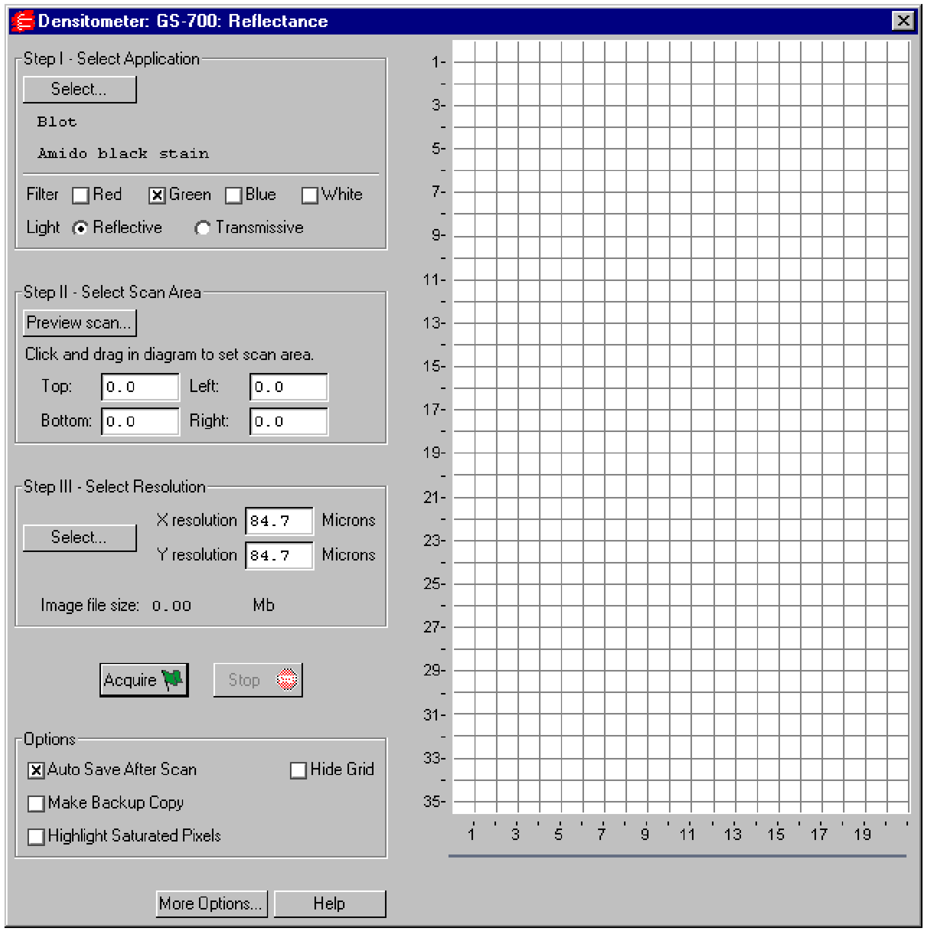

Fig. B-2. GS-700 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–35 top to bottom and 1–21 left to right if the light source is uncalibrated reflective, and 1–25 top to bottom and 1–20 left to right if the light source is uncalibrated transmissive (see below for details).

To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the GS-700:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Acquire the image.

B.2. Step 1. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

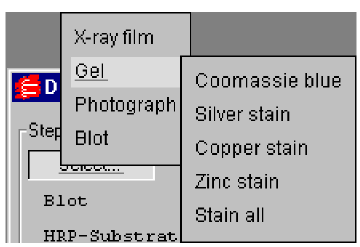

Fig. B-3. Example of the application tree in the GS-700 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

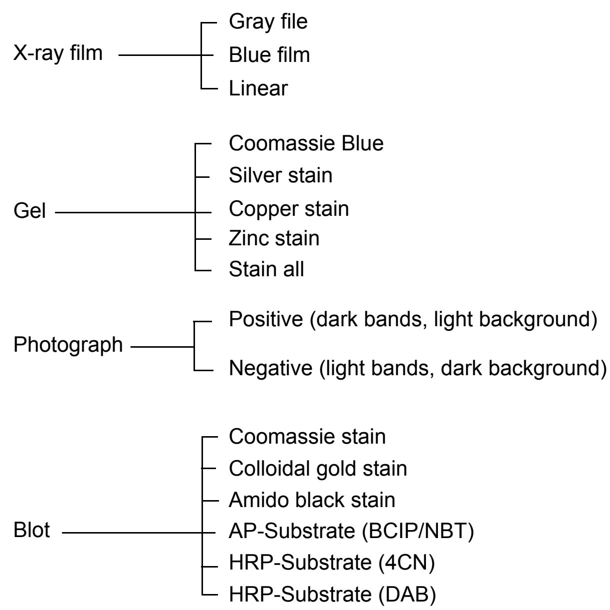

Fig. B-4. Applications available in the GS-700 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. B-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, TLC plates, and photographs. The scanning dimensions in this mode are 21 cm x 35 cm (uncalibrated).
- Transmissive mode is for scanning transparent mediums such as films, gels, and slides. The scanning dimensions in this mode are 20 cm x 25 cm (uncalibrated).

B.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

B.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

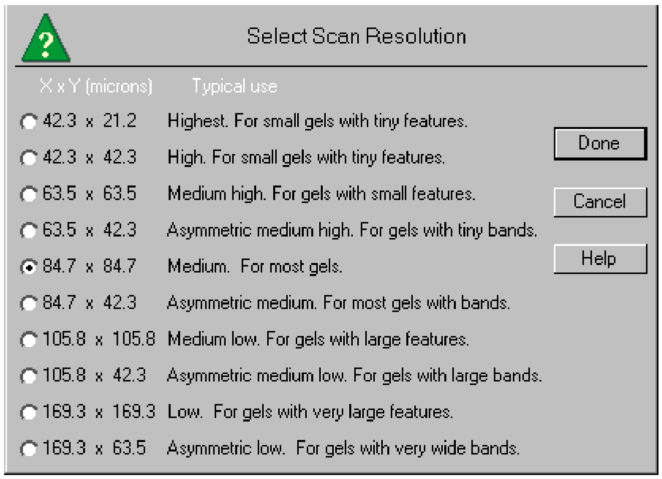

Fig. B-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section B.7. for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

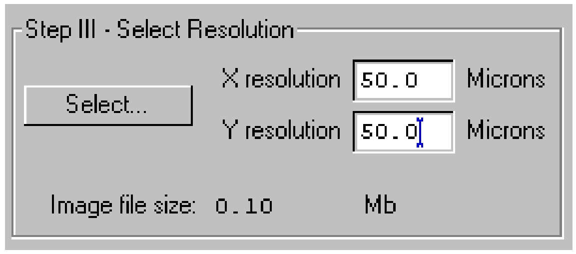

Fig. B-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

B.5. Acquire the Image

If you want to calibrate your scans, read the following section on calibration before scanning.

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a PDQUEST image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

B.6. Calibration

If you have installed a calibration strip on the GS-700 scanner plate, you can automatically calibrate your scans. To order a calibration strip from Bio-Rad, contact your Bio-Rad sales representative.

See the calibration manual for details on installing the calibration strip.

To set the automatic calibration settings, click on the More Options button in the GS-700 acquisition window. This will open the Densitometer Options dialog box.

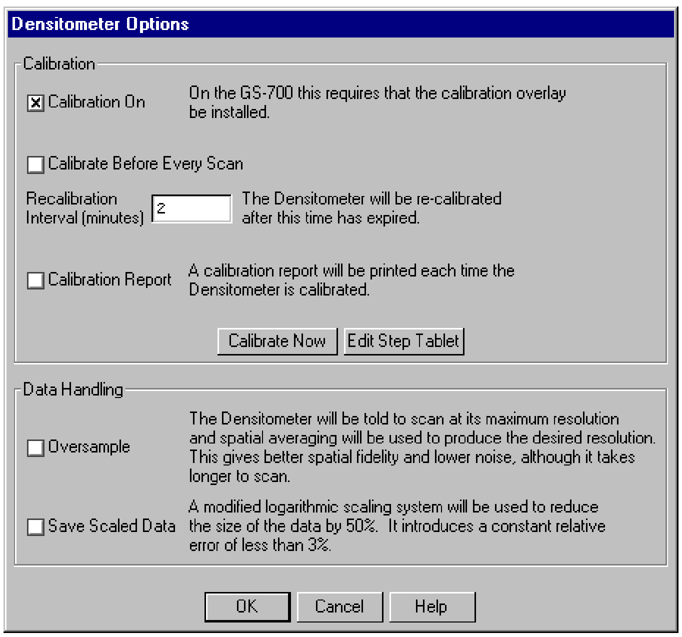

Fig. B-8. More Options in the GS-700 acquisition window.

To enable automatic calibration, click on the Calibration On checkbox.

Note: Your calibration strip must be installed for this feature to work!

With Calibration On selected, the other calibration settings become active.

Calibration Strip Window

When calibration is turned on, a calibration strip window will appear below the main scanning window and the length of the main scanning window will be reduced to 29 cm reflective and 23 cm transmissive.

Every time the densitometer calibrates, an image of the calibration strip will appear in the calibration strip window.

B.6.a. Editing the Step Tablet

Before you can calibrate, you must enter the step tablet values for your calibration strip into the Step Tablet Values form.

In the package with your calibration strip, you will find the manufacturer's printout of the diffuse density values for each segment of your step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Scanning with incorrect step tablet values entered into the form can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-700 will open.

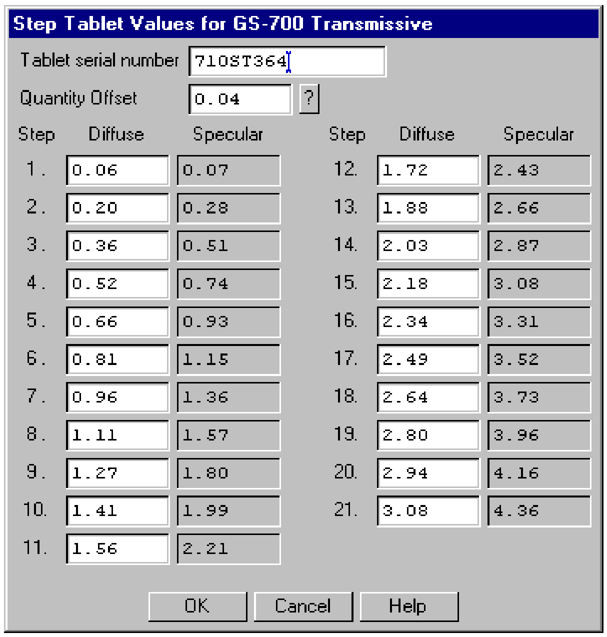

Fig. B-9. Step Tablet dialog box.

The dialog box will indicate whether the displayed form is for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

Note: In transmissive scanning, the scanner is calibrated against a transmissive step tablet. In reflective scanning, the scanner is calibrated against a reflective step tablet, unless the reflective calibration strip is uncalibrated. If the reflective calibration strip does not have step tablet values associated with it, you can use the default values in the form.

In the Tablet Serial Number field, you can type in the serial number for the step tablet values you are entering. (The reflective step tablet will probably not have a serial number.)

Your calibration strip will have a clear plastic cover. If the quantity offset value for the clear cover is included in your calibration strip package, you can enter this number in the Quantity Offset field. Otherwise, use the default value.

Finally, enter the values in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned. You only need to enter new values if you use a new step tablet.

When you finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

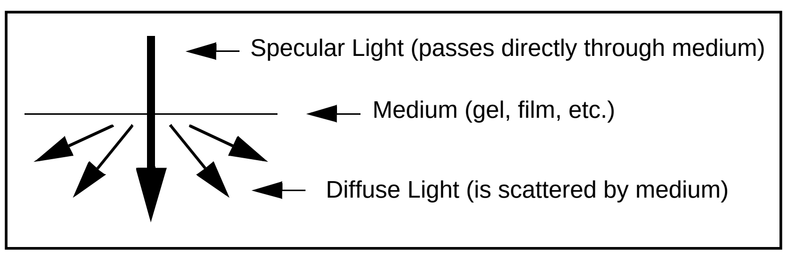

Fig. B-10. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$Specular\ OD = 1.4\ \ Diffuse\ OD$$

B.6.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the densitometer to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: With calibration turned on, the scanner will automatically recalibrate each time you change your filter or your reflective / transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

B.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-700 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix C. GS-710 Imaging Densitometer

Fig. C-1. GS-710 Imaging Densitometer

Before you can begin scanning images with the GS-710 Imaging Densitometer®, your instrument must be properly installed and connected with the host computer. See the hardware manual for installation, startup, and operating instructions.

PC Only: A Note About SCSI Cards

The GS-710 is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the GS-710, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure the imaging device is disconnected or turned off, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

C.1. Opening the GS-710 Acquisition Window

To acquire images using the GS-710, go to the File menu and select GS-710.... The acquisition window for the densitometer will open, displaying a control panel and a scanning window.

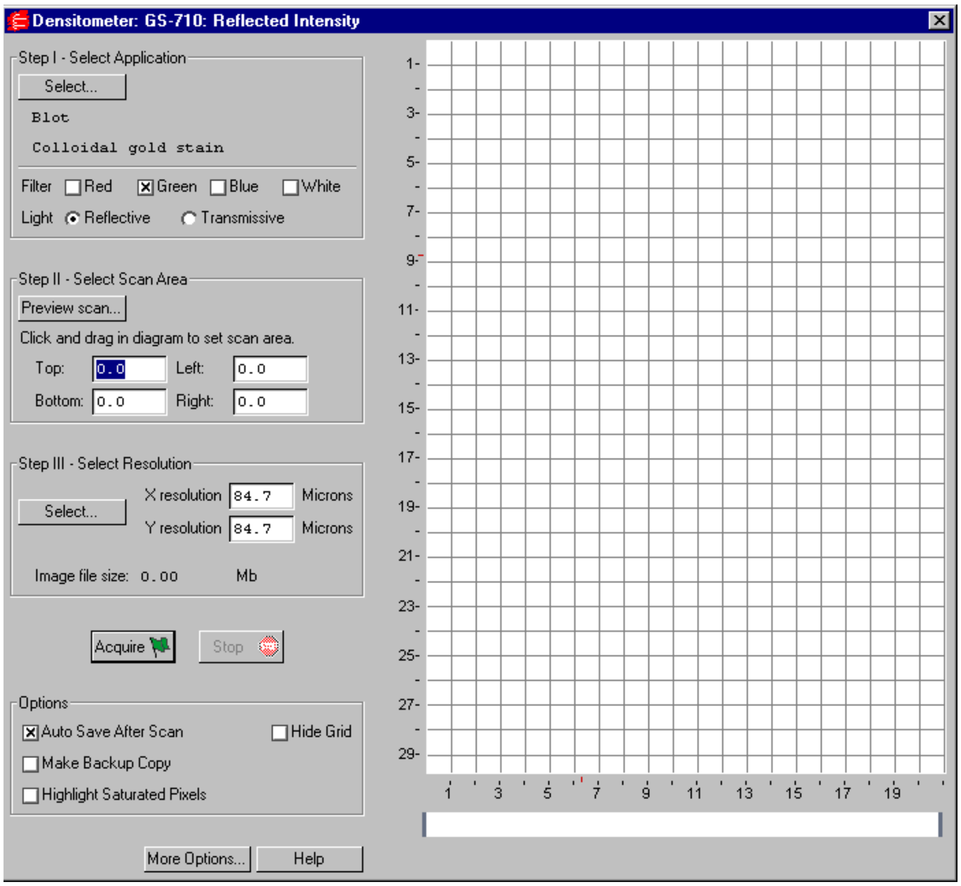

Fig. C-2. GS-710 acquisition window

The scanning window is marked by grid lines that divide the area into square centimeters. These are numbered 1–29 top to bottom and 1–21 left to right if the light source is reflective, and 1–23 top to bottom and 1–20 left to right if the light source is transmissive (see below).

To hide the gridlines, click on the Hide Grid checkbox under Options.

Below the main scanning window is the calibration strip window. Every time the densitometer calibrates, an image of the calibration strip will appear in this window.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are five basic steps to scanning an image using the GS-710:

1. Select the application.
2. Select the scan area.
3. Select the resolution of the scan.
4. Calibrate the instrument. (This is automatic, after you enter the step tablet values before you first scan after installation.)
5. Acquire the image.

C.2. Step 1. Select Application

To set the parameters for your particular scan, you can:

1. Select from a list of possible applications, or
2. Choose your own filter and light source settings.

Selecting from the List of Applications

When you select from the list of applications, the software automatically sets the appropriate filter(s) and other parameters for that particular application.

To select from the list of applications, click on the Select button under Select Application.

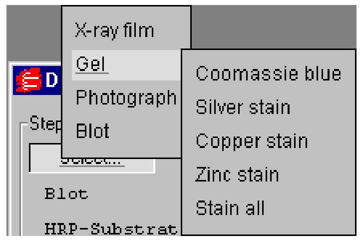

Fig. C-3. Example of the application tree in the GS-710 dialog box.

The applications are listed in a tree that expands from left to right. First you select the category of your application, then you select your particular application.

To exit the tree without selecting, press the ESC key.

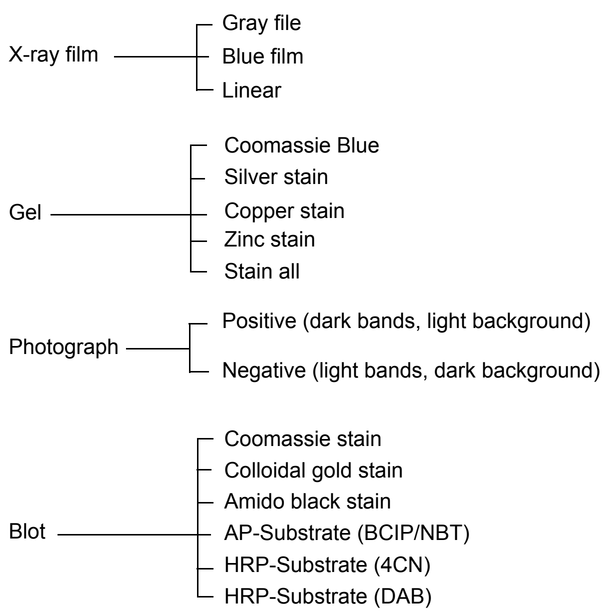

Fig. C-4. Applications available in the GS-710 acquisition window.

Choosing Your Own Settings

If you know the filter and light source settings you want, or want to experiment with different settings, you can choose them yourself.

Next to Filter, click on either the Red, Green, Blue, or White checkbox, or a combination of two of the first three (Red-Green, Green-Blue, Red-Blue).

Filter Color	Wavelength	Application Examples
Red	595–750 nm	Coomassie G-250, BCIP/NBT, Fast Green FCF, Methylene Blue
Green	520–570 nm	Coomassie R-250, Basic Fuchsin
Blue	400–530 nm	Crocein Scarlet
Gray Scale	400–750 nm	Silver Stains, Copper Stains, Film

Fig. C-5. Examples of filter colors and applications.

Next to Light, select the appropriate light source.

- Reflective mode is for scanning opaque mediums such as dried gels on filter paper, arrays, or photographs. The scanning dimensions in this mode are 21 cm x 29 cm.
- Transmissive mode is for scanning transparent mediums such as films, gels, or slides. The scanning dimensions in this mode are 20 cm x 23 cm.

C.3. Step II. Select Scan Area

Preview Scan

Before selecting the particular area to scan, you can preview the entire scanning area to determine the exact position of your sample.

Click on Preview Scan. A quick, low-resolution scan of the entire scanning area will appear in the scanning window. Your sample should be visible within a portion of this scan.

Selecting an Area

Using the preview scan as a guide, select your scan area by dragging your mouse within the scanning window. The border of the scan area you are selecting will be marked by a frame.

Note: The scan area you select must be at least 1 cm wide.

When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning area box you have created, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). As you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

C.4. Step III. Select Resolution

To select from a list of possible scanner resolutions, click on the Select button under Select Resolution. This will open the Select Scan Resolution dialog box.

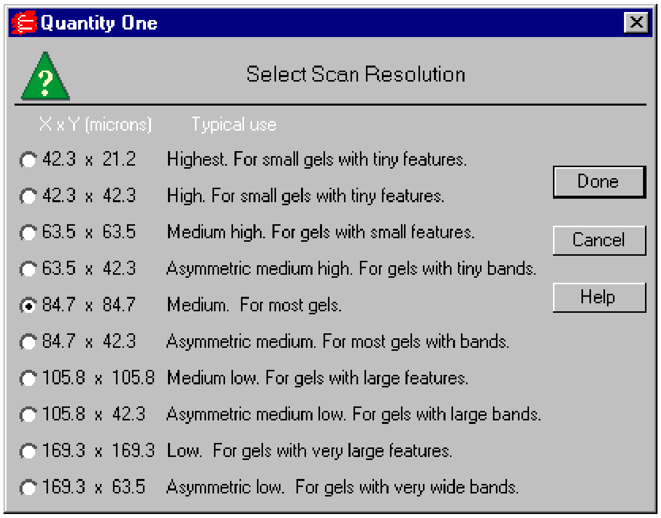

Fig. C-6. Select Scan Resolution dialog box.

Available resolutions are listed from highest to lowest in terms of the dimensions of the resulting pixels (in microns). Smaller pixels equal higher resolution. Each resolution is listed with its typical use.

In general, the size of your pixels should be one-tenth the height of your smallest object.

Some of the resolutions are asymmetrical, meaning that the resolution is higher in the vertical dimension (i.e., the pixels in the resulting image are larger in the horizontal dimension than in the vertical dimension). This is useful for gels with bands, where you are more interested in resolving in the y dimension to determine the size of bands and the spacing between them.

Specifying Your Own Resolution

If you select Oversample under More Options (see section C.7., Other Options, for details), you can specify your own resolution within the range of 43–169 microns (micrometers). With Oversample selected, enter values directly in the fields next to X resolution and Y resolution in the main acquisition window.

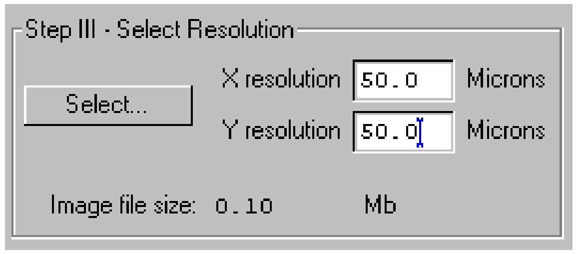

Fig. C-7. Entering a custom resolution (with Oversample selected).

Image File Size

Image File Size (under Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

C.5. Calibration

Images scanned using the GS-710 are automatically calibrated. The scanner plate has a built-in calibration strip for both transmissive and reflective scanning.

- The transmissive calibration strip calibrates the densitometer from 0 to 3.0 O.D.
- The reflective calibration strip calibrates the densitometer to 2.0 O.D.

The first time you use the GS-710, you must select some calibration settings to ensure that your calibration is accurate.

To set the automatic calibration settings, click on the More Options button in the GS-710 acquisition window. This will open the Densitometer Options dialog box.

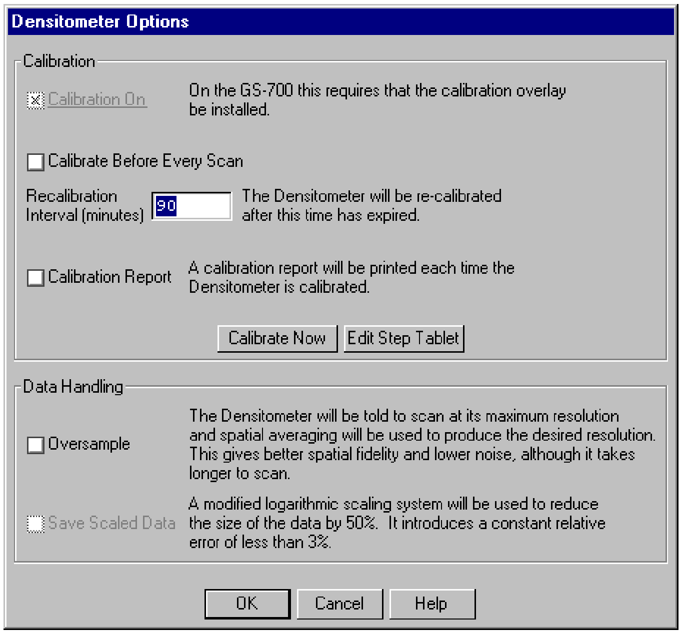

Fig. C-8. Densitometer Options dialog box.

C.5.a. Editing the Step Tablet

Before you scan for the first time, you must enter the step tablet values for your transmissive calibration strip into the Step Tablet Values form.

Note: In reflective scanning, the scanner is calibrated against the reflective step tablet. However, you do not need to enter the values for this step tablet; you can use the default values in the Step Tablet Values form.

Attached to the outside of your GS-710, you will find a copy of the manufacturer's printout of the diffuse density values for each segment of your transmissive step tablet, as well as a serial number that identifies the step tablet to which those density values belong. Those exact density values must be entered into the computer for the software to associate a correct density value with each step on the step tablet.

Note: Scanning in transmissive mode with incorrect step tablet values entered into the computer can cause significant errors in the reported densities of your scans.

To enter the density values for your step tablet, click on the Edit Step Tablet button in the Densitometer Options box. The Step Tablet Values dialog box for the GS-710 will open.

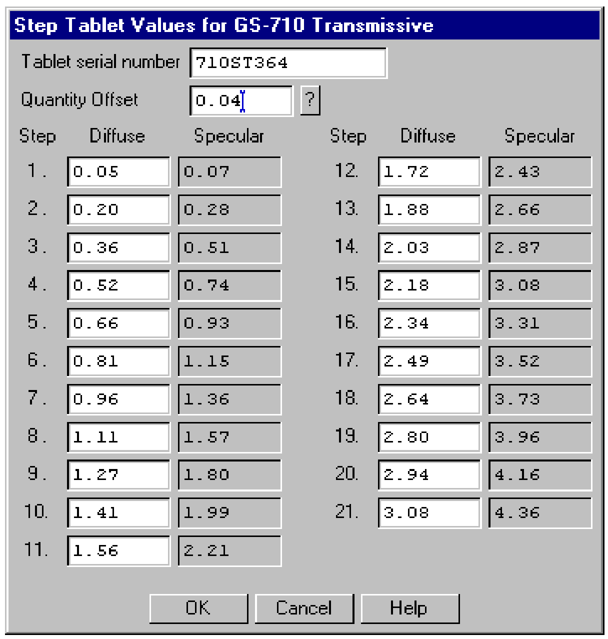

Fig. C-9. Step Tablet dialog box.

The dialog box will indicate whether the values are for the transmissive or reflective step tablet, depending on whether you are in transmissive or reflective mode.

If you are in reflective mode, you <u>do not</u> need to change the default step tablet values.

If you are in transmissive mode, you do need to enter the correct step tablet values that were shipped with your GS-710. You can type the serial number for these values into the Tablet Serial Number field.

The Quantity Offset field does not apply in the GS-710. This value should be set to zero.

Next, enter the values for the transmissive step tablet in the appropriate fields under the Diffuse column. After the step tablet is scanned, the software will associate each density value with its corresponding segment on the step tablet.

The step tablet density values do not need to be entered each time you calibrate. Once they have been entered and saved, they will be automatically recalled when the calibration strip is scanned.

When you are finished filling out the Step Tablet Values form, click on OK.

Diffuse Versus Specular O.D.

In the step tablet form, you enter O.D. as diffuse density, and then the software automatically calculates the specular density.

Specular density is a measure of the light that passes directly through a medium. Diffuse density includes light that is scattered as it passes through the medium. Step tablet values are given in diffuse density, but are measured by the scanner in specular density, and therefore must be converted according to the specular / diffuse O.D. ratio. This conversion does not affect quantitation.

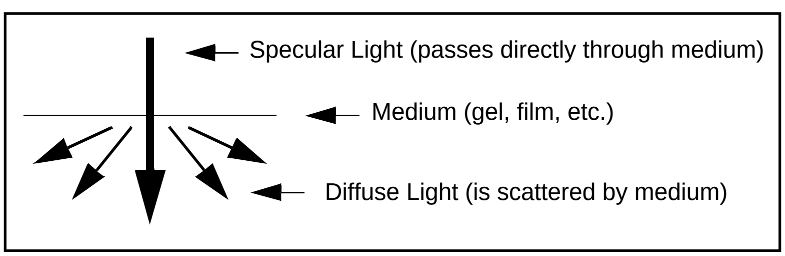

Fig. C-10. Specular and diffuse density

Diffuse density values are converted to specular optical density units according to the following formula:

$$Specular\ OD = 1.4\ \ Diffuse\ OD$$

C.5.b. Calibration Settings

After you have entered the step tablet values, you can immediately calibrate by clicking on the Calibrate Now button (in the Densitometer Options dialog box).

You can also specify how often you want the GS-710 to automatically recalibrate. Either click on the Calibrate Before Every Scan checkbox or enter a Recalibration Interval (in minutes) in the appropriate field.

Note: The scanner will automatically recalibrate each time you change your filter or your reflective / transmissive setting. (If you select a different application with the same filter and light settings, it will not auto recalibrate.)

Calibration Report

To print out a calibration report each time the densitometer calibrates, click on the Calibration Report checkbox.

C.6. Acquire the Image

To begin to scan, click on the Acquire button. The scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

After the scan is complete, a window will open displaying the scan image, at which point you can save it as a PDQUEST image.

Note: The image will open with a default file name that includes the date, time, and (if applicable) user name. However, unless you have selected Auto

Save After Scan, the file will not be saved until you select Save or Save As from the File menu.

C.7. Other Options

Oversample

This feature allows you to scan at the maximum resolution of the GS-710 (42.3 x 42.3 microns) and then use spatial averaging to create an image with lower resolution. This can result in better images at lower resolution—however, it takes longer to scan.

To turn on oversampling, click on the More Options button in the acquisition window and click on the Oversample checkbox.

With oversampling on, you can specify your own resolution within the range of 43–169 microns by entering values directly in the fields next to X resolution and Y resolution in the main acquisition window.

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

If you have checked Auto Save After Scan, you can also automatically create a backup copy of any scan you create.

Click on the Make Backup Copy checkbox. With this checkbox selected, when a scan is created and saved, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an “.sbk” extension. Macintosh backup files will have the word “backup” after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix D. Fluor-S MultiImager

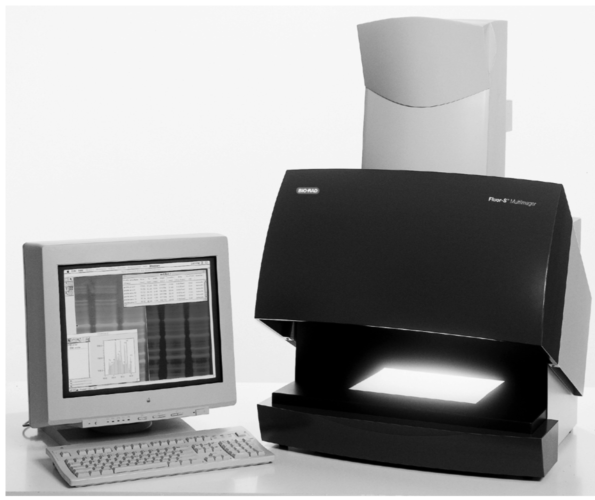

Fig. D-1. Fluor-S MultiImager.

Before you can begin acquiring images using the Fluor-S® MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S is green before attempting to capture an image. If you are using a PC, the Fluor-S should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

D.1. Opening the Fluor-S Acquisition Window

To acquire images using the Fluor-S, go to the File menu and select Fluor-S.... The acquisition window for the imager will open, displaying a control panel and a video display window.

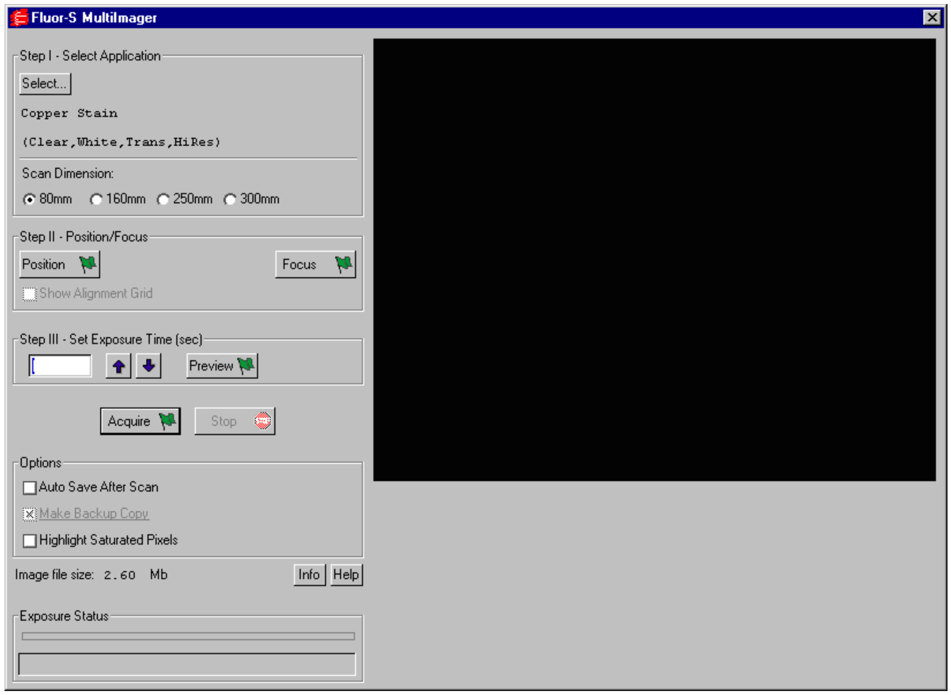

Fig. D-2. Fluor-S acquisition window

When the Fluor-S window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

D.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

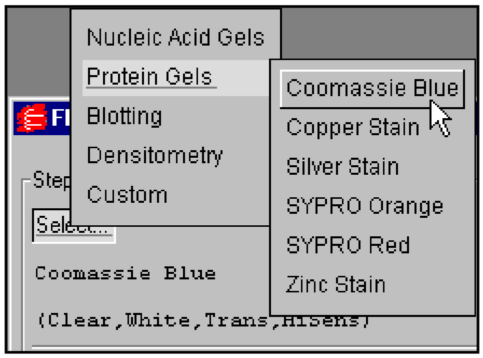

Fig. D-3. The application tree in the Fluor-S acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under "Blotting," you also have the option of selecting High Resolution or High Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

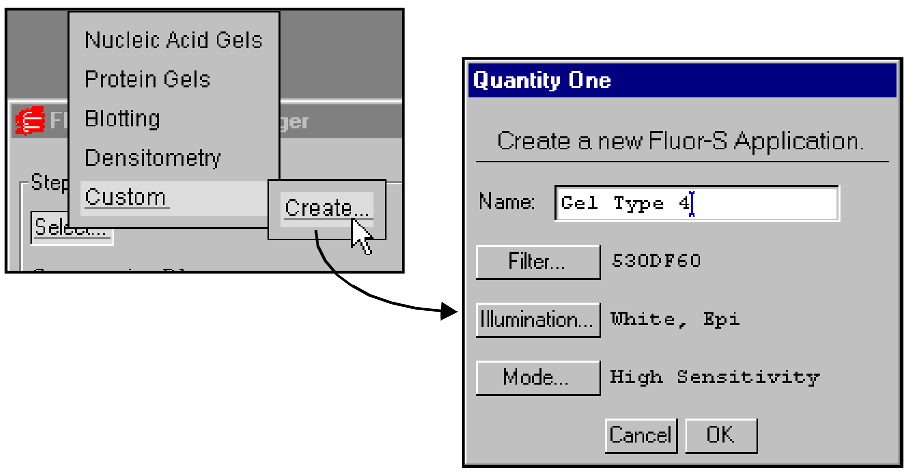

Fig. D-4. Creating a new custom application.

To select the filter (including user-defined), type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

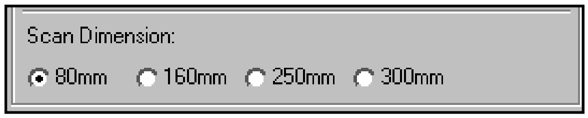

Fig. D-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Resolution Versus High Sensitivity

High Resolution and High Sensitivity are mutually exclusive options. High Resolution is the normal operating mode for the Fluor-S. High Sensitivity provides optimal sensitivity for low-light applications such as chemiluminescence.

In High Resolution mode, images are captured at the maximum resolution of the camera.

In High Sensitivity mode, the pixels in the camera are “binned” (i.e., four pixels are combined into one) to increase the amount of signal per pixel. However, combining the pixels results in a reduction in the resolution of the image.

D.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S box automatically turns on. To turn the light off while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The first step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

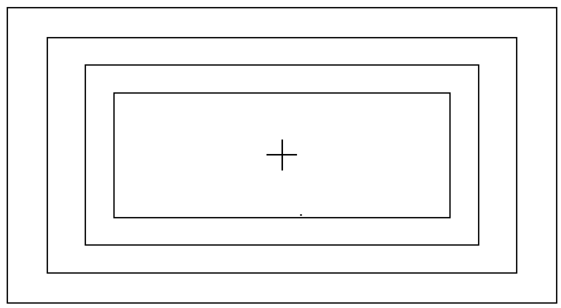

Fig. D-6. Fluor-S alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting. (See the following table on recommended f-stops.)

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

D.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of the image on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application. The table provides recommended exposure times for various applications.

Table 1: Recommended f-stops and Exposure Times

Application	f-stop[1] Zoom lens	f-stop[1] 50mm lens	Exposure time[2]
Ethidium bromide	f-4	f-2.8	5–30 sec.
SYBR Green	f-4	f-2.8	5–30 sec.
SYPRO Orange	f-4	f-2.8	5–30 sec.
Radiant Red	f-4	f-2.8	5–30 sec.
Fluorescein	f-4	f-2.8	30 sec.–2 min.
Texas Red	f-4	f-2.8	30 sec.–2 min.
Colorimetric Stains (e.g., Coomassie or Silver)	f-11	f-11	1–5 sec.
X-ray Film	f-11	f-11	1–5 sec.
Photographs	f-11	f-11	1–5 sec.
Chemi[3]		f-1.8	1–10 min.

[1]For sharper focusing, close the f-stop down 1–2 stops from full open while focusing.
[2]Increase exposure time two fold for each step increase in f-stop.
[3]For chemi applications, the 50mm lens is recommended. Always remove the 660 filter.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemiluminescent samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.

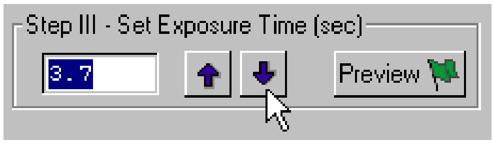

Fig. D-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a "dark" image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

D.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

PDQUEST will first acquire the image, and then acquire a "dark" image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

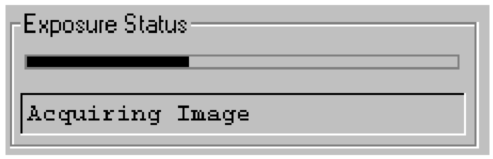

Fig. D-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using PDQUEST's menu and toolbar functions.

D.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field.

The dimensions of your imaging area are determined by how much you are zooming in or out on your sample. The dimensions determine the size of the pixels in your image (i.e., resolution). When you adjust one imaging area dimension, the other dimension will change to maintain the aspect ratio of the camera lens.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. It is also necessary if you are comparing the quantities of objects (e.g., using the Volume Tools) between images.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create. This size is determined by whether you selected High Resolution or High Sensitivity when you selected an application.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix E. Fluor-S MAX MultiImager

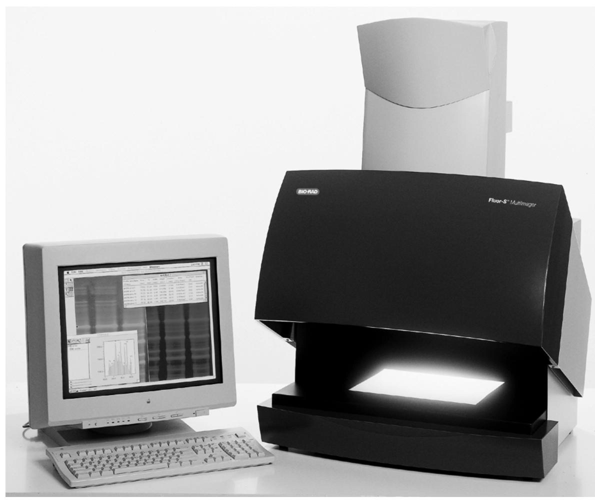

Fig. E-1. Fluor-S MAX MultiImager.

Before you can begin acquiring images using the Fluor-S® MAX MultiImager, the imaging system must be properly installed and connected with the host computer. See the Fluor-S MAX hardware manual for installation, startup, and operating instructions.

Note: Make sure that the temperature light on the Fluor-S MAX is green before attempting to capture an image. If you are using a PC, the Fluor-S MAX should be turned on and the initialization sequence completed *before* the host computer is turned on. See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Fluor-S MAX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Fluor-S MAX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in "simulation mode." In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create "dummy" images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

E.1. Opening the Fluor-S MAX Acquisition Window

To acquire images using the Fluor-S MAX, go to the File menu and select Fluor-S MAX.... The acquisition window for the imager will open, displaying a control panel and a video display window.

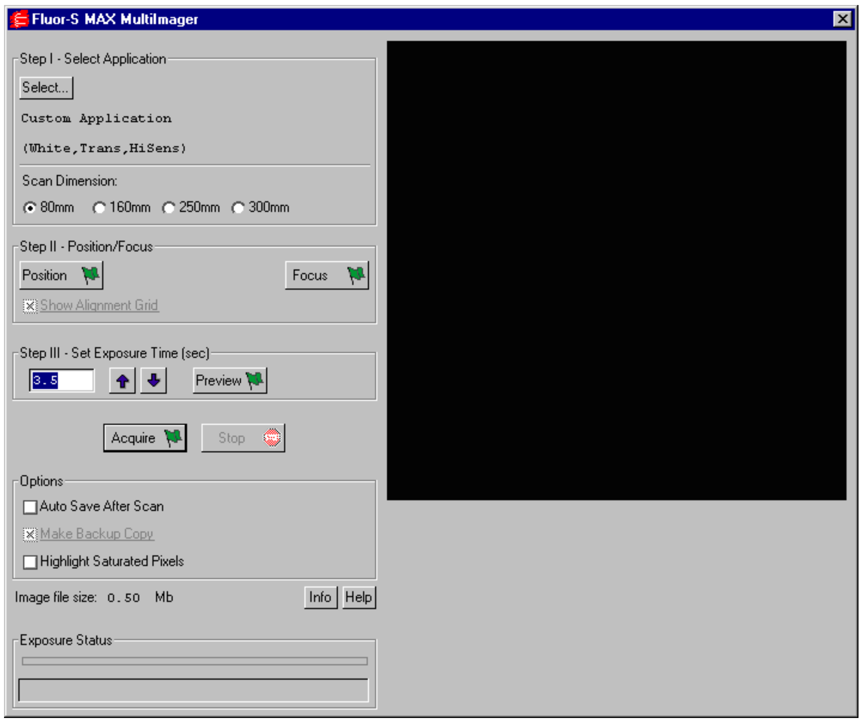

Fig. E-2. Fluor-S MAX acquisition window.

When the Fluor-S MAX window first opens, no image will be displayed.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to acquiring an image using the Fluor-S MAX:

1. Select the application.
2. Position and focus the gel or other object to be imaged.
3. Set the exposure time.
4. Acquire the image.

E.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application. The available applications and their associated settings are listed in a tree that expands from left to right.

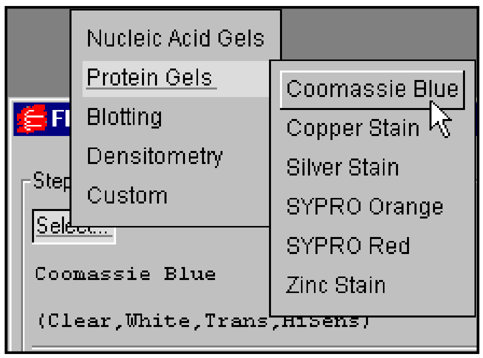

Fig. E-3. The application tree in the Fluor-S MAX acquisition window.

First select your general application, then select the particular stain or medium you are using. If you select one of the chemi applications under "Blotting," you also have the option of selecting High Sensitivity or Ultra Sensitivity (see below).

When you select an application, the software automatically sets the appropriate filter (520LP, 530DF60, 610LP, clear, or none) and light type (UV, white, or none) in the Fluor-S MAX for that particular application. With the exception of copper and zinc stains, the software also selects the light source (trans, epi, or neither). With copper and zinc stains, you are prompted to select Epi or Trans as the light source.

For applications involving trans illumination, you must also specify a scan dimension (see below).

Your selection will be displayed below the Select button. To exit the tree without selecting, press the ESC key.

Custom Applications

If your application is not listed and you know the filter and light settings you want to use, you can create and save your own custom application.

From the application tree, select Custom, then Create. This will open a dialog box in which you can name your application and select your settings.

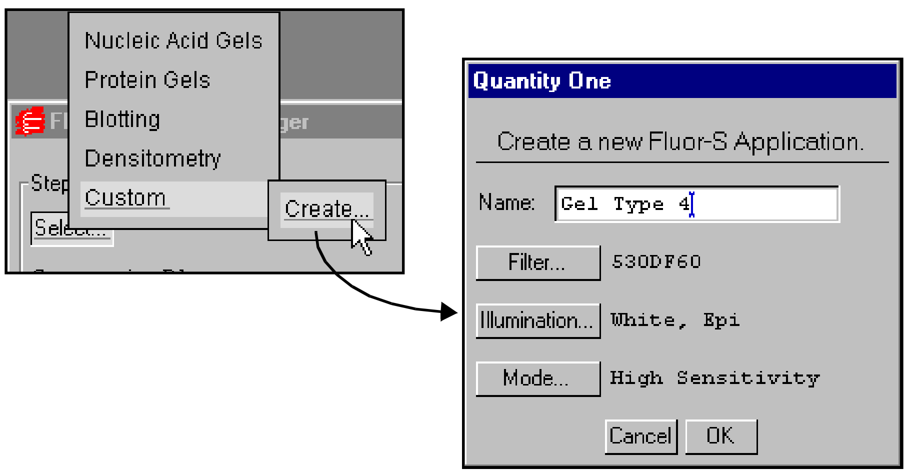

Fig. E-4. Creating a new custom application.

To select the filter, type of illumination, and camera mode, click on the appropriate buttons. Enter a name for your application in the Name field. Click on OK to implement your changes.

After you have created an application, you can select it from the application tree by selecting Custom and the name you created. You can delete the application by selecting Custom, Delete, and the name of the application.

Scan Dimension

If an application uses trans illumination, the Scan Dimension buttons become active. The scan dimension is the distance traveled by the transilluminating light source as it scans horizontally across the platen.

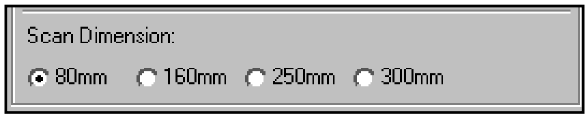

Fig. E-5. Scan dimension settings for trans illumination.

The full scanning range is 300 mm. Select a smaller range if your sample is small and you do not want to wait while the light source travels over the maximum scan width.

High Sensitivity and Ultra Sensitivity

High Sensitivity and Ultra Sensitivity are different camera modes. In Ultra Sensitivity mode, the Fluor-S MAX camera is cooled to –35.0 degrees C. In High Sensitivity, the camera is cooled to –20.0 degrees C.

High Sensitivity is the normal operating mode of the Fluor-S MAX. Ultra Sensitivity provides optimal sensitivity for low-light applications. It may be selected for the chemiluminescence and chemifluorescence applications, or for a custom application.

Note: When you change from High to Ultra Sensitivity or visa versa, there will be a delay of several minutes while the Fluor-S MAX camera cools down or warms up. If you attempt to acquire an image during this period, you will be notified of the changing temperature. If you do not want to wait, you can cancel the mode change.

E.3. Step II. Position/Focus

Note: When you click on the Position or Focus button, the light inside the Fluor-S MAX box automatically turns off. This is because if you focus with the camera at maximum aperture (see note below), leaving the light on would make it difficult to view the image. To turn the light on while positioning or focusing, hold down the SHIFT key when clicking on the button.

Position

The next step in acquiring an image is centering your gel or other object within the camera frame. To do this, click on the Position button. The Fluor-S MAX will begin capturing a “live” image and updating it every second.

With the Position button selected, look at the image in the acquisition window while you position your object in the center of the platen. If you have a zoom lens on the camera, you can adjust the magnification while you position.

You can select the Show Alignment Grid checkbox to facilitate positioning.

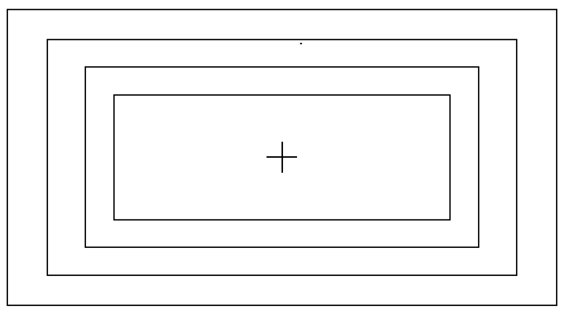

Fig. E-6. Fluor-S MAX alignment grid.

When you are finished positioning, click on the Stop button.

Focus

Note: Before focusing, you should adjust the f-stop on the camera to the lowest setting (i.e., the maximum aperture). This reduces the depth of field, allowing you to more accurately focus the camera. Then, after focusing, increase the f-stop to the desired setting.

After you have positioned your sample, click on the Focus button and look at the image in the acquisition window while aligning the two focusing arrows on the camera lens. While focusing, the camera will limit its focus to a small portion of the sample (this will not affect any zoom lens adjustments you may have made.)

When you are finished focusing, click on the Stop button.

E.4. Step III. Set Exposure Time

When you are ready to capture an image, you will need to select an exposure time. “Exposure” refers to the integration of image captures on the CCD over a set period of time. The effect is analogous to exposing photographic film to light.

The exposure time you select should be based on your application and your “best guess” as to what exposure will give you the best image.

Note: For most applications, you can select an exposure time, capture an image, study it, then adjust the exposure time accordingly. Repeat this procedure as many times as necessary to obtain a good image. For chemi samples, which degrade over time and emit low levels of light, select a high exposure time initially.

You can enter an exposure time (in seconds) directly in the field, or use the Arrow buttons to adjust the exposure time in 10 percent increments.

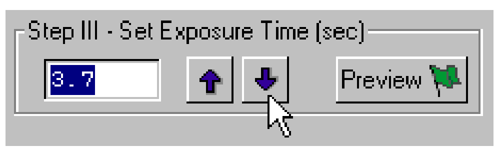

Fig. E-7. Selecting an exposure time.

Preview

For shorter exposures, you can use Preview to test different exposure times. Click on the Preview button create a preview exposure and display it in the acquisition window.

A preview scan takes only half as long to create as a real scan, because the preview scan does not capture a “dark” image (see below). The progress of the exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

You cannot save preview scans.

If you want to stop a preview scan that is in progress, click on the Stop button.

E.5. Acquire the Image

When you are ready to acquire an image, click on the Acquire button.

PDQUEST will first acquire the image, and then acquire a “dark” image for the purpose of filtering out dark noise. (Dark noise is a type of image noise that is generated by CCD cameras.)

The progress of each exposure will be displayed in the Exposure Status bar at the bottom of the dialog box.

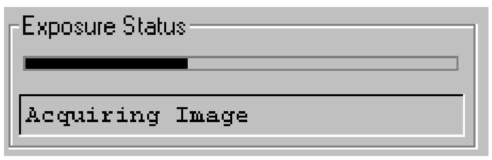

Fig. E-8. Exposure Status bar when acquiring an image.

The type of exposure (image or dark) that is being performed will be noted below the status bar.

Note: A dark image is only acquired the first time you perform a scan with particular application and exposure settings. If you perform subsequent scans with the same settings, no dark image will be acquired.

If you want to stop a scan that is in progress, click on the Stop button.

Saving the Image

After an image has been acquired, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using PDQUEST's menu and toolbar functions.

E.6. Options

Auto Save After Scan

To automatically save any image you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Imaging Area Size

The imaging area is the area of a sample that is captured by the camera and displayed in the video display window. To specify the size of your imaging area, click on the Info button and enter each dimension in the appropriate field. When you adjust one imaging area dimension, the other will change to maintain the aspect ratio of the camera lens.

The imaging area will change depending on your zoom factor. For example, if you have zoomed in on a sample that is 4.5 x 4.5 cm, then you would enter 4.5 for the width and height of your imaging area.

Note: Your imaging area settings must be correct if you want to do 1:1 printing. They must also be correct if you want to compare the quantities of objects (e.g., using the Volume Tools) between images.

The imaging area dimensions also determine the size of the pixels in your image (i.e., resolution). A smaller imaging area will result in a higher resolution.

File Size of Images

Image File Size (below Options) shows the size of the image file you are about to create.

If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to acquire an image. (Macintosh users can increase the application memory partition. See your Macintosh computer documentation for guidance.)

Appendix F. Personal Molecular Imager FX

Fig. F-1. Personal Molecular Imager FX

Before you can begin acquiring images using the Personal Molecular Imager® FX, the instrument must be properly installed and connected with the host computer. See the Personal FX hardware manual for installation, startup, and operating instructions.

Note: The Personal FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The Personal FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the Personal FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

F.1. Opening the Personal FX Acquisition Window

To acquire images using the Personal FX, go to the File menu and select Personal FX.... The acquisition window for the imager will open, displaying a control panel and the scanning area window.

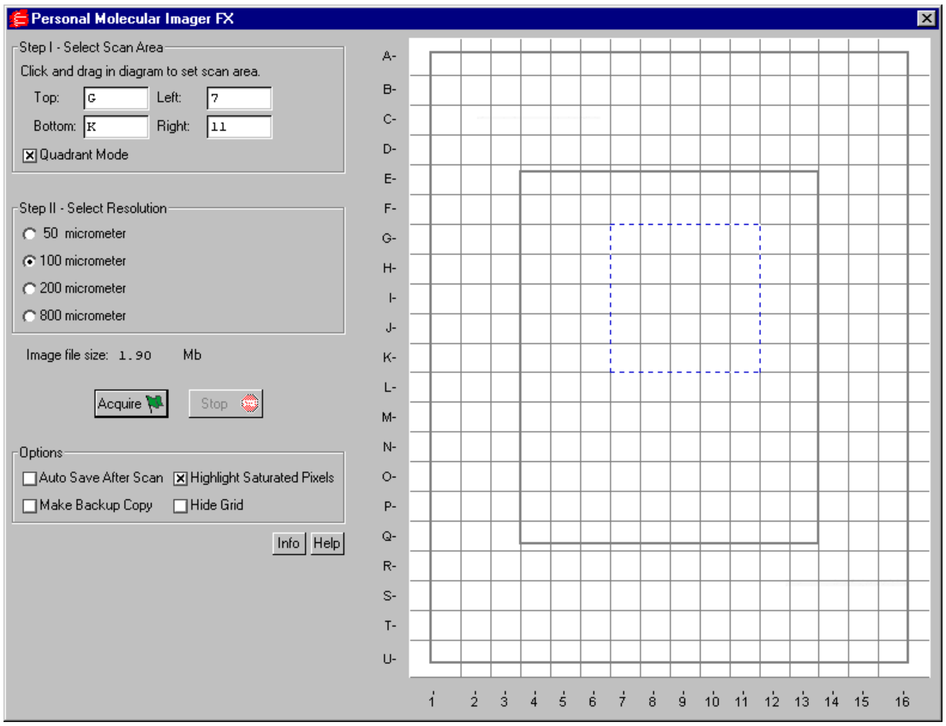

Fig. F-2. Personal FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the Personal FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are three basic steps to scanning an image using the Personal FX:

1. Select the scan area
2. Select the resolution
3. Acquire the image

F.2. Step 1. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

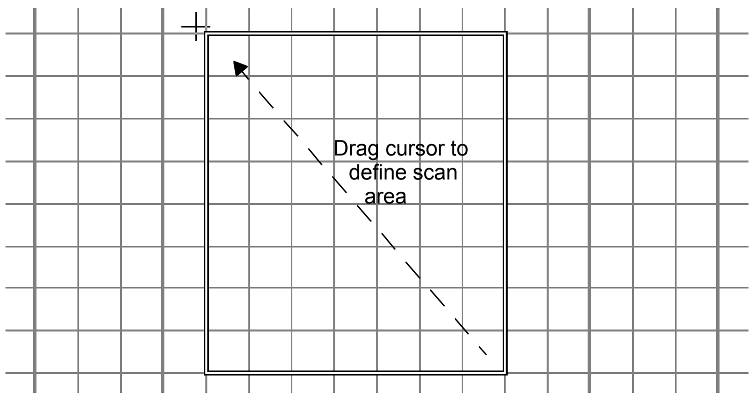

Fig. F-3. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.
- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

F.3. Step II. Select Resolution

The Personal FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

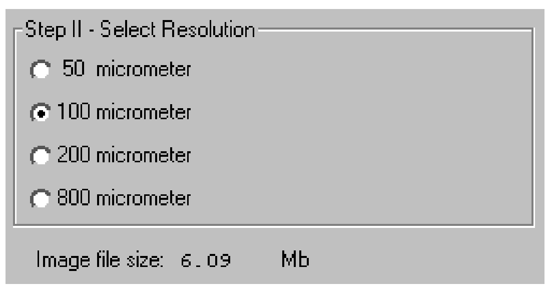

Fig. F-4. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution should be used for typical gels and arrays.
- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

F.4. Acquire the Image

Once you have selected your scan area and resolution, you ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using PDQUEST's menu and toolbar functions.

F.5. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix G. Molecular Imager FX

Fig. G-1. Molecular Imager FX

Before you can begin acquiring images using the Molecular Imager® FX, the instrument must be properly installed and connected with the host computer. See the FX hardware manual for installation, startup, and operating instructions.

Note: The FX should be turned on and the initialization sequence completed *before* the host computer is turned on (except in the case of certain Power Macintosh configurations). See the hardware manual for more details.

PC Only: A Note About SCSI Cards

The FX is connected to your computer by a Small Computer System Interface (SCSI) cable. To use the FX, you must have a SCSI card installed in your PC. If you have an older PC, you may also need to load the SCSI and WinASPI drivers that came with your card. Newer releases of Windows 95 and Windows NT will already have those drivers installed.

Simulation Mode

Any of the imaging device acquisition windows in PDQUEST can be opened in “simulation mode.” In this mode, an acquisition window will open and the controls will appear active, but instead of capturing real images, the window will create “dummy” images of manufactured data.

You do not need to be connected to an imaging device to open a simulated acquisition window. This is useful for demonstration purposes or practice scans.

To enter simulation mode, make sure you are not connected to the imaging device, then select its name from the File menu. You will be asked if you want to open the simulated window. The title of the window will indicate that it is simulated.

Note: There is no simulated calibration for the GS-700 and GS-710 Imaging Densitometers.

G.1. Opening the FX Control Panel

To acquire images using the FX, go to the File menu and select FX.... The acquisition window for the imager will open, displaying the control panel for the imager and the scanning area window.

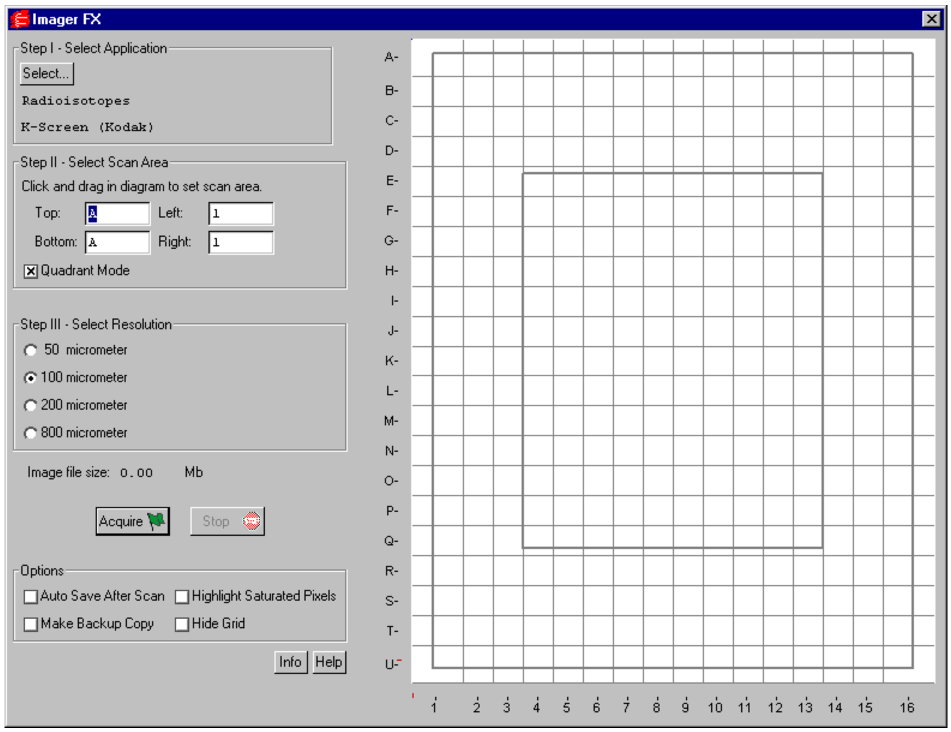

Fig. G-2. FX acquisition window

The default scanning window is marked by grid lines that divide the area into quadrants. There is also an outer box and inner box marked by thicker lines. This conforms to the sample pad for the standard Bio-Rad Exposure Cassette that is supplied with the FX. The quadrants are numbered 1–16 left to right and lettered A–U top to bottom.

If you prefer a scanning window measured in centimeters, deselect the Quadrant Mode checkbox in the control panel by clicking on it. To hide the gridlines, click on the Hide Grid checkbox under Options.

The control panel has been arranged from top to bottom to guide you through the acquisition procedure. There are four basic steps to scanning an image using the FX:

1. Select the application.
2. Select the scan area.
3. Select the resolution.
4. Acquire the image.

G.2. Step 1. Select Application

To set the appropriate filter and other parameters for the type of object you are imaging, click on the Select button under Select Application.

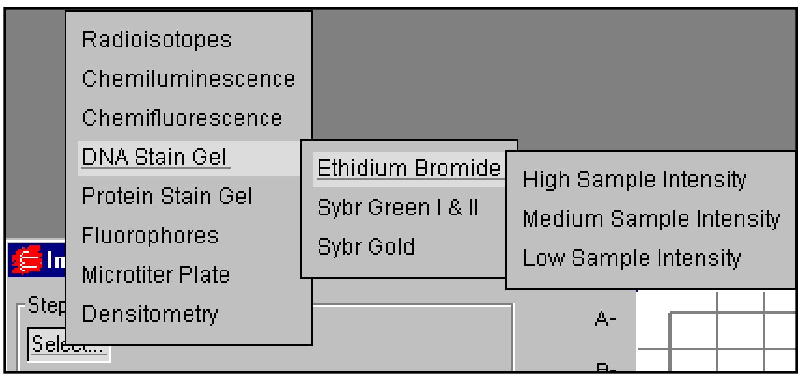

Fig. G-3. Example of an application tree: Ethidium Bromide gel.

The applications and associated settings are listed in a tree that expands from left to right. When you select an application, the software automatically sets the appropriate filter(s) in the FX for that particular application.

First you select the category of your application, next you select your particular application, and finally (if appropriate) you select the intensity of your samples.

To exit the tree without selecting, press the ESC key.

Your selection will be displayed below the Select button.

- Radioisotopes
  - CS- or BI-Screen (Bio-Rad)
  - K-Screen (Kodak)
  - Fuji-Screen
- Chemiluminescence
  - Chemi-Screen (Bio-Rad)
- Chemifluorescence
  - Attophos
- DNA Stain Gel
  - Ethidium Bromide
  - Sybr Green I & II
  - Sybr Gold
- Protein Stain Gel
  - Sypro Orange
  - Sypro Red
  - Nile Red
- Fluorophores
  - FITC
  - R6G
  - TAMRA
  - CY3
  - Texas Red
- Microtiter Plate
  - DNA (Sybr Green I)
  - Protein (Nano Orange)
  - ssDNA (Oligreen)
  - DNA (Picogreen)
  - B-Gal (FDG)
  - GUS (FDG)
- Densitometry
  - Coomassie Blue Gel/Blot
  - Copper Stain Gel/Blot
  - Silver Stain Gel/Blot
  - X-Ray Film (Grey Type)

Fig. G-4. Applications available in the FX acquisition window.

Sample Intensity

When selecting any application except radioisotopes, you will need to specify a sample intensity. This is simply a rough estimate of how much sample is apparent in your gel or other object.

Do not worry if you are not sure about the level of intensity of your sample—if you make a mistake, you can always change the setting and capture another image.

For example, if you specify a low sample intensity and the resulting image has too many saturated pixels, you will receive a warning message. Simply change the setting to medium sample intensity and rescan. If you specify a high sample intensity and the resulting image is too faint, select medium or low sample intensity and rescan.

G.3. Step II. Select Scan Area

To select a scan area, drag your mouse within the scanning window. (In the scanning window, your cursor appearance will change to a cross.) The border of the scan area you are selecting is marked by a frame.

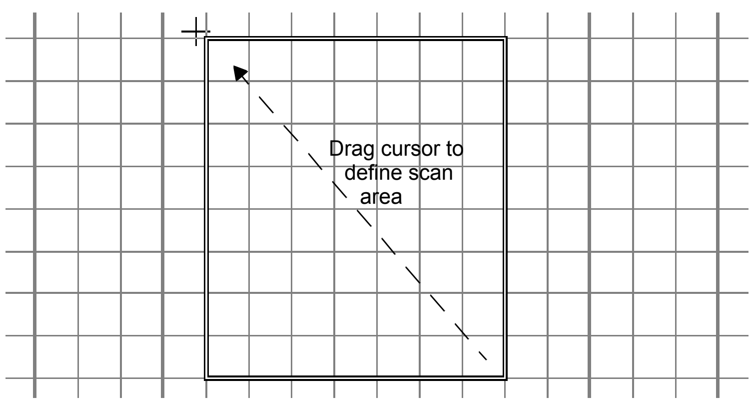

Fig. G-5. Selecting a scan area.

If you are in quadrant mode, note that the frame “locks” onto the next quadrant as you drag. When you release the mouse button, the border changes to a dashed blue line, indicating a selected area.

- To *reposition* the scanning box you have selected, position your cursor inside the box and drag. The entire box will move.

- To *resize* the box, position your cursor on a box side and drag. The side you have selected will move.
- To *redo* the box entirely, position your cursor outside the box and drag. The old box will disappear and a new box will be created.

You can also select the scanning area by entering coordinates in the appropriate fields (Top, Bottom, Left, Right). After you enter a coordinate, the position of the scanning area box will change accordingly.

When selecting, be sure to include the entire area of interest, and be generous with borders. You can always crop the image later.

G.4. Step III. Select Resolution

The FX acquisition window allows you to scan at 50, 100, 200, or 800 micrometers. These resolutions are listed as option buttons in the control panel.

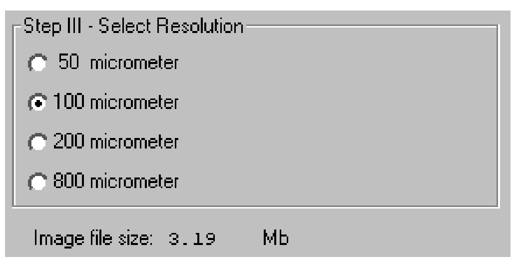

Fig. G-6. Resolution option buttons.

The resolution you select should be based on the size of the objects (e.g., bands, spots) you are interested in. For example:

- 50 micrometer resolution should be reserved for images requiring the highest level of detail, e.g., high density in situ samples, 1,536-well microplates, high density arrays, samples with very closely spaced bands. Files scanned at 50 micrometers can be very large.
- 100 micrometer resolution is useful for typical gels and arrays.

- 200 micrometer resolution is useful for gels with large bands and dot blots.
- 800 micrometer resolution should be reserved for very large objects, such as CAT assays.

File Size of Images

Image File Size (below Select Resolution) shows the size of the scan file you are about to create. If you do not have enough computer memory for the specified file size, an error message will appear when you attempt to scan. If this happens, select a lower resolution or decrease the size of the area to be scanned. (Macintosh users can also increase the application memory partition. See your Macintosh computer documentation for guidance.)

G.5. Acquire the Image

Once you have selected your application, scan area, and resolution, you are ready to acquire an image.

Click on the Acquire button. There may be a short delay while the image laser warms up; then the scanned image will begin to appear in the scanning window, line by line.

To interrupt a scan, click on the Stop button. A message will ask you to confirm the interrupt, and then you will be asked if you want to keep the partial scan. This feature is useful if you overestimated the size of the area you selected.

Note: If the image you are scanning has more than 10 saturated pixels, you will receive a warning message. If this happens, you can go back and select a higher sample intensity in the application tree.

Saving the Image

After the scan is complete, a message will appear asking you if you want to keep the scan. If you select Yes, a separate window will pop up containing the new image. The image will have a default file name that includes the date, time, and user (if known). To save the image, select Save or Save As from the File menu.

You can then analyze the image using PDQUEST's menu and toolbar functions.

G.6. Options

Auto Save After Scan

To automatically save any scan you create, click on the Auto Save After Scan checkbox.

With this checkbox selected, when you click on Acquire, a Save As dialog box will open asking you to specify a file name and location for the image you are about to create. The scan will begin when you click on the Save button.

Make Backup Copy

You can automatically create a backup copy of any scan you create. To do so, first select Auto Save After Scan (see above), then select the Make Backup Copy checkbox.

With this checkbox selected, when you save a scan, a backup copy will be placed in the same directory as the scanned image. Windows backup files will have an ".sbk" extension. Macintosh backup files will have the word "backup" after the file name.

This backup copy will be read-only, which means that you cannot make changes to it. You can open it like a normal file, but you must save it under a different file name before editing the image or performing analysis.

Highlight Saturated Pixels

When this is checked, any saturated pixels in the image will appear marked in red.

Hide Grid

To hide the gridlines in the scanning area window, click on the Hide Grid checkbox.

Appendix H. Keyboard Functions

Keyboard Layout

Key	Unmodified	Shift
Left	Pan Left	All Pan Left
Up	Pan Up	All Pan Up
Right	Pan Right	All Pan Right
Down	Pan Down	All Pan Down
Page Up	Pan Up	Graph Retreat
Page Down	Pan Down	Graph Advance
F1	View Entire Image	All Entire Images
F2	Zoom Box	Grab
F3	At Cursor	All Same
F4	Hide Overlays	Hide All Overlays
F5	Spot Crosshairs	All Spot Crosshairs
F6	Spot Parameters	Quantity Table
F7	Match Symbols	Match Offsets
F8	Mark Landmarks	Mark Unmatched Spots
F9	Show Match	Find Spot
F10	Pop-up Graph	Margin Graph
F11	Interchange Images	Interchange All Images
F12	--	Save All
F11	--	--
F12	--	--

Keyboard Layout

Key	Alternate	Control
F1	Configure Sub-Windows...	Configure Sub-Windows...
F2	Zoom In	Zoom Out
F3	All At Cursor	Previous Image
F4	Close	Auto-assign
F5	Spot Ellipses	All Spot Ellipses
F6	Spot Quantity	Quantitation Mode
F7	Saturated Spots	Cancelled Spots
F8	Mark Added Spots	Combined Spots
F9	Display SSPs	Plot SSP Grid
F10	Graph Advance	Graph Retreat
F11	--	--
F12	--	--
F11	--	--
F12	--	--
,	Graph Retreat	--
.	Graph Advance	--
A	--	Assign
B	--	Transform...
G	--	Grab
H	--	Help Contents and Index
I	--	File Info...
O	--	Load...
P	--	Print Image...
Q	--	Exit
S	--	Save
T	--	Plot Cross-section
U	--	Unload
V	--	About PDQUEST

Appendix I. Techniques

The techniques described in this appendix apply to detecting spots and normalizing for quantity among matchset members.

I.1. Background Normalization

Background normalization (under Edit Auto-Detect Parameters) is a useful method for correcting the background of calibrated gels which have been exposed to some source of radiation outside the gel. Applying calibration to such an image gives incorrect results, since calibration establishes a fixed relationship between film density and counts. Background normalization uses a ‘rolling ball’ algorithm to determine the background of the image of both the 2-D scan and the calstrip scan and uses that information to correct both images to have the same background value everywhere. This ensures that the calibration curve is valid everywhere on the image.

An example: once some ^{35}S-labelled methionine was spilled in our freezer, causing the side of the film closest to it to be darker than the size which was farther away. The spilled methionine caused the background of the left edge of the film to be 0.05 O.D. greater than that of the right edge of the film, with a continuous variation between those extremes across the film. Since the lightest segments of the calstrip were toward the right edge of the film, the background 0.0 DPM level was set at an O.D. value below that of background at the right edge. Applying the calibration curve to the whole film caused the entire left side of the film to become darker and many spots to be detected where there were none. Applying normalization to this image causes the dark-to-light variation from left-to-right to disappear.

Background normalization takes considerable computation and will add several minutes to spot detection. Also, on films with large areas of saturation or large areas of elevated background, it will introduce artifacts which are worse than the erroneous calibration. Therefore, it must be used with discretion.

I.2. Calibration and Spot Detection for 8-bit Scanners

When using a scanner or imager that produces 8-bit images (e.g., the Gel Doc), detection of spots can vary on the same 2-D scan depending on whether or not it is calibrated. The following is a discussion of calibration and spot detection's dependency on it.

Note: Calibration generally is not required for scanners or imagers that produce 16-bit images.

I.2.a. Calibration

Calibration consists of the following steps: determine the table of values relating film density to counts, calculate a curve which maps every possible density to a count value, translate the 2-D scan using the calibration curve.

Conditions

Calibration is an attempt to compensate for the nonlinear response of the film to exposing radiation. The calstrip must undergo the same processing as its associated film since the film's response curve and background noise are dependent on the film processing. If processed correctly, X-ray film will not saturate until over 2 O.D., although the top of the linear range of its response is at approximately 2 O.D. ,which is why we recommend that as a ceiling.

Density patches on the calstrip must have uniform counts over their entire area.

Determine the table of values relating density to counts

Density is a value from 0–255 (8-bit binary number). When scanning, the user selects the O.D. value to which 255 corresponds. Typically this is 2 O.D. Each patch on the calibration strip is averaged within the specified box to obtain a density value for that patch. The counts are obtained from a database which the user must enter for each batch of calstrips.

Calculate a curve which maps density to counts

The lowest point of the curve relates background density to 0.0 counts. Background is obtained from the calstrip scan by finding the mode of density values which are below the mean of all density values.

A linear interpolation is performed between adjacent sample points to obtain values there. If the density value of any segment is within 1% of 255, it is considered saturated. The highest two nonsaturated samples are used to extrapolate the curve to saturation. This establishes the effective counts at saturation.

If the slope of a curve segment falls below 10% of the highest value ever seen, saturation is set at the sample point at the lower end of the offending segment. This is only necessary when the film response decreases below 2 O.D. This normally indicates a failure of film processing. The user should diagnose the problem in the film exposure/development process and reprocess the gel if possible. The linearity of the film response has been compromised.

Translate the 2-D scan

Every pixel in the 2-D scan file is mapped to a count value using the calibration table of interpolated and extrapolated values.

A scale factor is calculated which attempts to map 0–2 O.D. to the first 255 steps in the range. This is to accommodate the spot detection slope detector discussed below. This is obtained in the following way:

The average slope of the curve is obtained by a least squares regression of the nonsaturated calibration samples.

The scale factor is calculated from the following:

hgt_to_den = (255 * avg_slope * max_OD) / (2.0 * pix_at_max_OD – bkgd_dens * max_OD).

This scale factor is applied to the mapped values above.

I.2.b. Spot detection slope detector

A slope threshold technique is used to recognize spot edges. Slope values are obtained as Y/X where Y is density (uncalibrated) or counts (calibrated) and X is distance between scanner pixels in 0.1 mm units.

When using uncalibrated data, the data are scaled so that 0–255 represents 0–2 O.D. to give uniform results when detecting different images.

When using calibrated data, the equivalent scaling is difficult to obtain due the nonlinear response of the film. A linear regression through the density-counts sample points is used to approximate this relationship.

I.3. Auto-Detect Parameters

Spot Detection is a series of steps which results in a list of spots which have been located and quantified. Due to the imperfections in gel and imaging technology, it is impossible to construct a single series of steps which optimally performs this task on all images. Therefore, the facility has been added to PDQUEST to allow a user to specify what processing steps should be performed by autodetection.

An image is obtained from an imaging device. This image is referred to as a 2-D scan. If available, calibration information is applied to the 2-D scan or in the case of multiple exposures of the same gel, the 2-D scans. The 2-D scan or scans are then merged to produce the Gel Image. This image includes all information available from all the 2-D scans on a single gel, hence the name Gel Image. Image processing to remove streaks and background is performed on the Gel Image. Spot detection locates spots on the image and spot fitting quantifies them. The resulting quantified spots are stored in a Gel Spots file.

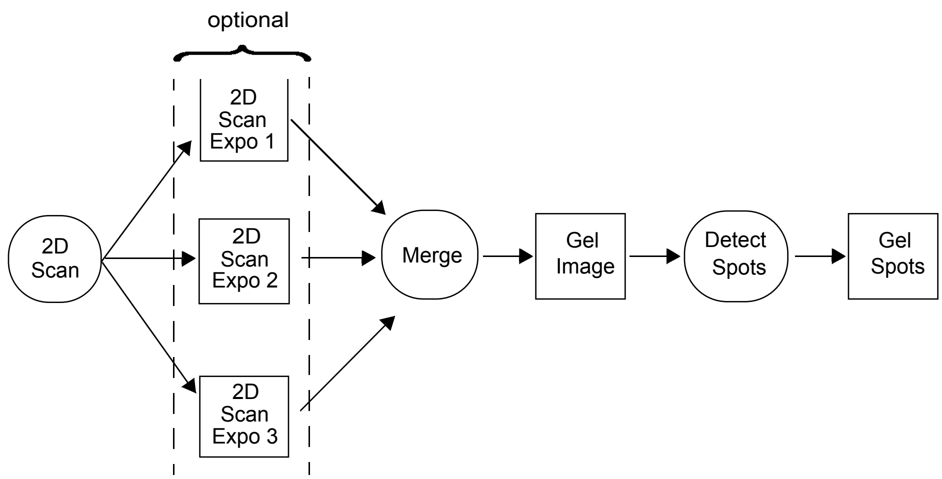

Fig. I-1. Auto-Detect procedure.

I.3.a. Auto-Detect Glossary

2-D scan —The image file produced by scanning.

Background —Film fog and unfocused protein on the image.

Background Normalization —The process that makes the background of the gel scan and the calstrip scan agree over the entire image.

Calstrip —Either a calibration strip or the scan of a calibration strip

Gel Image —The image file produced by calibrating and merging scans.

Gel Spots —The list of spots from a single gel.

Radius —The radius of a disk used to determine background for the purpose of background subtraction or background normalization. Its size is a number of pixels in the image. Typical values are 50–150 for gels and over 800 for calstrips.

Spot Detection —The process that locates spots on the Gel Image.

Spot Fitting —The process that finds the optimal gaussian parameters in X and Y to represent the spots found by spot detection.

Streak —A protein that has focused only in one dimension.

I.3.b. Scan Processing Parameters

Averaging Smooth

This is a procedure which will slightly reduce high-frequency noise or small jagged features. It is useful after merging since merging artifacts will sometimes cause the detection of too many spots.

Median Smooth

This removes ‘shot’ noise from the image. It is used when the scanning device is likely to introduce single-pixel errors. This is largely true only of CCD or photodiode-array detectors. You can recognize the need for this type of smooth if the scan at very high magnification shows single-pixel values which deviate significantly from the neighboring pixels.

Fourier Smooth

Fourier smoothing is a process that will remove small features on the gel (like noise) with little effect on larger ones. To apply this operation, you need to enter a “smoothing window” size. The smoothing window moves over the image, averaging the intensities of the pixels within it according to a specified formula.

You set the window size by entering its dimensions in terms of numbers of pixels in the X and Y directions. The larger the window, the greater the effect on the image. A typical window size is 2–4 pixels (square). A window size of less than 1 will not change your image. A window size of 5 or greater can remove features that you do not want to lose.

Since this smooth requires a relatively large amount of computing time and can significantly alter the image, it should be used only when clearly needed. Images from a phosphor-imager are often candidates for Fourier Smoothing, since the imaging plates frequently collect significant background radiation.

Normalize Background

Prior to calibration, this process determines the background image for a scan and for its calstrip and adjusts both images so that the background is the same over the extent of the images. Since the calibration curve establishes a fixed relationship between density and DPM/area, it is important that the background everywhere on the film be constrained to be that of the calstrip. This function should only be used if the films are calibrated. Its use is not advised if there are significant regions of streaking. The affect of normalization should be subtle. If you observe significant features (large areas of streaking) being removed by normalization, either try increasing the radius used to detect background, or do not perform normalization.

Convert 2-D Scan to Gel Image

This is either Merge Exposures if there is calibration information, or Make OD Gel if there is none. The system determines this automatically.

I.3.c. Gel Image Processing Parameters

Averaging Smooth

The same as above. Perform at least one pass of this after merging.

Median Smooth

The same as above. Not normally used on Gel Images.

Fourier Smooth

The same as above. Used after merging on very noisy images. A window value of 3 to 5 is typical.

Remove Background

Removal of background, vertical and horizontal streaks uses a process called Rolling Disk Background Subtraction. The smaller you set the radius of the disk, the more density will be detected as background.

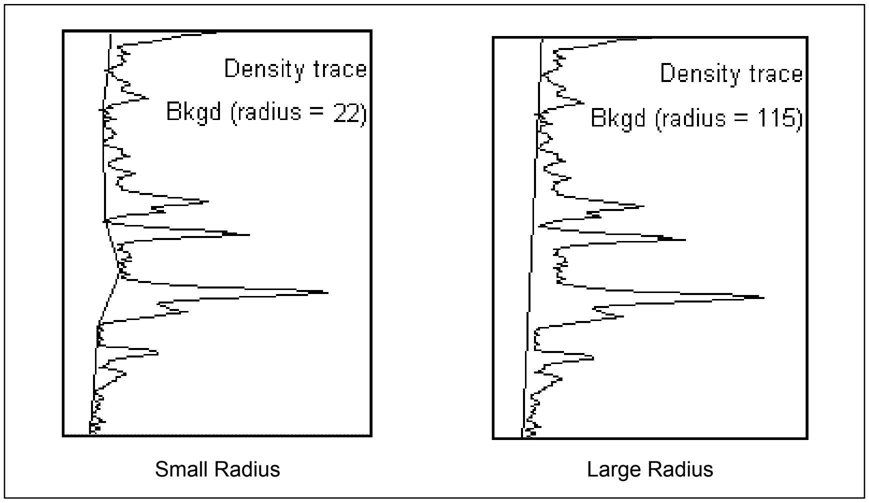

Fig. I-2. Different amounts of background removed by rolling disk background subtraction using a small disk radius as compared to a large disk radius.

Use the Plot Bkgd Trace function (Quantify > Manual Spot Detection > Subtract Background > Plot Bkgd Trace) to observe the baseline established with the current disk radius. Change the radius and replot the background trace to observe the effect of changing the radius.

The radius should be set large enough so that the baseline does not extend up into spots. Use Plot Bkgd Trace over the largest object on the darkest exposure you want to retain and find the lowest disk value which gives a proper baseline. Typical values are from 50 to 150.

Remove Vertical Streaks

This removes both background and vertical streaks. Note that the film must be carefully aligned when scanning since the streaks must be truly vertical to be cleanly removed.

Remove Horizontal Streaks

This removes both background and horizontal streaks. Note that the film must be carefully aligned when scanning since the streaks must be truly horizontal to be cleanly removed.

Detect Spots

This makes a list of the position of all spots in the image. The three sensitivity levels adjust both the density and size of spots which will be recognized.

I.3.d. Fit All Spots

This quantifies all spots found in the image using two-dimensional Gaussian fitting.

I.3.e. Correct Faint Spots

This corrects spots below a certain density over background (0.2 OD) by making their X and Y sigma values the same as those of the spots in the same Y-coordinate region of the film. This overcomes the difficulty of obtaining an accurate fit when very few pixels are available.

I.3.f. Process 2-D Scan vs. Gel Image

As a general rule, the scans should be processed as little as possible while still providing accurate merging. All streak removal, background subtraction, and aggressive smoothing should be performed on the Gel Image prior to spot detection. This is provided as a 'rule of thumb' and not gospel.

I.3.g. Examples of Auto-Detect Parameter Settings

Default Processing

1. Scan Processing

   Averaging Smooth
2. Convert to Gel Image
3. Gel Image Processing

Remove Vertical Streaks

Averaging Smooth

4. Detect Spots (Medium)
5. Fit All Spots
6. Correct Faint Spots

The scans are smoothed as a matter of course. The averaging smooth changes the image very little and improves the accuracy of merging and spot detection.

Vertical Streaks are removed since they are common gel artifacts and cause erroneous quantitation of spot buried in the streaks.

Another averaging smooth is preformed to suppress the detection of superfluous spots.

The spots are detected, fit (quantified), and the faint spots are corrected.

Clean, calibrated ^{35}S films

1. Scan Processing

   Averaging Smooth

   Background Normalization

2. Convert to Gel Image
3. Gel Image Processing

   Remove Vertical Streaks

   Averaging Smooth

4. Detect Spots
5. Fit All Spots
6. Correct Faint Spots

This is the same as the default example, except for the addition of Background Normalization.

Noisy, streaky calibrated ^{35}S films

1. Scan Processing

   Averaging Smooth

   Normalize Background (only if image is not damaged by this)
2. Convert to Gel Image
3. Gel Image Processing

   Remove Vertical Streaks

   Remove Horizontal Streaks

   Fourier Smooth (4, 4)
4. Detect Spots (Low or Medium)
5. Fit All Spots
6. Correct Faint Spots

This adds vertical and horizontal streak removal and a fourier smooth pass to suppress detection of superfluous spots. If too many spots are still found, you can either increase the fourier window or decrease the sensitivity of spot detection.

Note that background normalization may damage the scans and must be run manually before setting-up the auto-detect parameters to determine whether this is the case.

Uncalibrated Clean Images

1. Scan Processing

   Averaging Smooth
2. Convert to Gel Image
3. Gel Image Processing

   Remove Vertical Streaks

   Averaging Smooth
4. Detect Spots (Medium or High)

5. Fit All Spots

6. Correct Faint Spots

This is the default procedure file.

Uncalibrated noisy images

1. Scan Processing

    Median smooth

    Averaging smooth

2. Convert to Gel Image

3. Gel Image Processing

    Remove Vertical Streaks

    Remove Horizontal Streaks

    Fourier Smooth (4, 4)

4. Detect Spots (Medium or Low)

5. Fit All Spots

6. Correct Faint Spots

This adds a pass of median smoothing right up front to clean up the scans. Vertical and horizontal streaks are removed, followed by a fourier smooth, spot detection, and fitting.

I.4. Rolling Disk Background Determination

Determining background is a difficult problem, given that a typical user is not willing to wait very long for the answer and different samples exhibit different background characteristics. Also, a related problem is that of streak removal, which ideally is accomplished at the same time.

The method we use for background subtraction and streak removal is based on a simple geometric model that is efficient enough to be performed in a relatively short time.

Consider the following diagram. The density trace represents O.D. values taken from a single vertical column of pixels in the image. (or several columns averaged together). A disk of radius ‘r’ is used to obtain ‘touch points’ on the density trace. These ‘touch points’ are connected by straight line segments to obtain the background trace. The touch points are defined as those points on the trace where the topmost point of the circle can touch without any other part of the circle intersecting the density trace.

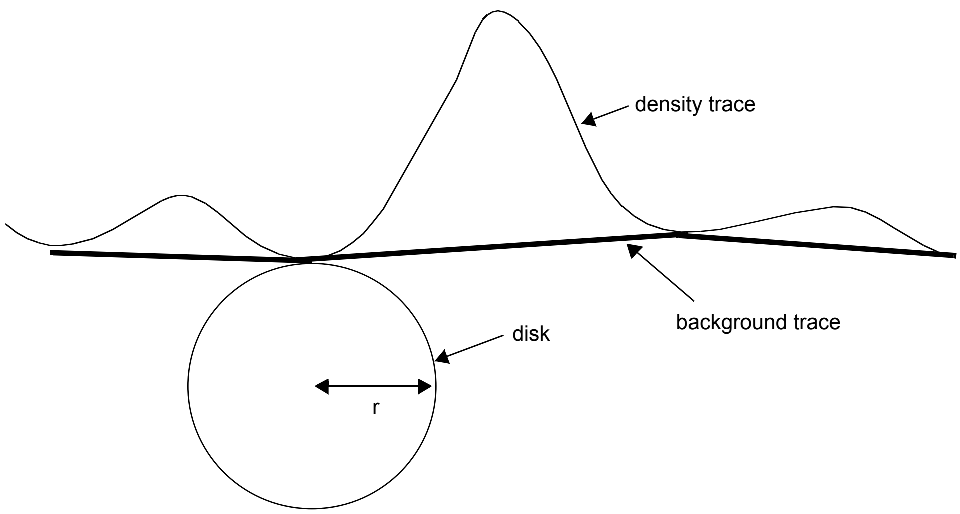

Fig. I-3. Rolling Disk background determination

Using a vertically-oriented disk results in vertical streak removal since the disk is able to contact the underside of the streak over its length.

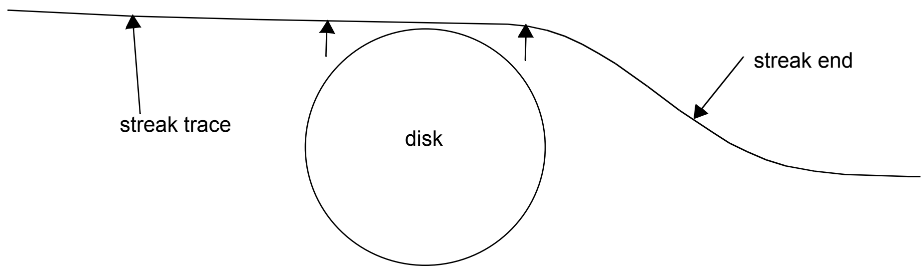

Fig. I-4. Disk contacting underside of streak trace

The background trace is nearly identical to the streak trace. There are some artifacts where the streak ends which usually do not introduce new spots into the Gel Spots file. The user must be aware that this may occur. If the streaks fade gradually the artifacts will not be present. If the streaks are saturated this often introduces pronounced artifacts around the saturated areas. The user must remove any spots detected in this region.

There is some disagreement about the validity of streak removal. Most researchers we have consulted share the opinion that the protein causing the streak and protein in spots embedded in the streak can co-exist in the gel and density measured at any point is the sum of the various contributing factors.

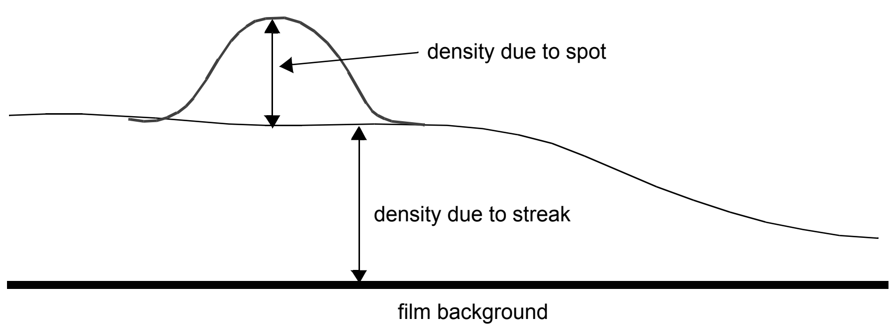

Fig. I-5. Summed streak and spot densities.

If you accept this model, then it is proper to remove the streak before quantitating the spot. This is the default method of spot detection.

The contradictory view is that proteins in a gel are mutually exclusive: no two proteins can co-exist at the same location in the gel. The implication is that the density of the spot does not add to the density of the streak.

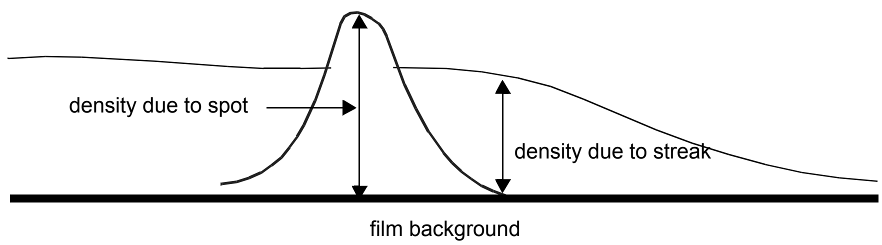

Fig. I-6. Contradictory view: spot and streak proteins do not co-exist.

If you accept this view, then change the auto-detect procedures to use Subtract Background (Ball) rather than Subtract Background (Vert. Disk).

Note that these considerations apply only to spots buried in streaks and does not affect the quantitation of most spots which are either isolated or overlap other spots.

I.5. Pipetting Error Compensation

Pipetting Error Compensation (in the Quantity Normalization dialog) can be used to correct systemic errors such as pipetting or counting errors which affect the quantitation of all spots in the gel equally. It result in a small correction factor which is applied to all spots when calculating quantity in PPM. When properly applied, it improves the accuracy of quantitative comparisons between gels.

However, it relies on two key assumptions: the log of spot quantity data is normally distributed, and changes to the spots are random.

In samples we have studied which are derived from whole cell lysates, the assumption of normal distribution has been true. The user must verify this for his own samples.

Changes to the spots will not be random in experiments where overall protein synthesis is affected, e.g. the introduction of a toxic substance. In this case the Compensation will obscure the experimental result and apply a ‘correction’ factor which removes it.

The Compensation factors are obtained by calculating the average ratios of the log of the quantitation for all spots between all gels in a matchset. From the second assumption, the overall average ratio should be 1.0, indicating no net change. The correction factor for each gel is the reciprocal of its average change.

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